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Jauert et al.(10) **Pub. No.: US 2025/0277244 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **GENETICALLY MODIFIED YEAST AND
FERMENTATION PROCESSES FOR THE
PRODUCTION OF ARABITOL**(71) Applicant: **Cargill, Incorporated**, Wayzata, MN
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9, 2022.**Publication Classification**(51) **Int. Cl.****C12P 7/18** (2006.01)**C12N 9/04** (2006.01)(52) **U.S. Cl.**CPC **C12P 7/18** (2013.01); **C12N 9/0006**
(2013.01); **C12Y 101/01301** (2013.01)

(57)

ABSTRACT

Disclosed herein are genetically engineered yeast cells capable of producing arabitol. The engineered yeast cell may comprise an exogenous polynucleotide sequence encoding an arabitol phosphate dehydrogenase (APDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at CA least 99%, or 100% identical to SEQ ID NO: 11.

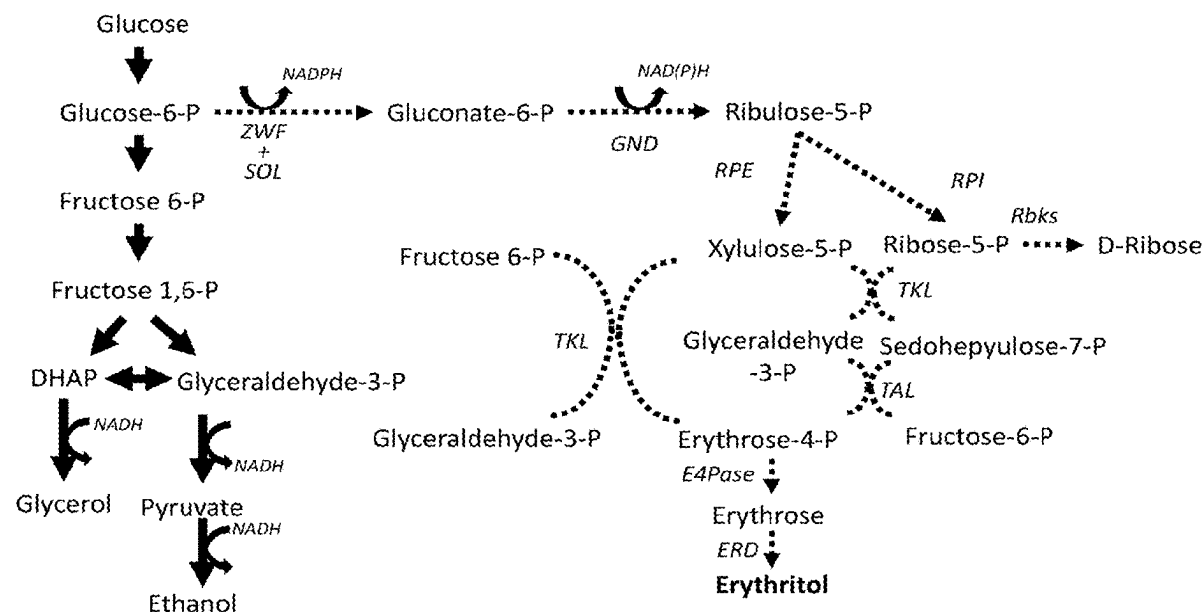
Specification includes a Sequence Listing.

FIG. 1

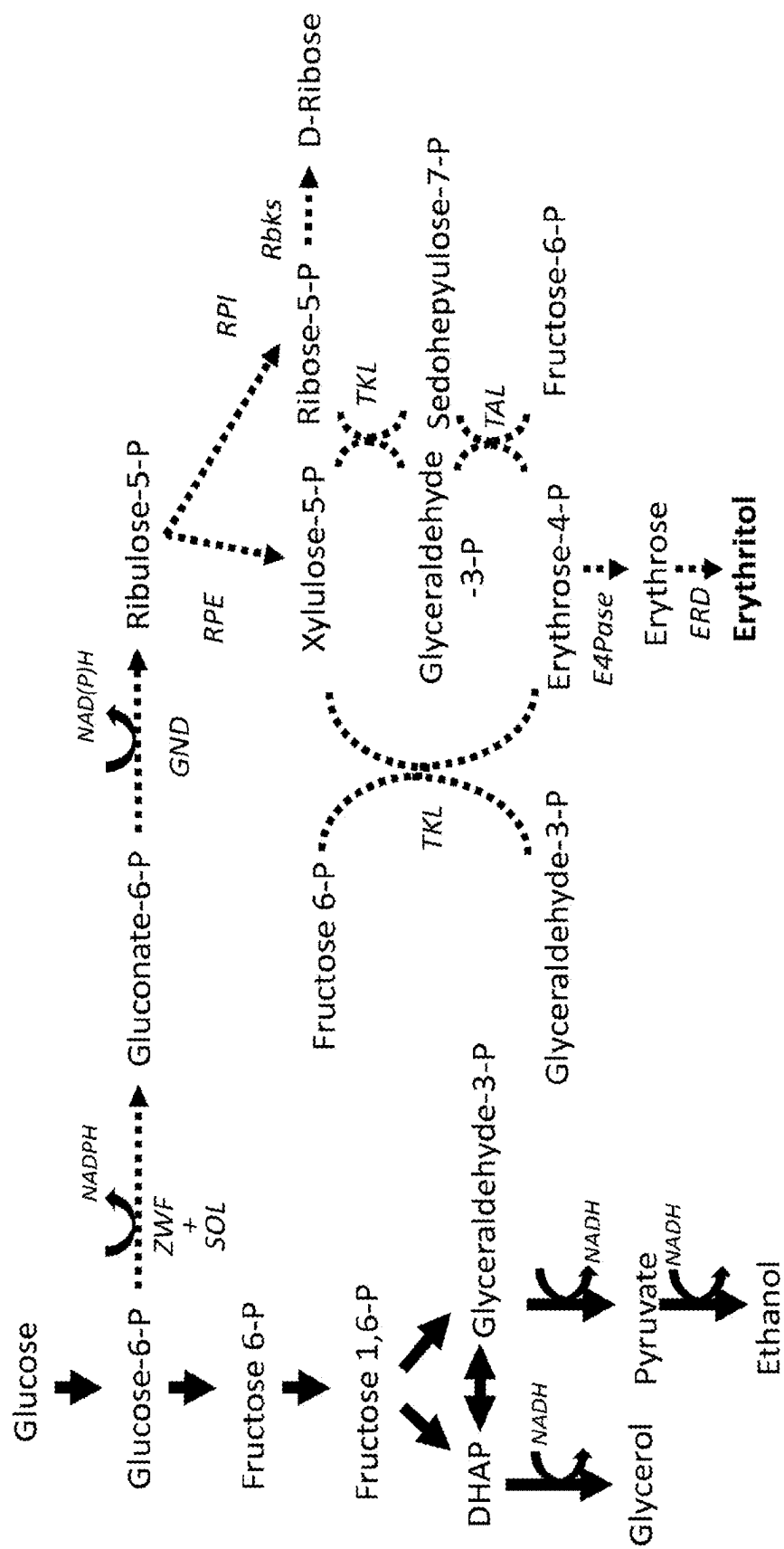


FIG. 2

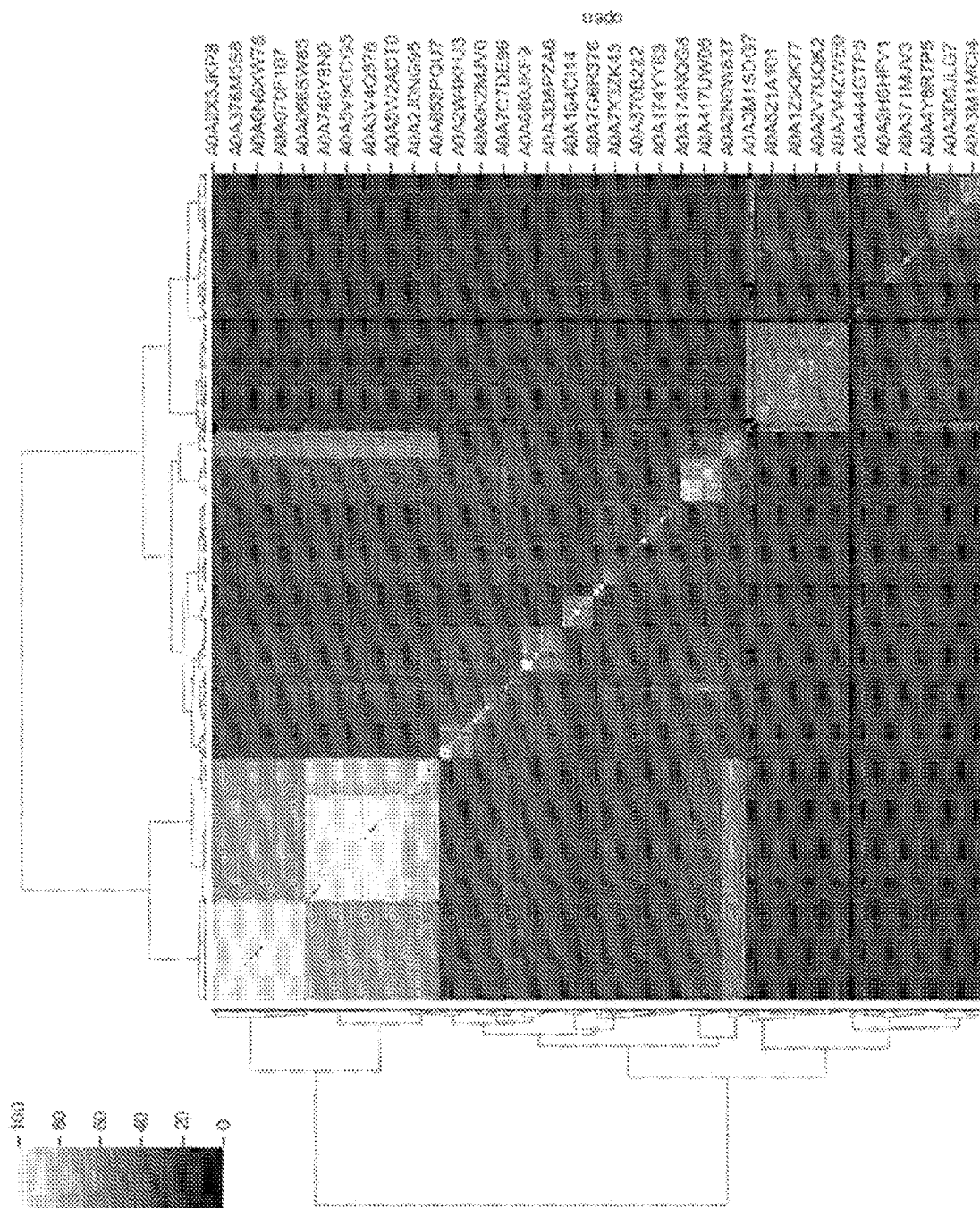


FIG. 3

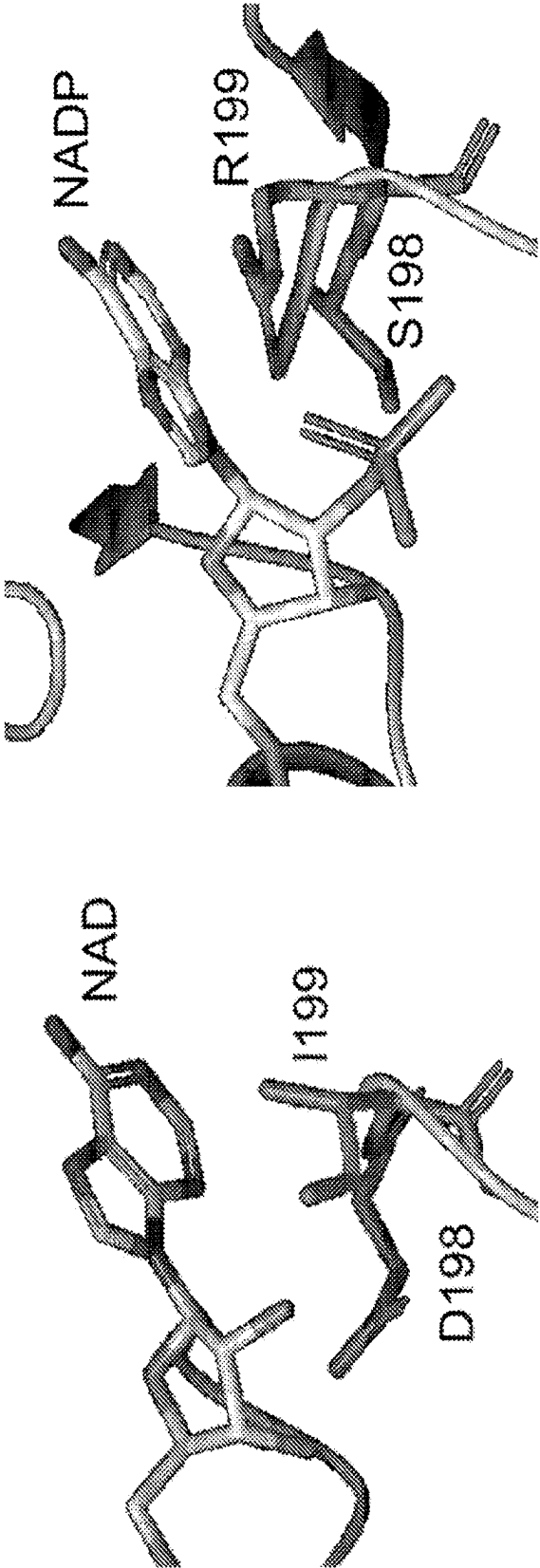


FIG. 4

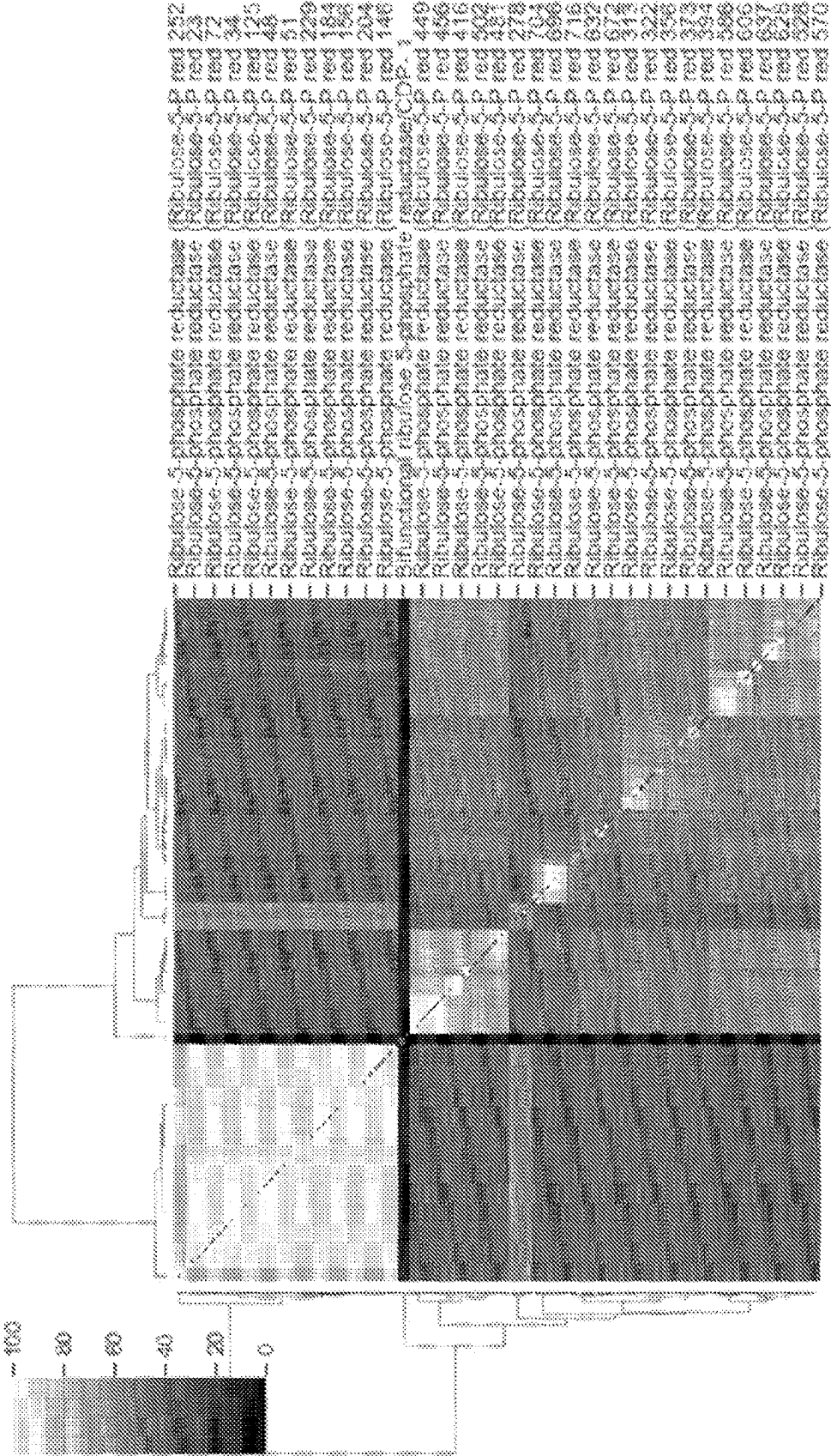


FIG. 5

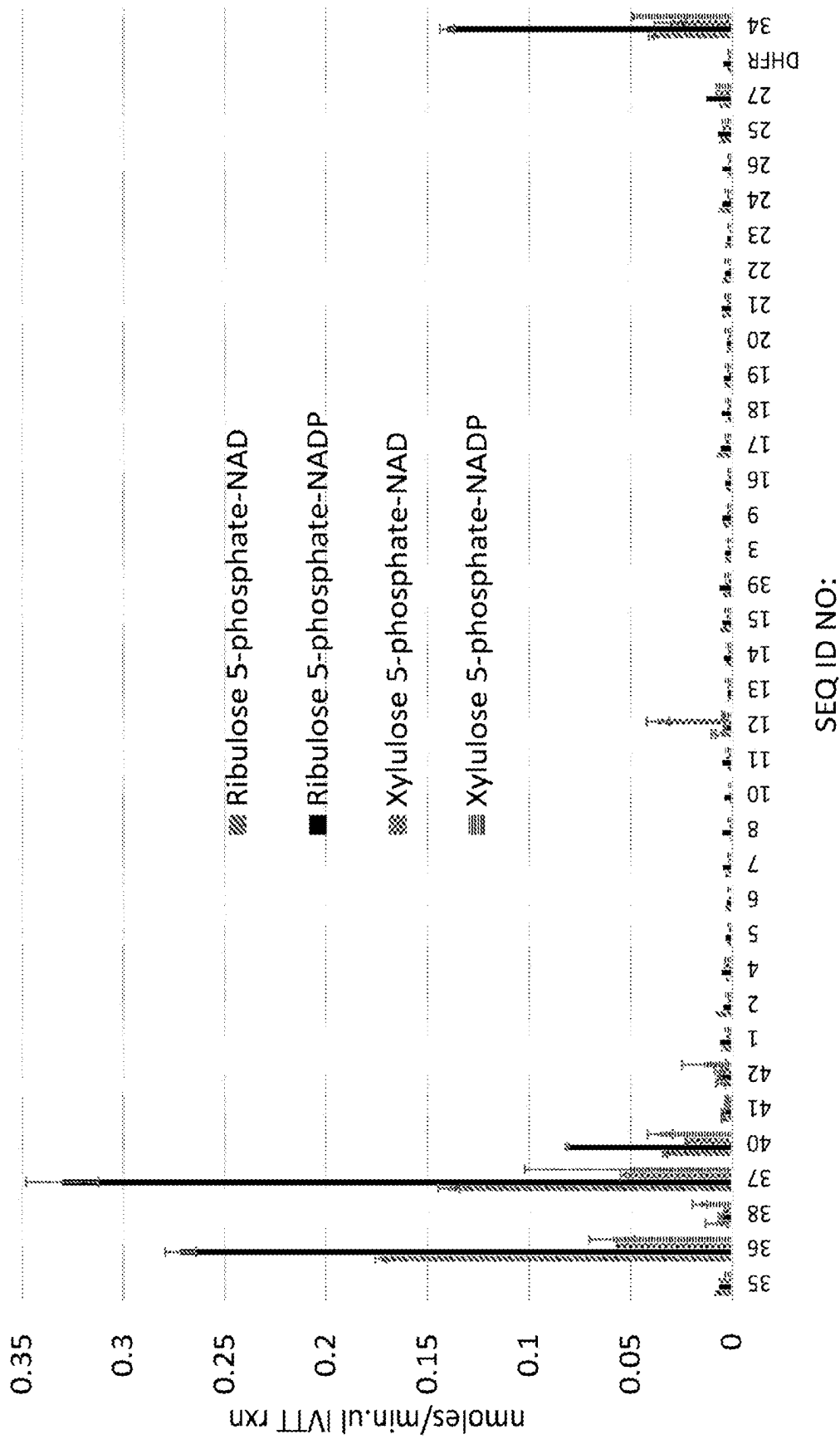


FIG. 6

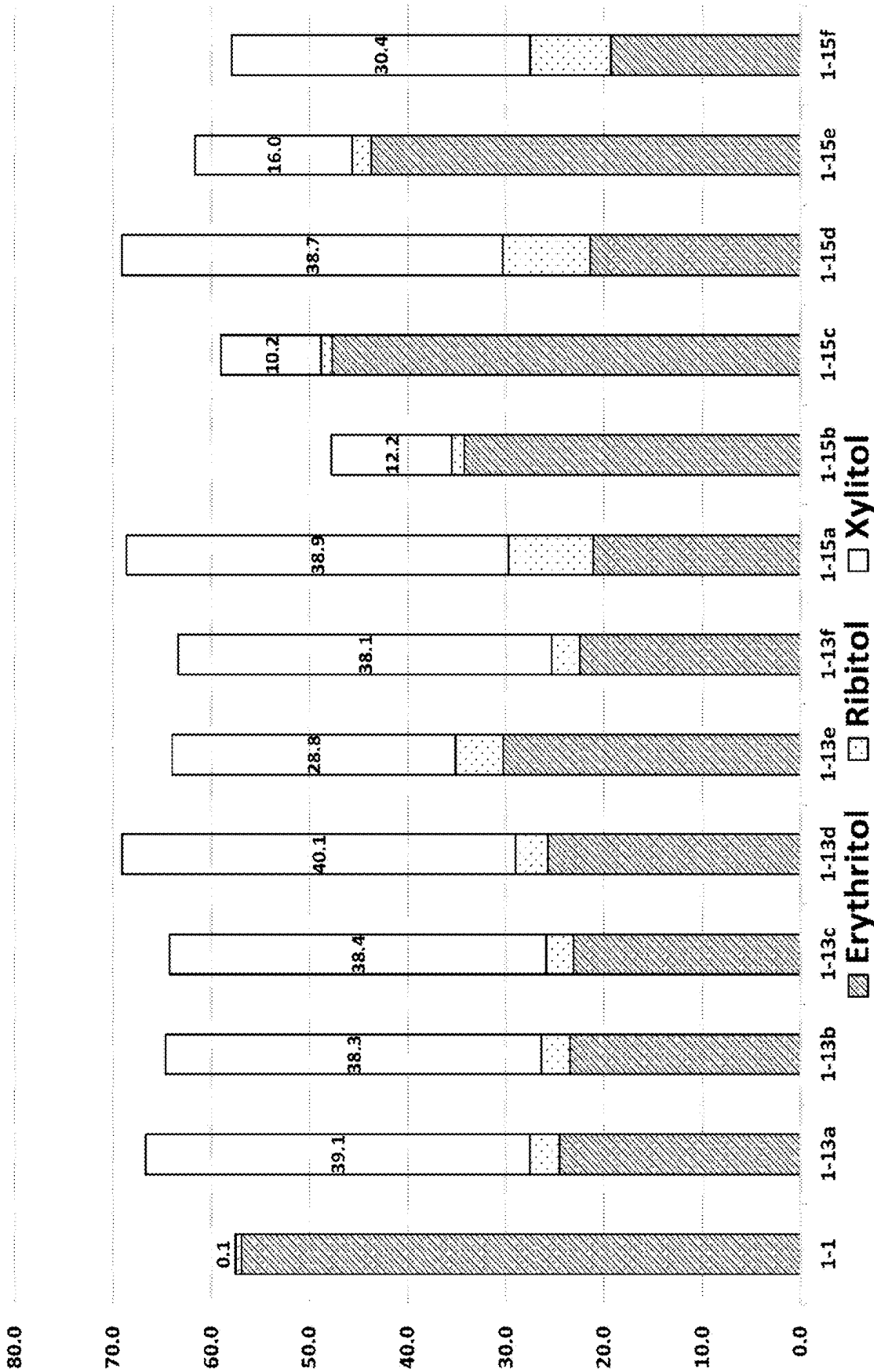


FIG. 7

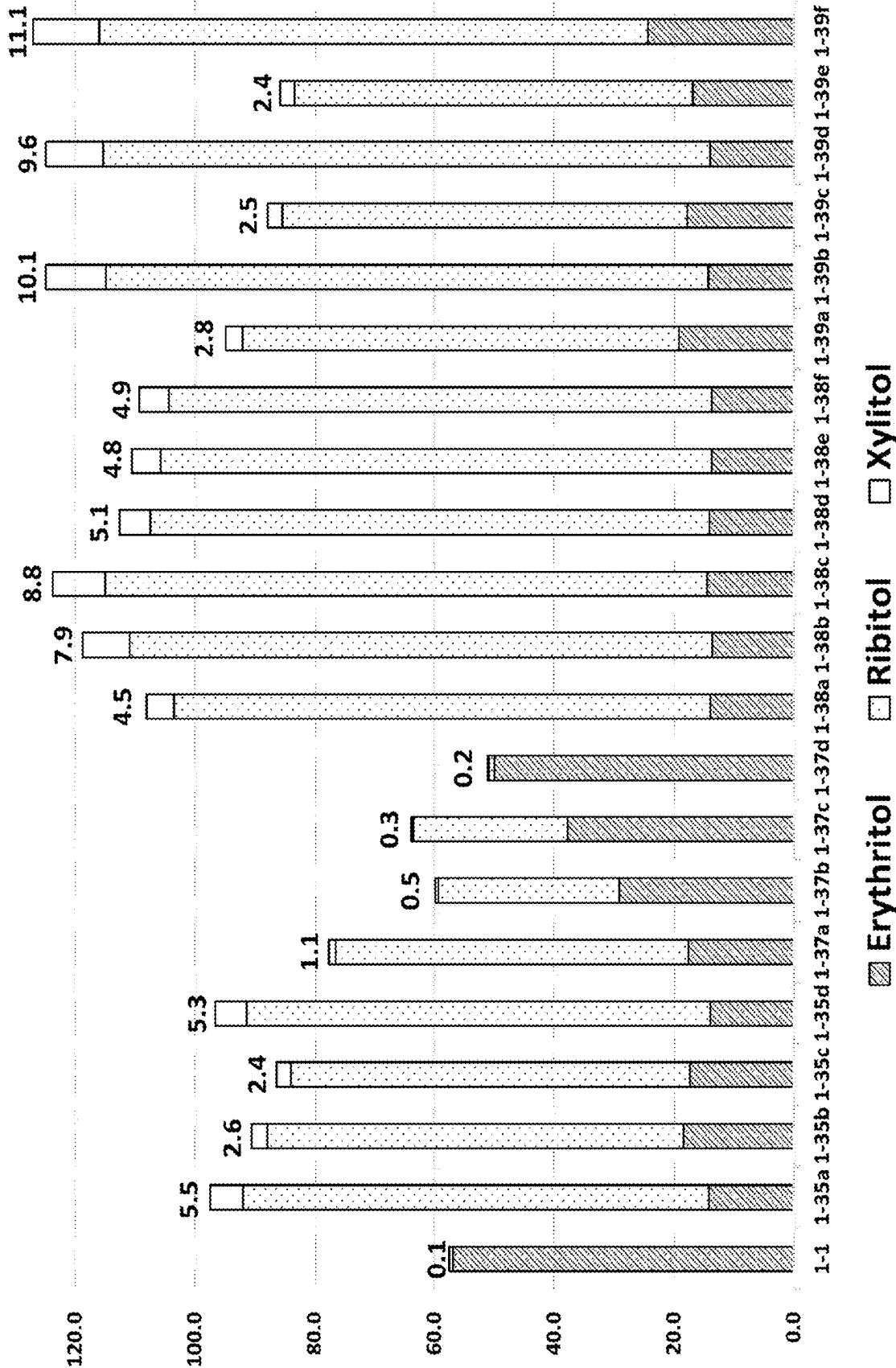


FIG. 8

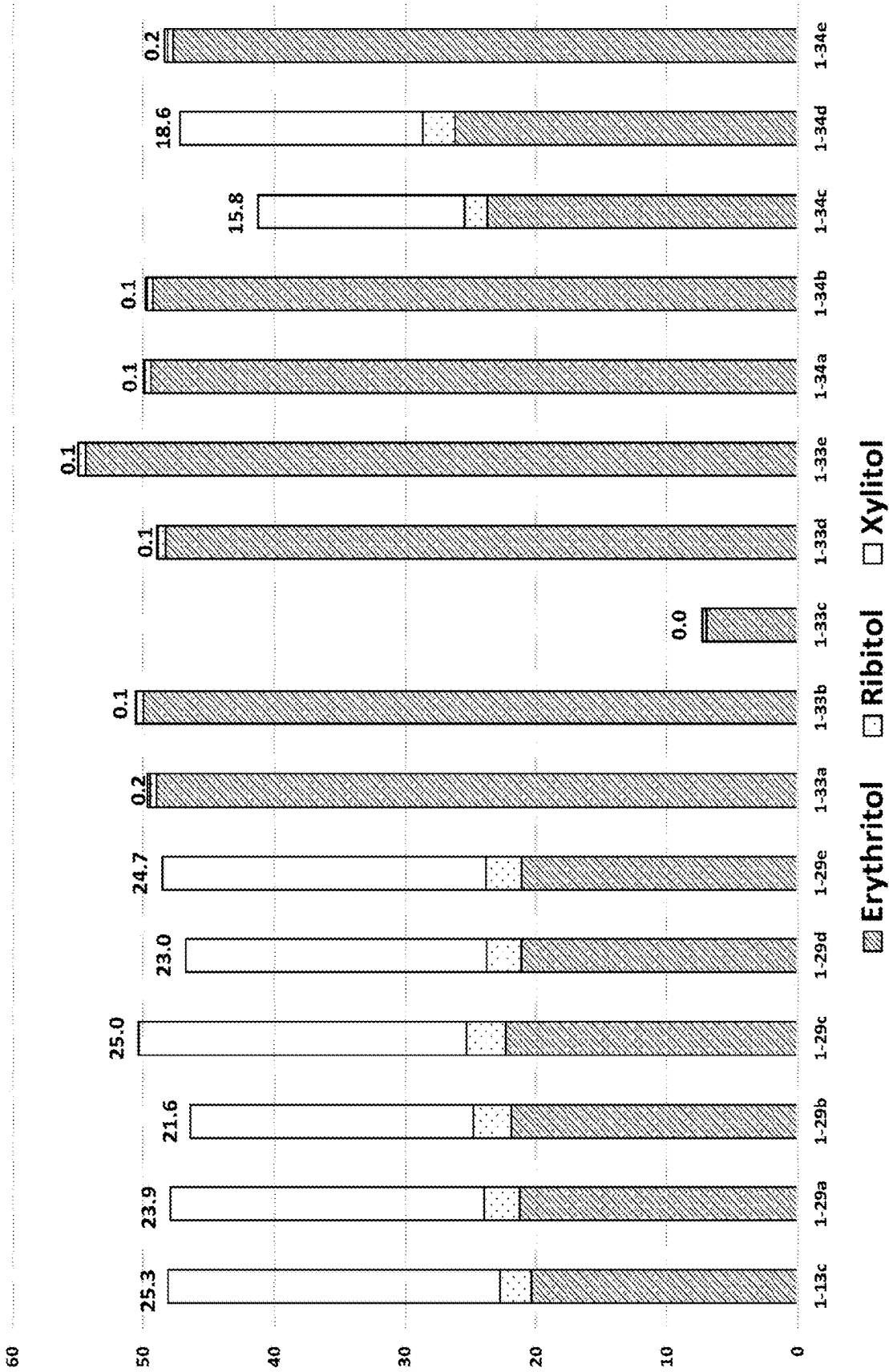


FIG. 9

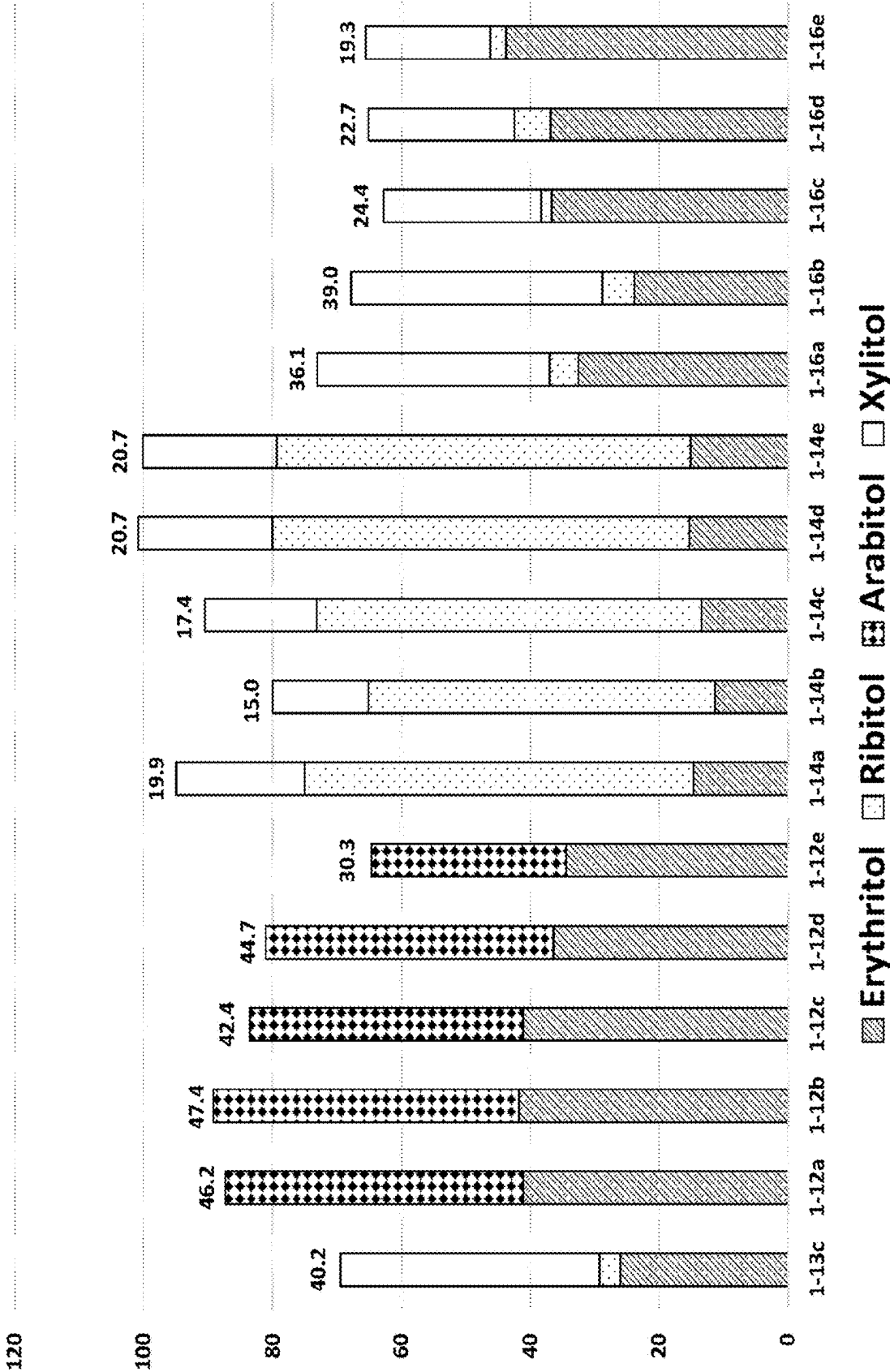


FIG. 10

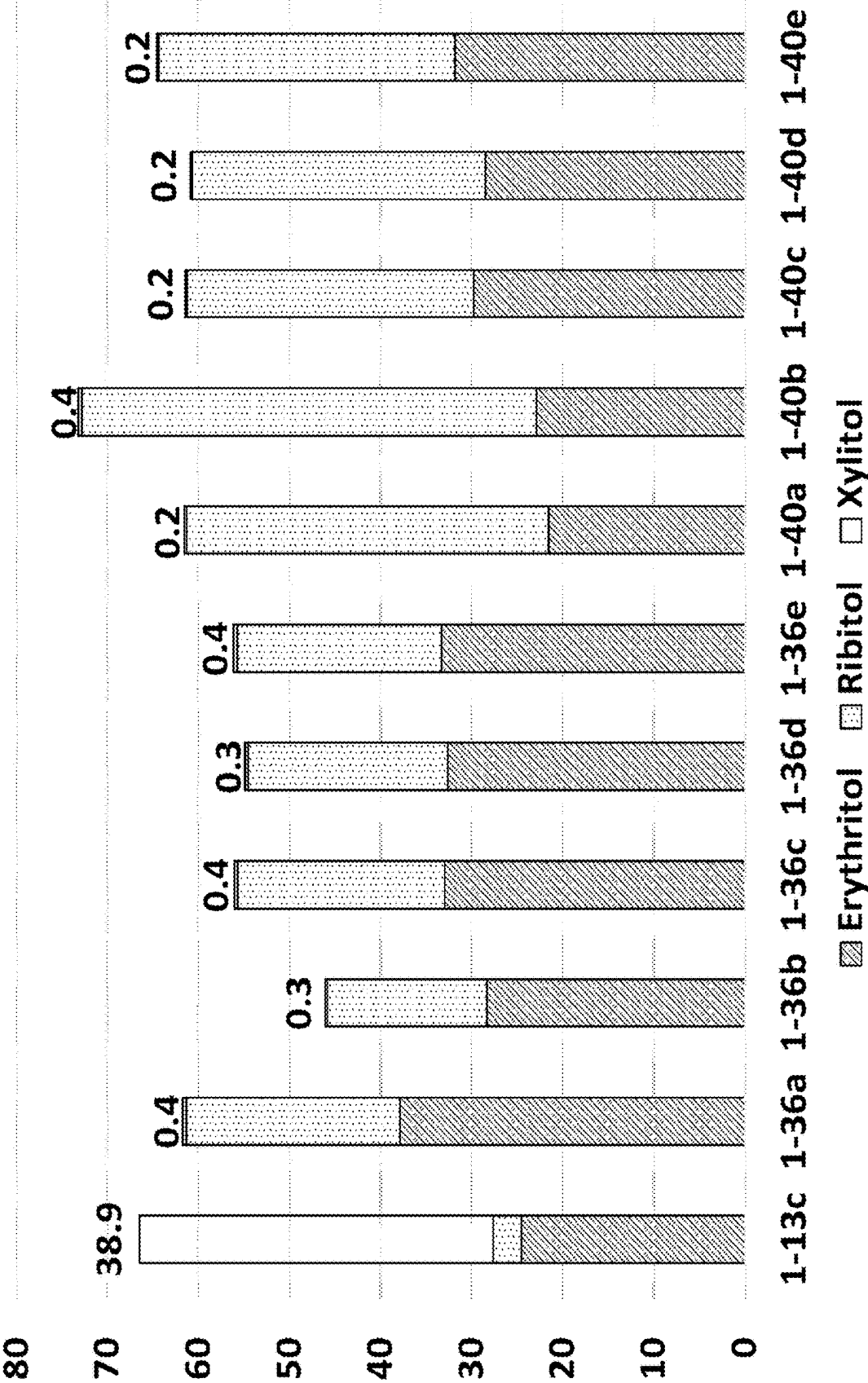
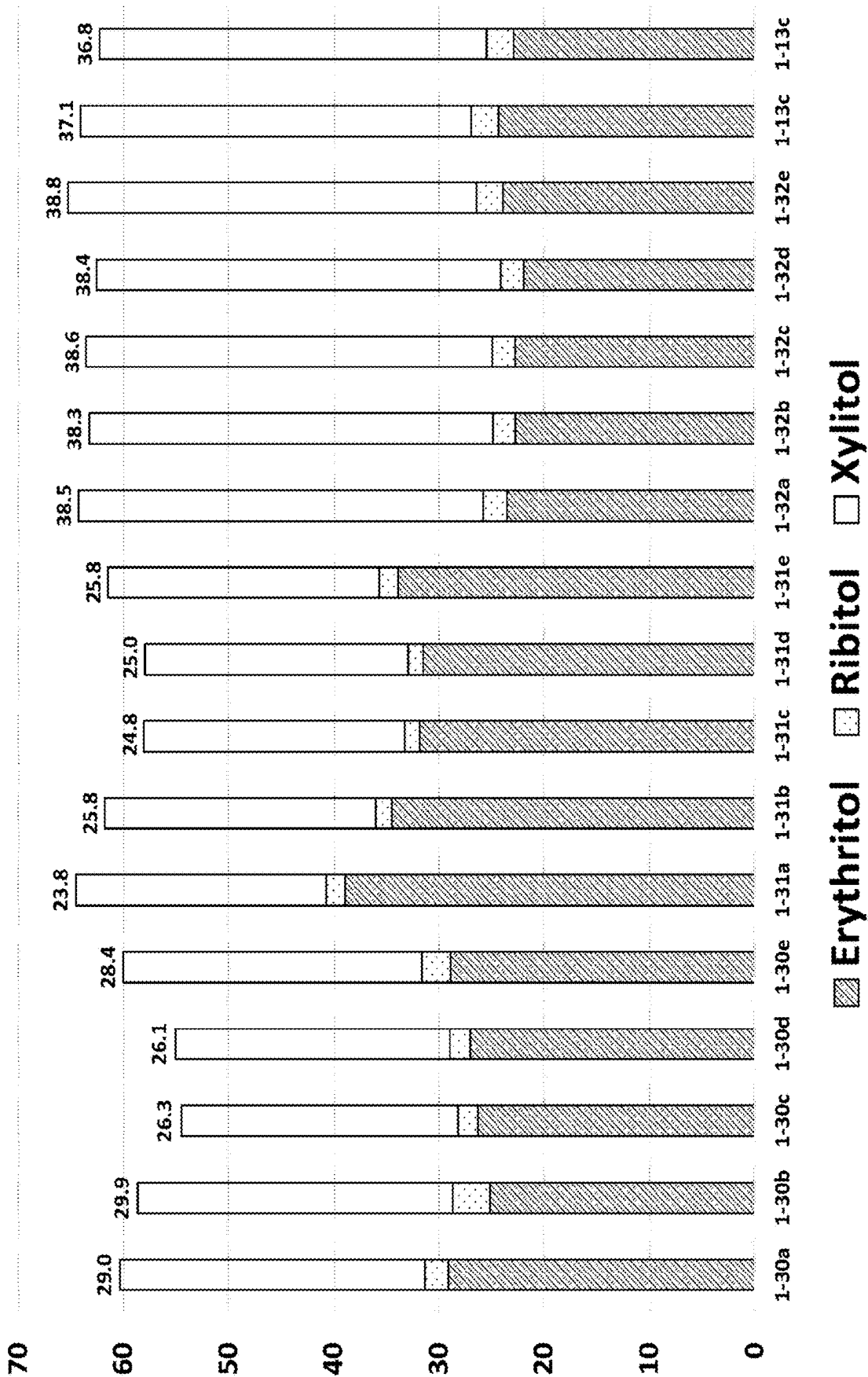


FIG. 11



GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF ARABITOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of US Provisional Application No. 63/364,380, filed May 9, 2022, which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

[0002] The content of the Sequence Listing XML of the sequence listing named "PT-1424-WO-PCT.xml" which is 448,928 bytes in size created on May 4, 2023 and electronically submitted via Patent Center herewith the application is incorporated by reference in its entirety.

BACKGROUND

[0003] Xylitol is a low-calorie sweetener used as a food additive and sugar substitute. Commonly used in drug, dietary supplement, confectionary, and toothpaste compositions, xylitol has also been associated with anticariogenic properties when used in chewing gums. Traditional methods of xylitol production, including chemically catalyzed hydrogenation of xylose hydrolyzed from biomass extracted xylan, are both monetarily and environmentally costly. These methods require high temperatures and pressures, large amounts of water, and metal catalysts that must be mined. In contrast, fermentation processes have been used commercially at large scale to produce other organic molecules, such as ethanol, citric acid, lactic acid, and the like, and may offer a cost effective and sustainable alternative to traditional xylitol processing methods.

[0004] In the development of microorganism-based fermentation strategies for the production of xylitol, production of metabolic pathway intermediates and alternative fermentation products are important considerations. For example, metabolic pathways active in the production of xylitol may have overlap with the metabolic pathways for the production of arabitol, erythritol, ribitol, and the like. The intermediates and products have their own uses and markets that make their fermentation commercially relevant. Accordingly, provided herein are genetically modified yeast and fermentation methods for the production of arabitol while reducing production of erythritol.

SUMMARY

[0005] The present disclosure provides a genetically engineered yeast cell capable producing arabitol, the engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol-phosphate dehydrogenase (APDH) enzyme. The APDH enzyme may comprise a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO:11. The yeast cell may be an osmotolerant yeast cell. The yeast cell may be a cell of the subphylum Ustilaginomycotina. The yeast cell may be selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia*

lipolytica, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*. The yeast cell may be a yeast cell of the genus *Moniliella*.

[0006] The disclosure also provides a genetically engineered *Moniliella* cell capable of producing arabitol, the engineered *Moniliella* cell comprising an exogenous polynucleotide sequence encoding an arabitol-phosphate dehydrogenase enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO:11.

[0007] The arabitol-phosphate dehydrogenase enzyme may have a sequence at least 85% identical to SEQ ID NO: 11. The arabitol-phosphate dehydrogenase enzyme may have a sequence at least 90% identical to SEQ ID NO:11.

[0008] The engineered cell described herein may be a *Moniliella pollinis* cell. The yeast cell may be capable of producing ribitol at a titer of at least 20, 30, 50, 75, or 100 g/L when used in a fermentation process in the presence of dextrose at 35° C. for 96 hours. Erythritol production by the yeast cell may be reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence. The exogenous polynucleotide sequence may be integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus. The exogenous polynucleotide sequence may be operably linked to a heterologous or artificial promoter. The heterologous or artificial promoter may be selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:86), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO: 132), translational elongation factor 1 promoter (TEFp; SEQ ID NO: 133), modified TEFp (SEQ ID NO:131), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO: 134), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:135), enolase promoter (ENO1p; SEQ ID NO: 136), asparagine synthetase promoter (ASNSp; SEQ ID NO:137), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:138), and RPL16B (SEQ ID NO:139).

[0009] The disclosure also provides a method for producing arabitol using the engineered cells described herein, the method comprising contacting a substrate comprising dextrose with an engineered cell described herein, wherein fermentation of the substrate by the engineered yeast produces arabitol. The disclosure also provides a method for producing arabitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol-phosphate dehydrogenase (APDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO:11, wherein fermentation of the substrate by the engineered yeast produces arabitol. The fermentation temperature may be at or between 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C. The volumetric oxygen uptake rate (OUR) may be between 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). Arabitol may be produced at a rate of at least 0.1, 0.2, 0.3, 0.4, 0.5, 0.75, or at least 1.0 g L⁻¹ h⁻¹. Arabitol production may be at least at least 10, 20, 30, 40, 50, 75, or 100 g/L when the fermentation is run at 35° C. for 96 hours. Erythritol production may be reduced

relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Erythritol production may be less than 50, 40, 30, or less than 20 g/L when the fermentation is run at 35° C. for 96 hours. Glycerol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Ethanol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

BRIEF DESCRIPTION OF THE FIGURES

[0010] This patent or application contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawings will be provided by the Office upon request and the payment of the necessary fee.

[0011] The drawings illustrate generally, by way of example, but not by way of limitation, various aspects discussed herein.

[0012] FIG. 1 shows the native pentose phosphate pathway (dotted lines and arrows) and the native glycolysis pathways (solid lines and arrows) in *Moniliella pollinis*.

[0013] FIG. 2 shows diversity in the galactitol-1-phosphate-5-dehydrogenase (G1PDH) /xylitol-phosphate dehydrogenase (XPDH) sequence space.

[0014] FIG. 3 shows the structural characteristics of the NAD or NADP binding pocket located +23 amino acids from the characteristic GXGXXG motif (SEQ ID NO:133) of XPDH enzymes.

[0015] FIG. 4 shows diversity in the ribulose-5-phosphate reductase sequence space.

[0016] FIG. 5 shows in vitro activity of TarJ' and XPDH enzymes as outlined in Example 3.

[0017] FIG. 6 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-13a-f, and 1-15a-f as outlined in Example 5. Data labels report the concentration (g/L) of xylitol.

[0018] FIG. 7 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-35a-d, 1-37a-d, 1-38a-f, and 1-39a-f as outlined in Example 5. Data labels report the concentration (g/L) of xylitol.

[0019] FIG. 8 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-29a-e, 1-33a-e, and 1-34a-e as outlined in Example 6. Data labels report the concentration (g/L) of xylitol.

[0020] FIG. 9 shows erythritol, ribitol, arabitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-12a-e, 1-14a-e, and 1-16a-e as outlined in Example 8. Data labels report the concentration (g/L) of xylitol (strains 1-13c, 1-14a-e, and 1-16a-e) or arabitol (strains 12a-e).

[0021] FIG. 10 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-36a-e, and 1-40a-e as outlined in Example 9. Data labels report the concentration (g/L) of xylitol.

[0022] FIG. 11 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask

fermentations of strains 1-30a-e, 1-31a-e, 1-32a-e, and 1-13c as outlined in Example 10. Data labels report the concentration (g/L) of xylitol.

DETAILED DESCRIPTION

[0023] Reference will now be made in detail to certain aspects of the disclosed subject matter, examples of which are illustrated in part in the accompanying drawings. While the disclosed subject matter will be described in conjunction with the enumerated claims, it will be understood that the exemplified subject matter is not intended to limit the claims to the disclosed subject matter.

[0024] In this document, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference should be considered supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

[0025] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of “about 0.1% to about 5%” or “about 0.1% to 5%” should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “about X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “about X, Y, or about Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

[0026] Unless expressly stated, ppm (parts per million), percentage, and ratios are on a by weight basis. Percentage on a by weight basis is also referred to as wt % or % (wt) below.

[0027] This disclosure relates to various recombinant cells engineered to produce arabitol. In general, the recombinant cells described herein have an active pentose phosphate pathway and are characterized by expression of an exogenous arabitol-phosphate dehydrogenase (APDH) enzyme. The disclosure further provides fermentation methods for the production of arabitol from dextrose using the genetically engineered cells described herein.

[0028] In general, recombinant cells described herein are yeast cells. As used herein, “yeast” refers to eukaryotic single celled microorganisms classified as members of the fungus kingdom. Yeast are unicellular organisms which evolved from multicellular ancestors with some species retaining multicellular characteristics such as forming strings of connected budding cells known as pseudo hyphae or false hyphae. Yeast cells may also be referred to in the art as yeast-like cells, and as used herein “yeast cell” encompasses both yeast and yeast-like cells. Suitable yeast and yeast-like host cells for modification may include, but are not limited to, *Saccharomyces cerevisiae*, *Komagataella* sp., *Kluyveromyces* (e.g., *Kluyveromyces lactis*, *Kluyveromyces*

marxianus), *Yarrowia lipolytica*, *Issatchenkia orientalis*, *Pichia galeiformis*, *Pichia* sp. YB-4149 (NRRL designation), *Pichia pastoris*, *Candida* (e.g., *Candida magnoliae*, *Candida ethanolica*), *Pichia deserticola*, *Pichia membranifadens*, *Pichia fermentans*, *Aspergillus*, *Trichoderma*, *Myceliphthora thermophila*, *Moniliella* (e.g., *Moniliella pollinis*), *Pfaffia*, *Yamadazyma*, *Hansenula*, *Pichia kudriavzevii*, *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*), *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Penicillium*, and *Torula*. An ordinarily skilled artisan would understand the requirements for selection of a suitable yeast cell, and recombinant yeast cells of the present disclosure are not limited to those expressly recited herein. Methods for genetic engineering of yeast cells are known and described in the art and a skilled artisan would understand the methods necessary to transform and engineer a suitable yeast cell.

[0029] A suitable yeast cell may be a cell of the phylum *Basidiomycota* and the subphylum *Ustilaginomycotina*. Suitable yeast of the subphylum *Ustilaginomycotina* include, but are not limited to, *Ustilago* (e.g., *U. cynodontis*, *U. maydis*, *U. sphaerogena*, *U. cordal*, *U. scitaminea*, *U. coicis*, *U. syntherismae*, *U. esculenta*, *U. neglecta*, *U. crus-galli*, *Ustilago avenae*), *Sporisorium* (e.g., *Sporisorium exsertum*), *Moniliella* (e.g., *M. pollinis*, *M. tomentosa*, *M. acetoabutans*, *M. fonsecae*, *M. madida*, *M. megachiliensis*, *M. oedocephalis*, *M. nigrescens*), and *Pseudozyma* (e.g., *Pseudozyma tsukubaensis*), and *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*). Yeast of the subphylum *Ustilaginomycotina* have been known and described in the art as potential production organisms for valuable chemicals such as itaconate, malate, succinate, mannitol, and erythritol and other valuable biotechnological applications. See, for example, Geiser et al. (Prospecting the biodiversity of the fungal family Ustilaginaceae for the production of value-added chemicals,” *Fungal Biol Biotechnol*, 2014, 1:2), Feldbrugge et al., (“The biotechnological use and potential of plant pathogenic smut fungi,” *Appl Microbiol Biotechnol*, 2013, 97(8):3253-65), Guevarra et al., (“Accumulation of itaconic, 2-hydroxyparaconic, itatartaric, and malic acids by strains of the genus *Ustilago*, *Agric. Biol. Chem.*, 1990, 54(9), 2353-2358), and Moon et al., (“Biotechnological production of erythritol and its applications,” *Appl Microbiol Biotechnol*, 2010, 86:1017-1025).

[0030] A suitable yeast cell will have an active pentose phosphate pathway that produces ribulose-5-phosphate. As used herein “active pentose phosphate pathway” refers to expression of one or more functional enzymes which, together, convert glucose-6-phosphate, NADP⁺ or NAD⁺, and water to NADPH or NADH, CO₂, and ribulose-5-phosphate. Continuing in a non-oxidative phase, the pathway may also produce other pentose (i.e., 5-carbon) sugars. For example, the pentose phosphate pathway may produce ribulose-5-phosphate, ribose-5-phosphate, xylulose-5-phosphate, fructose 6-phosphate, combinations thereof, and the like, depending on the enzymatic activities present. The active pentose phosphate pathway may be native to the yeast cell or it may be introduced into the yeast cell by genetic engineering.

[0031] The yeast cell may be an osmotolerant yeast cell. As used herein, “osmotolerant” refers to a yeast capable of growth and reproduction under conditions of high osmolarity, such as at least 10% (w/v), at least 20% (w/v), at least

30% (w/v), at least 40% (w/v), at least 50% (w/v), or at least 60% (w/v) glucose and/or at least 6% (w/v), at least 10% (w/v), at least 12% (w/v), at least 13% (w/v), at least 15% (w/v) sodium chloride. Species and strains of osmotolerant yeast are known and described in the art, including many species of yeast used in industrial fermentation processes. Likewise, methods for assaying yeast osmotolerance are known and described in the art. See, for example, Tiwari, S., et al., (“Nectar yeast community of tropical flowering plants and assessment of their osmotolerance and xylitol-producing potential,” *Current Microbiology*, 2022, 79:28).

[0032] The recombinant yeast cell may be a recombinant *Moniliella* cell, for example, a *Moniliella pollinis* cell. FIG. 1 shows the predicted native pentose phosphate and glycolysis pathways in *Moniliella pollinis*. *Moniliella* has previously been used in the fermentation production of erythritol and methods for genetically modifying and fermenting *Moniliella* are known and described in the art. See, for example, Li et al. (“Methods for genetic transformation of filamentous fungi,” 2017, *Microb Cell Fact*, 16:168).

[0033] Various plasmids and methods for transformation of *Moniliella* are also described in the Examples below. For example, *Moniliella* may be transformed using a bipartite polynucleotide sequence(s) in which, following recombination, the exogenous polynucleotide of interest is integrated at the specified locus and the selection marker is expressible within the cell. Suitable selection markers are known and used in the art. The selectable marker may include, but is not limited to, *amdS* (for example broken into a 3' portion, SEQ ID NO: 167, and a 5' portion, SEQ ID NO:174), G418 resistance gene (for example broken into a 3' portion, SEQ ID NO: 172, and a 5' portion, SEQ ID NO: 175), zeocin resistance gene (for example broken into a 3' portion, SEQ ID NO: 168, and a 5' portion, SEQ ID NO: 169), nourseothricin N-acetyl transferase (NAT) (for example broken into a 3' portion, SEQ ID NO:171, and a 5' portion, SEQ ID NO:170), and invertase gene (SUC2) (for example a 3' portion of SEQ ID NO: 173 and a 5' portion of SEQ ID NO:176).

[0034] The recombinant cells described herein include one or more exogenous polynucleotide sequences encoding one or more exogenous polypeptides that, when expressed improve the fermentation of glucose to ribitol by the recombinant cells.

[0035] The terms “glucose” and “dextrose” are used interchangeably herein and refer to D-glucose except where expressly indicated otherwise.

[0036] As used herein, “exogenous” refers to genetic material or an expression product thereof that originates from outside of the host organism. For example, the exogenous genetic material or expression product thereof can be a modified form of genetic material native to the host organism, it can be derived from another organism, it can be a modified form of a component derived from another organism, or it can be a synthetically derived component. For example, a *K. lactis* invertase gene is exogenous when introduced into *S. cerevisiae*.

[0037] As used herein, “native” refers to genetic material or an expression product thereof that is found, apart from individual-to-individual mutations which do not affect function or expression, within the genome of wild-type cells of the host cell. For the purposes of this application, the *Moniliella pollinis* cell “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is

incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by Eridania Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385, is considered the wild-type *Moniliella pollinis* cell.

[0038] As used herein, the terms “polypeptide” and “peptide” are used interchangeably and refer to the collective primary, secondary, tertiary, and quaternary amino acid sequences and structure necessary to give the recited macromolecule its function and properties. As used herein, “enzyme” or “biosynthetic pathway enzyme” refer to a protein that catalyzes a chemical reaction. The recitation of any particular enzyme, either independently or as part of a biosynthetic pathway is understood to include the co-factors, co-enzymes, and metals necessary for the enzyme to properly function. A summary of the amino acids and their three and one letter symbols as understood in the art is presented in Table 1. The amino acid name, three letter symbol, and one letter symbol are used interchangeably herein.

TABLE 1

Amino Acid three and one letter symbols		
Amino Acid	Three letter symbol	One letter symbol
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

[0039] Variants or sequences having substantial identity or homology with the polypeptides described herein can be utilized in the practice of the disclosed recombinant cells, compositions, and methods. Such sequences can be referred to as variants or modified sequences. That is, a polypeptide sequence can be modified yet still retain the ability to exhibit the desired activity. Generally, the variant or modified sequence may include greater than about 45%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% sequence identity with the wild-type, naturally occurring polypeptide sequence, or with a variant polypeptide as described herein.

[0040] As used herein, the phrases “% sequence identity,” “% identity,” and “percent identity,” are used interchangeably and refer to the percentage of residue matches between at least two amino acid sequences or at least two nucleic acid sequences aligned using a standardized algorithm. Methods of amino acid and nucleic acid sequence alignment are well-known. Sequence alignment and generation of sequence identity include global alignments and local align-

ments which are carried out using computational approaches. An alignment can be performed using BLAST (National Center for Biological Information (NCBI) Basic Local Alignment Search Tool) version 2.2.31 software with default parameters. Amino acid % sequence identity between amino acid sequences can be determined using standard protein BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 6; Max matches in a query range: 0; Matrix: BLOSUM62; Gap Costs: (Existence: 11, Extension: 1); Compositional adjustments: Conditional compositional score matrix adjustment; Filter: none selected; Mask: none selected. Nucleic acid % sequence identity between nucleic acid sequences can be determined using standard nucleotide BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 28; Max matches in a query range: 0; Match/Mismatch Scores: 1, -2; Gap costs: Linear; Filter: Low complexity regions; Mask: Mask for lookup table only. A sequence having an identity score of XX % (for example, 80%) with regard to a reference sequence using the NCBI BLAST version 2.2.31 algorithm with default parameters is considered to be at least XX % identical or, equivalently, have XX % sequence identity to the reference sequence.

[0041] Polypeptide or polynucleotide sequence identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

[0042] The polypeptides disclosed herein may include “variant” polypeptides, “mutants,” and “derivatives thereof.” As used herein the term “wild-type” is a term of the art understood by skilled persons and means the typical form of a polypeptide as it occurs in nature as distinguished from variant or mutant forms. As used herein, a “variant,” “mutant,” or “derivative” refers to a polypeptide molecule having an amino acid sequence that differs from a reference protein or polypeptide molecule. A variant or mutant may have one or more insertions, deletions, or substitutions of an amino acid residue relative to a reference molecule.

[0043] The amino acid sequences of the polypeptide variants, mutants, derivatives, or fragments as contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant, mutant, derivative, or fragment polypeptide may include conservative amino acid substitutions relative to a reference molecule. “Conservative amino acid substitutions” are those substitutions that are a substitution of an amino acid for a different amino acid where the substitution is predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference polypeptide. Conservative amino acid substitutions generally maintain (a) the structure

of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge and/or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

[0044] As used herein, terms “polynucleotide,” “polynucleotide sequence,” and “nucleic acid sequence,” and “nucleic acid,” are used interchangeably and refer to a sequence of nucleotides or any fragment thereof. These phrases also refer to DNA or RNA of natural or synthetic origin, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. The DNA polynucleotides may be a cDNA (e.g., coding DNA) or a genomic DNA sequence (e.g., including both introns and exons).

[0045] A polynucleotide is said to encode a polypeptide if, in its native state or when manipulated by methods known to those skilled in the art, it can be transcribed and/or translated to produce the polypeptide or a fragment thereof. The anti-sense strand of such a polynucleotide is also said to encode the sequence.

[0046] Those of skill in the art understand the degeneracy of the genetic code and that a variety of polynucleotides can encode the same polypeptide. In some aspects, the polynucleotides (e.g., polynucleotides encoding an APDH polypeptide) may be codon-optimized for expression in a particular cell including, without limitation, a plant cell, bacterial cell, fungal cell, or animal cell. While polypeptides encoded by polynucleotide sequences found in various species are disclosed herein any polynucleotide sequences may be used which encodes a desired form of the polypeptides described herein. Thus, non-naturally occurring sequences may be used. These may be desirable, for example, to enhance expression in heterologous expression systems of polypeptides or proteins. Computer programs for generating degenerate coding sequences are available and can be used for this purpose. Pencil, paper, the genetic code, and a human hand can also be used to generate degenerate coding sequences.

[0047] The recombinant cells described herein may include deletions or disruptions in one or more native genes. The phrase “deletion or disruption” refers to the status of a native gene in the recombinant cell that has either a completely eliminated coding region (deletion) or a modification of the gene, its promoter, or its terminator (such as by a deletion, insertion, or mutation) so that the gene no longer produces an active expression product, produces severely reduced quantities of the expression product (e.g., at least a 75% reduction or at least a 90% reduction) or produces an expression product with severely reduced activity (e.g., at least 75% reduced or at least 90% reduced). The deletion or disruption can be achieved by genetic engineering methods, forced evolution, mutagenesis, RNA interference (RNAi), and/or selection and screening. The native gene to be deleted or disrupted may be replaced with an exogenous nucleic acid of interest for the expression of an exogenous gene product (e.g., polypeptide, enzyme, and the like).

[0048] The recombinant cells described herein may include one or more genetic modifications in which an exogenous nucleic acid is integrated into the genome of the host cell. One of skill in the art know how to select suitable loci in a yeast genome for integration of the exogenous nucleic acid. Suitable integration loci may include, but are not limited to, the PDC1, GPD1, CYB2A, CYB2B, g4240, YMR226, MDHB, ATO2, Adh9091, Adh1202, ADE2,

ADH2556, GAL6, MDH1, SCW11, ER1, ER3, pyrF, TRP3, gpdIIA, and gpdIIB loci. For example, in a *M. pollinis* host cells, suitable interaction loci may include, but are not limited to, the ER1 locus (defined as the locus flanked by SEQ ID NO:85 and SEQ ID NO:162), the ER3 locus (defined as the locus flanked by SEQ ID NO: 155 and SEQ ID NO:165), the PDC1 locus (defined as the locus flanked by SEQ ID NO:152 and SEQ ID NO:164), the pyrF locus (defined as the locus flanked by SEQ ID NO: 153 and SEQ ID NO: 163), the TRP3 locus (defined as the locus flanked by SEQ ID NO:156 and SEQ ID NO: 159), the gpdIIA locus (defined as the locus flanked by SEQ ID NO:157 and SEQ ID NO:161); and the gpdIIB locus (defined as the locus flanked by SEQ ID NO:158 and SEQ ID NO:166). The exogenous nucleic acid may also be integrated in an intergenic region or other location in the host cell genome not specifically specified herein. Other suitable integration loci may be determined by one of skill in the art. Furthermore, one of skill in the art would recognize how to use sequences to design primers to verify correct gene integration at the chosen locus.

[0049] The recombinant cell may have one or more copies of a given exogenous nucleic acid sequence integrated in a host chromosome(s) and replicated together with the chromosome(s) into which it has been integrated. For example, the yeast cell may be transformed with nucleic acid construct including a polynucleotide sequence encoding for a polypeptide described herein and the polynucleotide sequence encoding for the polypeptide may be integrated in one or more copies in a host chromosome(s). The recombinant cell may include multiple copies (two or more) of a given polynucleotide sequence encoding a polypeptide described herein. The recombinant cell may have one, two, three, four, five, six, seven, eight, nine, ten, or more copies of a polynucleotide sequence encoding a polypeptide described herein integrated into the genome. The multiple copies of said polynucleotide sequence may all be incorporated at a single locus or may be incorporated at multiple loci.

[0050] The recombinant cells described herein are capable of producing arabitol and include an exogenous polynucleotide sequence encoding an arabitol-phosphate dehydrogenase (APDH) enzyme. The exogenous polynucleotide sequence may be an exogenous arabitol-phosphate dehydrogenase gene. Following production of arabitol 5-phosphate or arabitol 1-phosphate, it is believed a native phosphatase enzyme removes the phosphate to produce arabitol.

[0051] An “arabitol-phosphate dehydrogenase gene” and an “APDH gene” are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with arabitol-phosphate dehydrogenase activity. As used herein “arabitol-phosphate dehydrogenase activity” refers to the ability to catalyze (i) the conversion of xylulose 5-phosphate and NADPH or NADH to arabitol-1-phosphate and NADP⁺ or NAD⁺ and/or (ii) the conversion of ribulose-5-phosphate and NADPH or NADH to arabitol-5-phosphate and NADP⁺ or NAD⁺. The APDH gene may be derived from any suitable source. For example, the ARDH gene may be derived from *Lactobacillus salivarius* cp400.

[0052] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Lactobacillus salivarius* cp400 gene encoding the amino acid of SEQ ID NO:11. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at

least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:11.

[0053] The exogenous polynucleotides in the recombinant cells described herein may be under the control of a promoter. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial promoter. Suitable promoters are known and described in the art. Promoters may include, but are not limited to, pyruvate decarboxylase promoter (PDC), translation elongation factor 2 promoter (TEF2), SED1, alcohol dehydrogenase 1A promoter (ADH1), hexokinase 2 promoter (HXK2), FLO5 promoter, pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:86); 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130); glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO: 132); translational elongation factor 1 promoter (TEFp; SEQ ID NO:133); modified TEFp (SEQ ID NO:131); phosphoglucosyltransferase 1 promoter (PGM1p; SEQ ID NO:134); 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO: 135); enolase promoter (ENO1p; SEQ ID NO: 136); asparagine synthetase promoter (ASNSp; SEQ ID NO:137); 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:138); and RPL16B (SEQ ID NO:139).

[0054] The exogenous nucleic acids in the recombinant cells described herein may be under the control of a terminator. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial terminator. Suitable terminators are known and described in the art. Terminators may include, but are not limited to, GAL10 terminator, PDC terminator, transaldolase terminator (TAL) 6PGD terminator (6PGDt; SEQ ID NO:140); ASNS terminator (ASNSt; SEQ ID NO: 141); ENO1 terminator (ENO1t; SEQ ID NO:142); hexokinase 1 terminator (HXK1t; SEQ ID NO: 143); PGK1 terminator (PGK1t; SEQ ID NO: 144); PGM1 terminator (PGM1t; SEQ ID NO:145); PYK1 terminator (PYK1t; SEQ ID NO:146); RPLA terminator (RPLAt; SEQ ID NO:147); transaldolase 1 terminator (TAL1t; SEQ ID NO:148); TDH3 terminator (TDH3t; SEQ ID NO:149); translation elongation factor 2 terminator (TEF2t; SEQ ID NO:150); and triosephosphate isomerase 1 terminator (TPI1t; SEQ ID NO:151).

[0055] A promoter or terminator is “operably linked” to a given polynucleotide (e.g., a gene) if its position in the genome or expression cassette relative to said polynucleotide is such that the promoter or terminator, as the case may be, performs its transcriptional control function.

[0056] The polypeptides described herein may be provided as part of a construct. As used herein, the term “construct” refers to recombinant polynucleotides including, without limitation, DNA and RNA, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. Recombinant polynucleotides are polynucleotides formed by laboratory methods that include polynucleotide sequences derived from at least two different natural sources or they may be synthetic. Constructs thus may include new modifications to endogenous genes introduced by, for example, genome editing technologies. Constructs may also include recombinant polynucleotides created using, for example, recombinant DNA methodologies. The construct may be a vector including a promoter operably linked to the polynucleotide encoding a polypeptide as described herein. As used herein, the term “vector” refers to a polynucleotide capable of transporting another polynucle-

otide to which it has been linked. The vector may be a plasmid, which refers to a circular double-stranded DNA loop into which additional DNA segments may be integrated.

[0057] The disclosure also provides fermentation methods for the production of arabinol using the recombinant cells described herein. The fermentation methods include the step of fermenting a substrate using the genetically engineered yeasts described herein to produce arabinol. The fermentation method can include additional steps, as would be understood by a person skilled in the art. Non-limiting examples of additional process steps include maintaining the temperature of the fermentation broth within a predetermined range, adjusting the pH during fermentation, and isolating the arabinol from the fermentation broth. The fermentation process may be a fully or partially aerobic process.

[0058] The fermentation method can be run using a suitable fermentation substrate. The substrate of the fermentation method can include glucose, sucrose, galactose, mannose, molasses, xylose, fructose, hydrolysates of starch, lignocellulosic hydrolysates, or a combination thereof. One skilled in the art will recognize what fermentation substrate is suitable for a given fermentation organism and system.

[0059] The fermentation process can be run under various conditions. The fermentation temperature, i.e., the temperature of the fermentation broth during processing, may be ambient temperature. Alternatively, or additionally, the fermentation temperature may be maintained within a predetermined range. For example, the fermentation temperature can be maintained in the range of 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C., preferably about 35° C. However, a skilled artisan will recognize that the fermentation temperature is not limited to any specific range or temperature recited herein and may be modified as appropriate.

[0060] The fermentation process can be run within certain oxygen uptake rate (OUR) ranges. The volumetric OUR of the fermentation process can be in the range of 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). In some embodiments, the specific OUR can be in the range of 0.05 to 10, 0.1 to 8, 0.15 to 5, 0.2 to 1, or 0.3 to 0.75 mmol O₂/(g cell dry weight·h). However, the volumetric or specific OURs of the fermentation process are not limited to any specific rates or ranges recited herein.

[0061] The fermentation process can be run at various cell concentrations. In some embodiments, the cell dry weight at the end of fermentation can be 5 to 40, 8 to 30, or 10 to 20 g cell dry weight/L. Further, the pitch density or pitching rate of the fermentation process can vary. In some embodiments, the pitch density can be 0.05 to 11, 0.1 to 10, or 0.25 to 8 g cell dry weight/L.

[0062] The initial dextrose concentration of the fermentation may be at least 100, 200, 250, 300, 350, or at least 400 g/L dextrose. The initial dextrose concentration may be between 100 to 400, 150 to 350, or 250 to 325 g/L.

[0063] The fermentation process can be associated with various characteristics, such as, but not limited to, fermentation production rate, pathway fermentation yield, final titer, and peak fermentation rate. These characteristics can be affected by the selection of the yeast and/or genetic modification of the yeast used in the fermentation process. These characteristics can be affected by adjusting the fermentation process conditions. These characteristics can be

adjusted via a combination of yeast selection or modification and the selection of fermentation process conditions.

[0064] The arabitol production rate of the process may be at least at least 0.2, 0.3, 0.5, 0.75, or at least 1.0 g L⁻¹ h⁻¹. The arabitol mass yield of the process may be at least 55 percent, at least 65 percent, at least 70 percent, at least 75 percent, at least 80 percent, or at least 85 percent. The final arabitol titer of the process may be at least 20, 30, 50, 75, or 100 g/L.

[0065] The fermentation process can be run as a dextrose-fed batch. Further, the fermentation process can be a batch process, continuous process, or semi-continuous process, as would be understood by a person skilled in the art.

EXAMPLES

[0066] The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Example 1—Xylitol-Phosphate Dehydrogenase Diversity

[0067] Roughly three thousand galactitol-1-phosphate-5-dehydrogenase (G1PDH)/xylitol-phosphate dehydrogenase (XPDH) enzyme sequences were obtained from Uniprot and analyzed. FIG. 2 illustrates the natural sequence diversity for this set of sequences. This set is diverse, with ~25% of the enzymes having no homologue more than 75% identical. As these enzymes tend to prefer NAD to NADP as a cofactor, the cofactor binding preferences of the homologs were assessed in a manner similar to that described by Duax et al., (“Rational proteomics I. Fingerprinting identification and cofactor specificity in the short-chain oxidoreductase (SCOR) enzyme family,” *Proteins*, 2003, 53(4):931-943). Cofactor binding pockets were identified by proximity to the Rossmann fold (+23 to +30 amino acids from the GXGXXG motif (SEQ ID NO:129)) and scored on the basis of total charge in an 8-residue window. The top 8 candidates that were predicted to use NADP were selected for further characterization, along with 4 candidates predicted to use NAD, and 3 controls.

[0068] Upon further review of the structural characteristics of the predicted binding pocket for factors that may influence cofactor preference, an important aspartate residue was identified. See FIG. 3. The polypeptide of SEQ ID

NO:34 and substitutions thereof were used to construct a structural homology model to predict cofactor binding pocket confirmations FIG. 3 shows the C-terminal end of the penultimate -strand on the outside of the Rossmann Fold domain. Without wishing to be bound by any particular theory, it is predicted that enzymes in which the first residue in this region (residue 198 relative to SEQ ID NO:34) is an aspartate and the second residue (residue 199 relative to SEQ ID NO:34) is a large hydrophobic amino acid (for example, isoleucine) will prefer an NAD cofactor due to the hydrogen bonding of the aspartate to the hydroxyl groups of the NAD ribose. However, enzymes in which that first residue (residue 198 relative to SEQ ID NO:34) is an alanine, glycine, or serine and the second residue (residue 199 relative to SEQ ID NO:34) is lysine or arginine will prefer an NADP cofactor as the positive charge on the lysine or arginine residue will interact with the negative charge of the phosphate of the NADP and the smaller residue in the first position allows space in the binding pocket for said phosphate. Based on this analysis, 12 additional enzymes were selected for their predicted preference for NADP. Finally, 6 additional enzymes with sequence similarity to active XPDH enzymes were selected for testing.

Example 2—TarJ’ Diversity

[0069] Roughly eight hundred ribulose 5-phosphate reductase sequences were obtained from Uniprot and analyzed. FIG. 3 illustrates the natural sequence diversity for this set of sequences. Overall, the diversity in this set is low, as only 10% of the enzymes have no sequence similarity more than 75% identical. As these enzymes tend to prefer NADP to NAD as a cofactor, no scoring was performed, and the sequences were simply aligned in Geneious (ClustalW, default settings). Eight enzymes were selected for further analysis based on sequence similarity.

Example 3—In Vitro Enzyme Assays

[0070] Polynucleotides encoding suspected XPDH homologs (Table 2) or TarJ’ homologs (Table 3) were cloned into a vector containing a T7 promoter and terminator for cell-free protein expression (New England Biolabs, PUR-Express® In Vitro Protein Synthesis). Cell-free synthesized proteins were analyzed for activity on four substrates (ribulose 5-phosphate, xylulose 5-phosphate, ribulose, and xylulose) with either NADP or NAD cofactors. Seven enzymes (XPDH of SEQ ID NOs:12 and 34, TarJ’ of SEQ ID NOs:36, 37, 38, 40, and 42), were able to catalyze the reduction of either ribulose 5-phosphate or xylulose 5-phosphate (FIG. 5) but not the reduction of xylulose or ribulose (data not shown).

TABLE 2

XPDH Homologs		
Source Organism	Gene Name/ Accession Numbers	Polypeptide SEQ ID NO:
<i>Mariniphaga sediminis</i>	D1164_16045	1
<i>Candidatus Dormibacteraeota</i> bacterium	DLM67_26945	2
<i>Candidatus Falkowbacteria</i> bacterium	COT93_01110	3
<i>Acidobacteria</i> bacterium	DMG39_01315	4
<i>Spirochaetaceae</i> bacterium	DCP56_03760	5
<i>Candidatus Hydrogenedentes</i> bacterium	ENW01_11985	6
<i>Rhizobium</i> sp.	E6Q76_20070	7
<i>Citrobacter freundii</i>	C2U38_22630	8

TABLE 2-continued

XPDH Homologs		
Source Organism	Gene Name/ Accession Numbers	Polypeptide SEQ ID NO:
<i>Escherichia coli</i>	gatD	9
<i>Escherichia coli</i> (strain K12)	rspB	10
<i>Lactobacillus salivarius</i> cp400	LSCP400_11731	11
<i>Clostridium difficile</i>	WP_004454628.1	12
<i>Clostridium difficile</i>	WP_011860836.1	13
<i>Lactobacillus rhamnosus</i>	AAT02414.1	14
<i>Bacillus halodurans</i>	WP_010896369.1	15
Lachnospiraceae bacterium	A0A3C1WRV0_9FIRM	16
<i>Faecalibacterium</i> sp. OF03-6AC	DXA66_03105	17
<i>Agathobacter ruminis</i>	CSX02_08830	18
<i>Alicyclobacillus acidoterrestris</i>	AAC03nite_18160	19
<i>Ruminococcus</i> sp. KGBM03662	FFK04_01895	20
Clostridiaceae bacterium	D3Z58_08610	21
<i>Thermoplasma</i> archaeon	A0A497HL05	22
Lachnospiraceae bacterium	A0A3B9I8G1	23
<i>Prevotella marshallii</i> DSM 16973 = JCM 13450	E0NTD1	24
<i>Blautia</i> sp. OM05-6	A0A417S5N4	25
<i>Candidatus Bathyarchaeota</i> archaeon	A0A7C4RGE8	26
Lachnospiraceae bacterium	A0A7U9REN4	27

TABLE 3

TarJ' Homologs		
Source Organism	Gene Name/ Accession Numbers	Polypeptide SEQ ID NO:
<i>Staphylococcus aureus</i>	AMV81374.1	34
<i>Staphylococcus aureus</i> subsp. <i>aureus</i> 71193	ST398NM01_0269	35
<i>Eubacterium ventriosum</i>	DW918_	36
<i>Pradoshia</i> sp. D12	F7984_17520	37
<i>Lactobacillus plantarum</i> EGD-AQ4	N692_02285	38
uncultured <i>Ruminococcus</i> sp.	DPQ25_03780	39
<i>Kandleria vitulina</i>	SAMN04487759_1396	40
<i>Lactiplantibacillus plantarum</i>	C7M34_02724 C7M35_00347 C7M36_02765 C7M47_01038 FET70_01115 LPJSA22_01709 Nizo2802_1951	41
Lachnospiraceae bacterium	IMSAGC019_01831	42

Example 4—Genetically Modified *Moniliella pollinis* Strains

[0071] Strain 1-1 is the *Moniliella pollinis* host strain “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l’Université Catholique de Louvain by *Eridania Béghin* Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385. Table 4 below lists various *Moniliella pollinis* strains, including information on the parent strain, the sequence with which the parent strain was transformed, and characterizations of the expression cassette(s) contained on the transformed sequence. Each “XPDH/TarJ’ Homolog Expression Cassette” contained, in order, a 5’ ER1 flanking sequence (SEQ ID NO:85), a MpPYK1 promoter (SEQ ID NO:86), a gene encoding the indicated XPDH or TarJ’ homolog (one of SEQ ID NOs: 87-128), a Mp6PGD terminator (SEQ ID NO: 140), and a 5’ portion of a G418 resistance gene expression cassette (SEQ ID NO: 175). Each “Selectable Marker Cassette” contained,

in order, a 3’ portion of a G418 resistance gene expression cassette (SEQ ID NO: 172), an MpTEF2 terminator (SEQ ID NO:150), and a 3’ ER1 flanking sequence (SEQ ID NO:160). Upon bipartite transformation with both the XPDH/TarJ’ Homolog Expression Cassette and the Selectable Marker Cassette, the two cassettes recombine for integration of both the nucleotide sequence encoding the XPDH or TarJ’ homolog and the G418 resistance marker at the ER1 locus.

[0072] The indicated *Moniliella pollinis* parent strain was transformed with the indicated sequence(s) by first protoplasting the parent strain by adding an enzyme mixture containing 0.6M MgSO₄, 7.5 g/L driselase, and 12.5 g/L *Trichoderma harzianum* lysing enzyme to a mycelial pellet of the parent strain. Protoplasts were then pelleted, washed with 0.6M MgSO₄, and resuspended in STC medium (0.6M sucrose, 50 mM CaCl₂), 10 mM Tris-HCl, pH 7.5). Bipartite transformations were prepared by adding 100 µg single stranded salmon sperm DNA and 1.5 to 5 µg each of the 5’ and 3’ DNA transformation fragments (3-10 µg total; see Table 4 for list of fragments) to approximately 200 µL protoplast mixture (10⁸ cells/mL). 1 mL 50% PEG in STC

medium was then added to the salmon sperm DNA, transformation DNA, and protoplast mixture and the resulting combination was incubated for 15 minutes at room temperature. Following incubation, recovery broth (0.4M sucrose, 1 g/L yeast extract, 1 g/L malt extract, 10 g/L glucose, pH 4.5) was added to the mixture and incubated at 27° C., 100 rpm, for 16 to 24 hours. Following the incubation, protoplasts were pelleted by centrifugation and resuspended in 1 mL PBS.

[0073] The resuspended protoplasts were plated on PDA+250 mg/L geneticin (G418) selection plates and incubated at 30-35° C. for at least 2-4 days until transformants grow. Resulting transformants were evaluated by colony PCR for integration of the indicated sequence. A PCR verified isolate was then designated as the indicated strain number. In some instances, more than one PCR verified isolate, e.g., “sister” isolates, are indicated by letters following the strain number. For example, strain 1-2 has 5 sister isolates, strains 1-2a, 1-2b, 1-2c, 1-2d, and 1-2e.

[0074] For example, Strain 1-1 was transformed with SEQ ID NO:43 and SEQ ID NO:44. SEQ ID NO:43 contains (i) 3' flanking DNA for targeted chromosomal integration into the ER1 locus (SEQ ID NO:162), and (ii) a 3' portion of the G418 resistance gene selectable marker (SEQ ID NO:172). SEQ ID NO:44 contains (i) an expression cassette for the XPDH homolog from *M. sediminis*, SEQ ID NO:87 encoding the amino acid sequence of SEQ ID NO:1, under the control of the PYK1 promoter of SEQ ID NO: 86 and the PGD terminator of SEQ ID NO: 140; (ii) 5' flanking DNA for targeted chromosomal integration into the ER1 locus (SEQ ID NO: 85); and (iii) a 5' portion of the G418 resistance gene selectable marker (SEQ ID NO: 175). Transformants were selected on PDA+250 mg/L geneticin (G418) selection plates and incubated at 30-35° C. for at least 2 days until transformants grow. Resulting transformants were streaked for single colony isolation on PDA+geneticin (G418) plates and single colonies were selected. Selected colonies were evaluated by colony PCR for integration of the indicated sequence. PCR verified isolates were designated strains 1-2a, 1-2b, 1-2c, 1-2d, and 1-2e.

TABLE 4

Strain	Parent Strain	XPDH/TarJ' Homolog Expression Cassette			Selectable Marker
		Transformation SEQ ID NO:	Nucleotide SEQ ID NO:	Encoded Polypeptide SEQ ID NO:	
1-2a-e	1-1	44	87	1	43
1-3a-e	1-1	45	88	2	43
1-4a-e	1-1	46	89	3	43
1-5a-e	1-1	47	90	4	43
1-6a-e	1-1	48	91	5	43
1-7a-e	1-1	49	92	6	43
1-8a-d	1-1	50	93	7	43
1-9a-e	1-1	51	94	8	43
1-10a-e	1-1	52	95	9	43
1-11a-e	1-1	53	96	10	43
1-12a-e	1-1	54	97	11	43
1-13a-f	1-1	55	98	12	43
1-14a-e	1-1	56	99	13	43
1-15a-f	1-1	57	100	14	43
1-16a-e	1-1	58	101	15	43
1-17a-e	1-1	59	102	16	43
1-18a-e	1-1	60	103	17	43
1-19a-e	1-1	61	104	18	43
1-20a-e	1-1	62	105	19	43
1-21a-e	1-1	63	106	20	43
1-22a-e	1-1	64	107	21	43
1-23a-e	1-1	65	108	22	43
1-24a-e	1-1	66	109	23	43
1-25a-e	1-1	67	110	24	43
1-26a-e	1-1	68	111	25	43
1-27a-e	1-1	69	112	26	43
1-28a-e	1-1	70	113	27	43
1-29a-e	1-1	71	114	28 (<i>Alkalihalobacillus ligniniphilus</i> XPDH WP_017729640.1)	43
1-30a-e	1-1	72	115	29 (<i>Jeotgalibacillus soli</i> XPDH WP_041086875.1)	43
1-31a-e	1-1	73	116	30 (<i>Heyndrickxia sporothermodurans</i> XPDH WP_066235909.1)	43
1-32a-e	1-1	74	117	31 (<i>Clostridium fungisolvans</i> XPDH WP_183276691.1)	43
	1-1	75	118	32 (<i>Alkalihalobacillus nanhaiisediminis</i> XPDH WP144449381.1)	43
1-33a-e					
1-34a-e	1-1	76	119	33 (<i>Neobacillus cucumis</i> XHPD WP_198313751.1)	43

TABLE 4-continued

Strain	Parent Strain	XPDH/TarJ' Homolog Expression Cassette			Selectable Marker
		Transformation SEQ ID NO:	Nucleotide SEQ ID NO:	Encoded Polypeptide SEQ ID NO:	
1-35a-d	1-1	77	120	34	43
1-36a-e	1-1	78	121	35	43
1-37a-d	1-1	79	122	36	43
1-38a-f	1-1	80	123	37	43
1-39a-f	1-1	81	124	38	43
1-40a-e	1-1	82	125	39	43
1-41a-e	1-1	83	127	41	43
1-42a-f	1-1	84	128	42	43

Example 5—Shake Flask Fermentation Assay

[0075] Strains 1-1, 1-35a-d, 1-37a-d, 1-38a-f, 1-39a-f, 1-42a-f, 1-13a-f, and 1-15a-f (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0076] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0077] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 6 and FIGS. 6 and 7.

TABLE 5

Production Medium	
Component	Concentration (units)
Glucose	300 (g/L)
KH ₂ PO ₄	1.27 (g/L)
(NH ₄) ₂ HPO ₄	0.13 (g/L)
(NH ₄) ₂ SO ₄	1.80 (g/L)
Urea	2.85 (g/L)
Citric Acid	200.0 (mg/L)
MgSO ₄ —7H ₂ O	690.2 (mg/L)
FeSO ₄ —7H ₂ O	32.35 (mg/L)
ZnSO ₄ —7H ₂ O	4.40 (mg/L)
MnSO ₄ —H ₂ O	0.53 (mg/L)
CuSO ₄ —5H ₂ O	0.45 (mg/L)
CaCl ₂ —2H ₂ O	91.60 (mg/L)
Na ₂ SO ₄	154.50 (mg/L)
Thiamine-HCl	10 (mg/L)
Choline-Cl	100 (mg/L)
Antifoam CF-32	1.00 (g/L)

[0078] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated

strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-35a, 1-37a-d, 1-38a-c, 1-39d-f, 1-42a-b, 1-42d, 1-13a-b, 1-13d-e, 1-15b-c, and 1-15e-f include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus.

TABLE 6

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-1	54.2270	0.5672	0.1156	56.9108	14.4101	45.2717
1-35a	19.0690	77.8018	5.5078	14.1262	11.6922	45.6143
1-35b	22.4343	69.5363	2.6051	18.4159	6.7667	51.1619
1-35c	31.4030	66.5826	2.4193	17.3705	7.5255	49.2545
1-35d	28.2827	77.4169	5.2674	13.9265	7.1954	49.1733
1-37a	42.3156	58.8415	1.1308	17.6218	8.1394	44.6216
1-37b	92.0525	30.1794	0.5249	29.1345	17.4800	28.4251
1-37c	77.9520	25.8442	0.3365	37.6779	28.5369	33.4608
1-37d	62.2726	0.9251	0.2062	50.0038	18.7245	45.7655
1-38a	19.5123	89.5643	4.5079	13.9258	6.0135	46.2436
1-38b	15.4952	97.2570	7.8962	13.6124	6.7302	35.1221
1-38c	9.1530	100.4624	8.8382	14.4767	6.4634	41.8789
1-38d	3.5262	93.3559	5.1135	14.0524	7.8424	46.9099
1-38e	5.8345	91.9564	4.7706	13.7265	7.3560	48.8018
1-38f	5.6278	90.6644	4.9442	13.6917	7.7348	47.5447
1-39a	18.6244	72.8649	2.8000	19.1466	8.1946	40.2287
1-39b	2.3933	100.5639	10.0874	14.2583	6.9797	44.4668
1-39c	31.7163	67.5513	2.4932	17.8548	7.1724	47.5427
1-39d	3.3290	101.3044	9.6496	13.9511	6.9558	46.4375
1-39e	38.9049	66.5485	2.4033	16.8345	7.1407	48.7270
1-39f	0.8569	91.5653	11.1004	24.3341	6.1712	37.6166
1-42a	53.2303	1.3617	0.1586	62.1601	16.1305	44.7835
1-42b	70.3640	0.5390	0.1319	52.0362	13.1695	44.2371
1-42c	54.3271	0.6740	0.1317	61.0264	17.1397	44.9797
1-42d	55.3052	0.7063	0.1484	63.2909	12.3513	43.1762
1-42e	61.0003	0.5738	0.1502	56.8237	13.3096	39.6208
1-42f	64.0496	0.5451	0.1352	55.2579	13.6874	42.7263
1-13a	58.2322	3.0713	39.0606	24.5223	9.9512	43.5850
1-13b	64.5326	2.8987	38.2994	23.4455	9.2364	42.1528
1-13c	62.8811	2.8168	38.3638	23.0771	9.2614	43.8806
1-13d	47.8236	3.3178	40.0543	25.7124	9.9847	40.8596
1-13e	59.3667	4.8671	28.8414	30.2839	9.5968	42.2960
1-13f	66.1336	2.8202	38.0642	22.4811	8.8765	42.3954
1-15a	58.6182	8.6323	38.8703	21.1152	9.3006	43.9871
1-15b	78.7837	1.2617	12.2069	34.2608	11.5806	43.9409
1-15c	55.9848	1.1212	10.2043	47.6605	12.0438	44.2130
1-15d	54.0847	8.9671	38.7047	21.4173	9.3022	44.1645
1-15e	60.7735	1.9279	16.0149	43.7036	11.1713	46.0607
1-15f	102.6556	8.2515	30.3843	19.3150	13.0177	32.7172

Example 6—Shake Flask Fermentation Assay

[0079] Strains 1-13c, 1-29a-e, 1-33a-e, and 1-34a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0080] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0081] A 250 mL non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 7 and FIG. 8.

[0082] As seen in FIG. 8, while sister strains 1-34c and 1-34d produced 15.8 and 18.6 g/L xylitol, respectively, strains 1-34a, 1-34b, and 1-34e did not produce significantly more xylitol than wild-type (strain 1-1, FIG. 6). While strains 1-34a, 1-34b, and 1-34e were initially PCR verified, it was later determined that the integrated polynucleotide, which should encode the *N. cucumis* XPDH homolog, contained a frameshift mutation and no functional XPDH was expressed. Therefore, while the results appear varied, they are in fact consistent given that strains 1-34a, 1-34b, and 1-34e did not contain a polynucleotide that encoded a functional XPDH.

TABLE 7

96-hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	104.7609	2.4041	25.3431	20.3872	8.3483	28.8597
1-29a	107.59	2.7375	23.9236	21.2592	8.327	29.7606
1-29b	117.0722	2.8833	21.624	21.9028	7.5941	30.1983
1-29c	92.8027	3.0122	25.0203	22.3197	8.607	34.1865
1-29d	113.7418	2.654	22.973	21.1195	7.992	30.2967
1-29e	106.1983	2.7505	24.6828	21.1105	8.3573	30.0138
1-33a	102.5508	0.4847	0.2175	48.9982	9.7203	31.4812
1-33b	101.4479	0.507	0.0751	50.0129	9.8179	31.3012
1-33c	231.4215	0.2896	0.0353	6.9913	16.2002	6.2899
1-33d	107.2532	0.5626	0.0542	48.3365	10.36	31.982
1-33e	89.76	0.5258	0.0847	54.4229	10.6124	27.6685
1-34a	97.4677	0.487	0.0849	49.4197	10.4031	31.8369
1-34b	99.6607	0.4724	0.0682	49.2765	9.7575	31.2798
1-34c	125.3303	1.7402	15.7865	23.7361	8.5658	28.3016
1-34d	109.6517	2.4549	18.5615	26.2014	8.1791	30.7102
1-34e	92.4856	0.434	0.22	47.7303	9.2374	32.5853

Example 7—Shake Flask Fermentation Assay

[0083] Strains 1-13c, 1-17a-e, 1-18a-e, 19a-e, 1-21a-e, 1-22a-e, 1-23a-e, 1-24a-e, 1-25a-e, and 1-27a-d (outlined in

Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0084] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0085] A 250 mL non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 8.

TABLE 8

96-hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13a	98.6422	2.6387	24.7712	25.2307	13.8005	28.8903
1-17a	74.915	0.5991	0.2543	60.8569	13.0048	33.3547
1-17b	79.2438	0.577	0.0725	58.9302	12.8216	35.3409
1-17c	151.0953	0.4237	0.0975	24.7655	21.0532	20.2202
1-17d	90.0013	1.4266	0.0645	55.808	18.8406	27.8706
1-17e	84.2381	0.7091	0.0428	59.7786	15.6082	29.9387
1-18a	187.0194	0.3163	0.1069	21.6446	15.8366	10.9522
1-18b	83.2939	0.6768	0.0524	63.3983	14.7387	31.6461
1-18c	88.21	0.6902	0.0625	59.74	14.7915	31.5115
1-18d	76.4267	0.8183	0.0692	57.1505	16.2282	34.0999
1-18e	89.0621	0.7336	0.0841	42.5495	26.7038	25.7627
1-19a	94.518	0.6166	0.0801	54.4468	18.8522	28.9871
1-19b	83.559	0.6632	0.0659	60.2352	14.9726	31.2065
1-19c	93.3168	0.7412	0.0646	60.6521	14.3042	29.9464
1-19d	165.946	0.4311	0.1236	24.1998	15.9581	19.0694
1-19e	78.3357	0.675	0.0637	61.3653	15.847	29.361
1-21a	77.8724	0.882	0.0789	65.2592	16.0186	31.302
1-21b	85.2124	0.6777	0.0619	56.1455	15.4518	33.3606
1-21c	85.008	1.0769	0.0802	58.6289	14.8934	32.8785
1-21d	83.4408	1.9855	0.0373	56.0976	14.2115	34.3856
1-21e	60.5121	2.039	0.0418	63.3641	15.5567	31.3166
1-22a	96.2241	0.8385	0.0644	62.4003	13.1106	29.8987
1-22b	90.5783	0.5506	0.0722	54.306	13.9584	34.1293
1-22c	85.3285	0.6442	0.0595	60.2244	15.0443	31.2101
1-22d	87.0015	0.6859	0.0824	59.9578	14.9889	31.3812
1-22e	78.5983	0.5432	0.0718	58.3192	14.9655	30.6604
1-23a	84.0794	0.4999	0.109	52.3855	13.872	35.266
1-23b	84.7042	0.5799	0.0696	58.9566	15.0869	33.9501
1-23c	87.9261	0.7983	0.0538	55.5429	17.3164	30.1523
1-23d	87.2081	0.6745	0.0623	59.4328	14.8259	31.3395
1-23e	82.9585	1.0568	0.0795	62.242	18.8816	23.5949
1-24a	86.7546	0.534	0.0725	53.0638	13.0757	34.4192
1-24b	104.4802	0.583	0.0821	48.7903	14.9732	29.8021
1-24c	104.5435	0.563	0.0872	48.1204	14.5086	31.0036
1-24d	169.2148	0.425	0.1078	25.2327	14.0784	19.6567
1-24e	78.1939	0.5172	0.082	55.4618	15.2743	36.423
1-25a	85.5913	0.8201	0.0711	57.2974	18.9442	29.1298
1-25b	83.3265	0.6891	0.0801	60.4083	15.034	31.4877
1-25c	139.0213	0.6194	0.1079	28.2606	22.9621	22.0197
1-25d	137.1956	0.6387	0.1254	28.3686	22.7144	22.2391

TABLE 8-continued

96-hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-25e	78.0893	0.6945	0.0792	61.3866	15.5101	31.7923
1-27a	54.0049	0.8002	0.0653	73.3789	15.1766	36.3078
1-27b	88.2258	0.5067	0.0543	53.4474	13.5761	34.0372
1-27c	114.4683	0.7334	0.0729	43.7274	17.9455	26.3899
1-27d	76.682	0.5437	0.0552	58.8882	12.8684	38.8414

Example 8—Shake Flask Fermentation Assay

[0086] Strains 1-13c, 1-3a-e, 1-10a-e, 1-11a-e, 1-12a-e, 1-14a-e, 1-16a-e, 1-28a-e, and 1-2a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0087] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0088] A 250 mL non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 9 and FIG. 9.

[0089] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strain 1-16b-e includes the transformed polynucleotide sequence, but it is not at the ER1 locus. Further analysis was inconclusive on the integration location in strains 1-2c and 1-2d.

TABLE 9

96-hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	52.365	3.1859	40.2435	26.0209	11.1145	37.6398
1-3a	58.7394	0.6129	0.3235	59.7362	15.12	40.891
1-3b	40.836	0.4677	0.1051	49.0415	15.6583	51.2491
1-3c	55.8652	0.5887	0.1253	59.2426	14.1511	42.761
1-3d	47.4122	0.5038	0.1279	50.8589	16.8177	41.1038
1-3e	45.8606	0.5127	0.1267	50.1914	16.3743	48.2382
1-10a	58.5369	0.5891	0.1201	57.9712	15.3885	40.4663
1-10b	58.9318	0.6075	0.1492	57.9368	14.8489	41.9797

TABLE 9-continued

96-hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-10c	67.992	0.4588	0.142	49.0939	14.307	40.394
1-10d	46.0003	0.6906	0.0995	62.4692	19.849	32.9163
1-10e	55.0568	0.655	0.136	61.0313	15.9185	38.2694
1-11a	51.9376	0.429	0.0526	60.4417	14.8425	40.9479
1-11b	45.6503	0.554	0.1326	61.3534	18.1726	44.3344
1-11c	5.6815	1.5471	0.0801	86.3607	27.719	41.9823
1-11d	40.3772	0.5489	0.1304	61.8324	18.171	41.9493
1-11e	53.0558	0.6124	0.115	60.6504	15.489	39.7766
1-12a	30.456	n.a.	n.a.	41.1173	9.6187	44.7618
1-12b	28.2127	n.a.	n.a.	41.7725	9.3555	45.1461
1-12c	45.2228	n.a.	n.a.	41.1422	9.0341	39.2179
1-12d	30.6334	n.a.	n.a.	36.381	10.0025	39.6166
1-12e	81.9896	n.a.	n.a.	34.348	12.8173	30.9024
1-14a	36.1578	60.3639	19.8883	14.677	6.9789	39.9443
1-14b	29.7316	53.6709	14.9948	11.3645	12.08	43.9787
1-14c	27.9273	59.5839	17.4099	13.471	7.8201	45.0742
1-14d	13.2601	64.6881	20.7494	15.3596	7.6729	41.5184
1-14e	24.8568	64.1917	20.7387	15.176	7.2937	39.5752
1-16a	53.8198	4.4367	36.058	32.5055	10.8652	39.1211
1-16b	52.0735	4.9579	38.9526	23.8416	10.1467	41.0993
1-16c	55.1727	1.6863	24.425	36.6549	11.7111	41.8186
1-16d	45.5429	5.6345	22.6578	36.7881	11.2977	41.0565
1-16e	41.996	2.3924	19.2866	43.8223	17.3556	40.5198
1-28a	57.3853	0.6449	0.2041	58.1115	15.5452	41.2408
1-28b	58.7477	0.5753	0.1016	56.9917	14.4135	41.1243
1-28c	59.1807	0.5957	0.1202	57.6647	14.5907	42.3829
1-28d	32.0943	0.6578	0.1353	62.5783	18.1666	40.946
1-28e	56.5966	0.5859	0.1193	57.5414	14.0459	41.5865
1-2a	50.0384	0.7556	0.1162	61.3047	15.3111	41.1167
1-2b	34.3286	0.79	0.1848	58.4062	19.6802	48.8752
1-2c	43.167	0.7455	0.0948	70.8234	11.4465	42.3114
1-2d	41.842	0.6423	0.1082	63.7359	15.0666	39.6665
1-2e	47.5807	0.6122	0.1415	60.543	13.8641	43.2135

Example 9—Shake Flask Fermentation Assay

[0090] Strains 1-13c, 1-8a-d, 1-26a-e, 1-36a-e, 1-41a-e, 1-40a-e, and 1-20a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0091] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0092] A 250 mL non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 10 and FIG. 10.

[0093] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-8c, 1-8d and 1-41c include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus. Further analysis was inconclusive on integration locus in strains 1-36a, 1-41b, 1-41e, and 1-20a-e.

TABLE 10

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	43.7126	3.1176	38.8613	24.5197	9.666	42.859
1-8a	46.3518	0.537	0.3381	57.0411	14.4597	46.2915
1-8b	47.5195	0.5334	0.1431	56.9461	13.7069	47.0977
1-8c	48.0894	0.5594	0.1063	55.9984	13.6091	47.8657
1-8d	37.4186	0.6121	0.1105	56.5403	13.8863	51.6109
1-26a	44.7559	0.6392	0.1211	58.7843	13.0302	45.1907
1-26b	26.8789	0.7454	0.1365	65.6702	14.7237	48.8557
1-26c	42.4506	0.7303	0.1018	65.269	12.7456	45.1127
1-26d	54.3955	0.5664	0.0934	50.543	16.7051	47.0098
1-26e	36.6369	0.7275	0.1162	70.0149	10.7288	48.273
1-36a	43.9038	23.4607	0.4003	37.8737	12.3842	43.023
1-36b	46.2296	17.4734	0.2509	28.3498	17.1992	51.7303
1-36c	33.5473	22.7172	0.384	32.9889	10.9137	53.8241
1-36d	30.3483	21.9108	0.3459	32.6397	12.4463	55.6903
1-36e	34.2625	22.4675	0.4111	33.3163	10.3248	53.7075
1-41a	50.84	0.6102	0.1238	53.2111	12.3563	47.0405
1-41b	27.0994	0.9407	0.0998	67.672	11.9654	54.0057
1-41c	54.6053	0.5387	0.1112	52.0424	14.0354	47.3747
1-41d	58.9086	0.6276	0.1701	51.36	21.0093	44.2126
1-41e	48.8262	0.5743	0.1143	57.4473	13.8819	45.8719
1-40a	38.7776	39.7726	0.2401	21.5304	12.246	44.721
1-40b	44.7092	49.927	0.3738	22.8583	10.4058	44.314
1-40c	43.6888	31.47	0.2197	29.7888	10.3401	50.5487
1-40d	43.0144	32.1802	0.1935	28.5121	13.9717	49.4035
1-40e	38.1811	32.5265	0.2004	31.8564	10.647	51.5381
1-20a	43.2545	0.7591	0.0973	60.0985	13.8425	44.7316
1-20b	44.7893	0.7386	0.1007	69.3394	13.0988	46.5606
1-20c	29.207	0.673	0.1105	63.3536	16.4597	49.4451
1-20d	39.0349	0.612	0.1224	57.5853	14.8878	49.7504
1-20e	48.061	0.7296	0.1017	61.7944	10.9017	47.5667

Example 10—Shake Flask Fermentation Assay

[0094] Strains 1-13c, 1-30a-e, 1-31a-e, 1-32a-e, 1-4a-e, 1-5a-e, 1-6a-e, 1-7a-e, and 1-9a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0095] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0096] A 250 mL non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production

culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 11 and FIG. 11.

[0097] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-30c and 1-30d include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus. Further analysis was inconclusive on the integration locus in strain 1-6c.

TABLE 11

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	65.2534	2.6255	37.1061	24.3306	10.3659	40.8187
1-13e	63.7146	2.5391	36.7742	22.9396	9.9483	41.7289
1-30a	64.53	2.2035	28.9854	29.1288	11.3525	39.7976
1-30b	58.571	3.5146	29.9395	25.1729	12.5878	42.2656
1-30c	61.7808	1.8694	26.2541	26.3438	15.1573	43.1151
1-30d	57.4209	1.9473	26.054	27.0466	13.4411	44.8634
1-30e	74.7224	2.7504	28.386	28.8879	14.2582	37.7887
1-31a	65.2966	1.7919	23.8031	38.9002	11.2516	37.7702
1-31b	58.0672	1.528	25.7531	34.4798	12.434	40.5374
1-31c	60.9593	1.4068	24.8131	31.8421	11.6425	41.8413
1-31d	60.5991	1.4246	25.033	31.5056	11.5117	42.3126
1-31e	65.0496	1.7865	25.7844	33.8874	12.8272	40.8807
1-32a	67.1283	2.236	38.4541	32.5687	11.1949	38.707
1-32b	70.2815	2.1497	38.3141	22.7553	10.5468	38.8618
1-32c	68.0248	2.2063	38.5515	22.7602	10.3843	38.5577
1-32d	66.3494	2.1502	38.3931	21.9976	11.0603	40.7174
1-32e	63.8648	2.5133	38.8442	23.9287	10.8515	39.9071
1-4a	70.2783	0.4834	0.3673	46.1316	14.32	43.4293
1-4b	72.2855	0.7763	0.1831	57.21	19.3193	35.128
1-4c	56.1849	0.5101	0.1291	47.8537	15.7451	47.9659
1-4d	77.2722	0.5838	0.1901	50.3863	14.1389	39.9826
1-4e	70.9407	0.4671	0.1525	46.2662	15.3398	44.2176
1-5a	56.7359	0.6505	0.1281	69.7951	13.2652	40.3786
1-5b	66.4853	0.4758	0.1138	47.5548	17.2347	44.8984
1-5c	56.2946	0.4561	0.1072	46.0599	15.9891	48.8712
1-5d	63.8624	0.5188	0.1223	55.0207	15.5579	41.5538
1-5e	53.5706	0.4438	0.0978	46.4767	15.7478	49.9892
1-6a	53.5487	0.4466	0.0857	46.0621	14.3516	50.2479
1-6b	56.3098	0.5689	0.0987	59.4119	14.2449	44.3129
1-6c	62.2098	0.4975	0.1159	50.7104	13.912	46.5117
1-6d	88.3493	0.392	0.0963	46.7061	15.7906	39.0682
1-6e	66.3903	0.513	0.1236	54.7569	15.6941	42.6935
1-7a	84.3102	0.7204	0.0924	52.2635	19.827	33.7597
1-7b	69.0519	0.4897	0.1187	51.9057	14.6164	43.9185
1-7c	67.7409	0.5193	0.1426	53.5386	14.7419	42.8362
1-7d	61.1123	0.5279	0.1161	54.4933	14.502	43.378
1-7e	58.0693	0.4778	0.1356	50.1206	18.9325	46.1013
1-9a	52.6189	0.5986	0.1156	66.6905	13.1314	43.3405
1-9b	59.3034	0.464	0.1102	58.2455	14.7562	43.3825
1-9c	53.516	0.4302	0.1246	51.9753	15.0032	48.9462
1-9d	53.9501	0.4221	0.0876	46.6648	15.398	49.7159
1-9e	54.575	0.4136	0.0957	46.5254	15.3139	49.828

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 note = Spirochaetaceae bacterium
 organism = unidentified

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TVPVNNIIPV PNGISGEISA ITEPCAVALH AVRKAQACVG DTVLITGAGT IGILIALWAR 180
AAGAEVFM TD IVVGKLKLAR SLGFQHL LDA KQTNVSEWVK QKCKHGV DIV FETTGH PAGL 240
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RELGAKSITA IDINPQKLAL AKELGATYVF NSREMSATEI HSDLENVQFD QLILETAGTP 240
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VALGAKSVTA IDISSEKLAL AKSFGAMLT NSLEMSAPQM QGVLR ELRFN QLILETAGVP 240
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NIKATNVLP	GDMDPFDAAM	IEPLAVAMHG	VLNIGVQVGD	TVAVMGSGTM	GQLVIQGLKI	180
AGAGTIIADV	ISDNKLRESK	ELGADIIINA	KDINPVEKIK	ELTGKGVDI	ALECAGSKIT	240
QEQCLLITKK	KSKIGFLGIA	YSDITLSEEA	FENIFRKELE	LKGFWNSYSA	PPPGQEWTKG	300
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YTKVKERNLI	KIPDEISYET	AAALEPVCIA	GHGLPRSEAK	VGDTVVLVLT	GPIGLFSIQW		180
AKIFGSTKII	AVDVFDEKLD	LAKELGADIC	INAKEKNIVE	EIKRLTDGDG	ADIVIESAGT		240
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 organism = unidentified

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CQGVVKTGAG	VDESIGRKA	SVFPMPLPCFE	CPSCKRGDYA	TCSNYKYYGS	RNDGGMGEYR	120
AIKAWNLLNL	DDSDVYCVGA	LSEPSAVGHH	ATEVGGVKAG	DKVFFVIGSGT	IGLLIGFFAQ	180
AKGAEVWIGA	RKESNDFAA	SLGFKTIDTN	NTEDEVAKAT	DGNMGMDVVE	AVGSNKCLIE	240
AIGSVTAGGT	LVTGQNPVGD	MNLPRDVYWR	ILRKQLTVRG	TWNSSYSDDKA	NDWHEVAALM	300
KEKKIPFEKL	ITRKYKLSQC	DEAFDFFRDH	SIQKSRVMFV	MHEKESEAYN	G	351

SEQ ID NO: 17 moltype = AA length = 336
 FEATURE Location/Qualifiers
 source 1..336
 mol_type = protein
 note = Faecalibacterium sp. OF03-6AC
 organism = Faecalibacterium sp.

SEQUENCE: 17

MVNMKALVFP	TAPHTFEYRD	VPRPEPAPDE	VLVEVKACAI	CGSDVHGSSG	ENGRRIPPLV	60
MGHEASGTIA	QLGEAVQSWA	VGDRVTFDST	EYCGYCWFT	HGMHNLCAQR	RIVGVACTEY	120
RKNGAMAQYI	AVKARTLYAL	PPAVSFQGC	LVEPFVGMH	AVLRSGLAAG	QTAAVIGDGT	180

-continued

IGLMTLIAAK	SQGAQVCISG	HHRTPMQLAA	QLGALDCAQL	PQLTAERGAD	VVYDAVGSQS	240
SFTQAMELVR	KGGRIMVIGN	HARTIDFPLQ	RCITEQIDVQ	FSYSSAGEYP	VCLQAIAGEGR	300
ADLTGLWQSF	PLAEGAQVFD	RLVHRDESLT	KAVLIP			336

SEQ ID NO: 18 moltype = AA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = protein
 organism = Agathobacter ruminis

SEQUENCE: 18

MKAYVLDGIG	QLNQHEYVLR	DLQPGEALVQ	VAAAGICGSD	LPRIFETGTY	HFPTIPGHEF	60
AGVVKELYAD	ADEEMLRQWM	GKRVAVFPLI	PCRTCRSCAN	HQYEMCSNYN	YLGSRDGGF	120
AEYCIVPVWN	LLQLPDEVSF	AQAAMMEPAA	VALHAIRRG	LHPDHPETGI	GKRVLMVGLG	180
TIGGLIAQWL	SQCGVTELT	TGHHAPQEEL	MHRYANENYR	FINTNQTAWT	DDENTYDLIF	240
DCINDSESVQ	QALRYVKPGG	QIVAVGNPKK	DLTLQKDIYW	KLLRKQVCMT	GTWNSSFSDH	300
SNDDWQILLR	YLQEQKLPQ	ALITHRLPFA	QLPKGLEIMR	LREEFYNKIM	IDLTVCS	357

SEQ ID NO: 19 moltype = AA length = 342
 FEATURE Location/Qualifiers
 source 1..342
 mol_type = protein
 organism = Alicyclobacillus acidoterrestris

SEQUENCE: 19

MKAVTYTPDQ	GFALVEVETP	QLGPDEVLLQ	VEACGVCSD	RQVVAGESVP	TGTAFPLVMG	60
HEIAGRIAL	GETVTDWHVG	DSVIVHFFVA	CGTCAPCVHG	QPNLCVRQVC	IGYORPGGFA	120
EQVAVPARQL	VRRPAELAPA	AAALVVDAPA	TPYHAMRAAG	VESGQTVLII	GTGGLGLAAL	180
MLAMGFHVGR	LGAVTRRQEG	IAIAEAHGAQ	MVVVTQDQER	TIARQLRRWS	GASGIDVVID	240
TVATADTITL	AAESVRPGGT	IAIVGMSADT	ALYPVAKTVR	RGLTFVASYG	SVREDVEQLV	300
SWVAEGRLLD	VRLVAGALPL	HEAARAFAPD	RPAGRWVIDP	SE		342

SEQ ID NO: 20 moltype = AA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = protein
 note = Ruminococcus sp. KGMB03662
 organism = Ruminococcus sp.

SEQUENCE: 20

MKSLINIEPK	MKALVLYGVG	DLRVTEVDVP	KLEKGTVLLK	IKACGICSSD	MPRCFVTGTY	60
HFPTIPGHEF	SGQIVAVGDD	VDESLLGKRS	CVFPMLPCKH	CKACKMEEYA	QCSGSYFSGS	120
RQDGAWAEYL	VVPVWNLVVF	DDTLDYKTAA	LCEPAAVSMH	AVNIGDIKEG	QNVVIIGTGT	180
IGFMIALFAK	ERTKTGKVVI	AGRSKNKLEY	AKKLGFDVVD	TSVDDLPAQV	KKIIGIDGAD	240
VTFEAVGSNE	AIASAIESTG	ALGSIVLVGN	PSTDILILPKN	IYWSILRKQI	TVKGSWNSSY	300
NSRVNDWKKA	LEVFEHSDVN	FADLITHTFT	IEENEKAFSA	IRNTGEFTLK	VMFTFDD	357

SEQ ID NO: 21 moltype = AA length = 348
 FEATURE Location/Qualifiers
 source 1..348
 mol_type = protein
 note = Clostridiaceae bacterium
 organism = unidentified

SEQUENCE: 21

MKAVVLHGKN	DICYEEWPEP	ILKTGTVKIH	VKACGICGSD	IPRVLGDSA	YYPILGHEF	60
SGCVTEVADD	VDKIQVGDH	AGVPLLPCKM	CDDCQQGNYS	QCKHYSFIGS	REQGGFADYV	120
VIPADNVVKV	DPTLPYEQVA	LFEPCTVALH	ALKVSSYKGG	KRVAILGGGT	IGLFALQWAK	180
IYGASQVTVF	GRSKEHLKIA	AALGADTTIS	TLDDNFMTEA	MRVTEKKGFD	YIYESAGSTI	240
TMKLAFELAA	NHAKICFIGT	PTENLTFSPK	LWENMNRKEF	QLMGSWMSCS	APFPGEWKE	300
TAHFFNTGQL	RYVPEMFPAQ	YPMDDVKKAF	EQFHNKNKVK	GRILLKNK		348

SEQ ID NO: 22 moltype = AA length = 337
 FEATURE Location/Qualifiers
 source 1..337
 mol_type = protein
 note = Thermoplasmata archaeon
 organism = unidentified

SEQUENCE: 22

MKVAVVYGDK	NIVLEDRIPI	SIGEKDVLVR	VKSSSICGAD	LHAYKGLSKR	RKPPLVMGHE	60
FSGMVERLGD	ETKGLKVGR	VVVEPIISCG	KCKACKKGKT	NVCENRKLIG	LNLDGAFAY	120
VSPVYEKCHE	LPRDLSFDEG	AIEPLAVAV	HAVNITPISS	NDNVLIIGSG	VIGLLITQLL	180
RSRTSGDILV	TGRRDYLLK	AREFGATDAI	NVIDCDAVKT	LLKKHGRVDV	VFEAVGSNET	240
VNQSISLVES	GGSTITLVGML	DREMTVDMME	VVAREITLRG	DYGYTTDDFK	EAEIASSGK	300
VDLKGLITHV	LPLDEIKKGF	DILAERRENV	IKVVINP			337

SEQ ID NO: 23 moltype = AA length = 343
 FEATURE Location/Qualifiers
 source 1..343
 mol_type = protein

-continued

note = Lachnospiraceae bacterium
organism = unidentified

SEQUENCE: 23

MKALVLEEIG	KLTLKEVETP	VPGRGEVLLR	IHACGICSSD	KARLFKNGTH	TMPLIPGHEI	60
SGEIVAGGPG	IREEIGRRA	AVFPLIPCRQ	CDSCRIGKYA	QCEHYDYFGS	RRDGGFAEYL	120
AVPVFNINLL	PDDMDYHTAA	LCEPATVASH	ALKRAKMKSG	DIVTIIGTGT	VGLLAAGIAR	180
IRRASKIIIV	GRSKEKMNA	AHPADHVINA	EGWDVNEAVA	SLTEGHMSDV	VLETGGNSDA	240
CTYALMAAGK	GTTVMFIGNP	EEPIVFNRI	YWKILNELT	LKGTWNSGYA	PEDNDWNDVI	300
RYMADGQIPA	DKIITDSFPL	EDYNKAFELL	LSDRFYTKIM	FDI		343

SEQ ID NO: 24 moltype = AA length = 348
FEATURE Location/Qualifiers
source 1..348
 mol_type = protein
 note = Prevotella marshii DSM 16973 = JCM 13450
 organism = Prevotella marshii

SEQUENCE: 24

MKANVLYHIG	DLRYTDVALP	ELHDGEALVR	VRACGICGSD	VARVFTTGT	HFPTIIGHEF	60
SGEVVETYS	ADAEWVGRRV	SVFPLIPCRH	CDACHHGQYE	MCTCYNVLGS	RTDGGFAEYV	120
AVPVWNLPR	PDSVSYEEGA	MMEPAAVAMH	ALKIARVGMG	DTVAITGPGT	IGMILTQLIQ	180
RIGGTKAVLI	GRTQAKLDYA	RRIGIRHVCN	SGTEHVADRV	GQLTGGRGAD	IAIEGTGASE	240
PMNLCLNLVR	SEGRIVAMGN	PQGDMLFRKD	AYWKLRLKQL	TLRGTWNSTY	GLGRDDWQDI	300
SALLAEGALN	LRGLITHRLP	LAEQQQLSL	MRDAAVYTNK	VMIVDEQG		348

SEQ ID NO: 25 moltype = AA length = 337
FEATURE Location/Qualifiers
source 1..337
 mol_type = protein
 note = Blautia sp. OM05-6
 organism = Blautia sp.

SEQUENCE: 25

MRAYELHGVN	DLRREDIEKP	EIPSGWVLVQ	VKASGICSSD	IPRIFTNGTY	HFPTIPGHEF	60
SGVVAAYGEG	VPEERVGKRV	GIFPLIPCRT	CQCRQKKYE	MCEHYDYLGS	RRDGGFAEYV	120
AVPDWNLLLE	PENVSFREA	MLEPLSVALH	AVKRSGVKPG	DTAAVIGTGM	IGFAAAAWAK	180
ALGAETVFI	GRGEAKRAIA	EQIPGISYVT	EEESKAIQAD	VVVEAVGTPQ	AVSSAVLLAR	240
PGGSIVLMGN	PSGLAMEKN	VYWRILRKQL	NLMGTWNSSY	EKDEACDWTE	ALDALSKKKI	300
PAQALITHCF	PSEKLMGLE	LMKNHKEPYC	KVMVSWE			337

SEQ ID NO: 26 moltype = AA length = 354
FEATURE Location/Qualifiers
source 1..354
 mol_type = protein
 note = Candidatus Bathyarchaeota archaeon
 organism = unidentified

SEQUENCE: 26

MKMRRAAILY	GPKDIRIENI	EMPKVKPEWV	LIKVRATGIC	GSDLHFYKEK	SFIPISDVG	60
KGRYVPGHEF	SGEVYELGSK	VKGLKKGDRV	GVEPIIGCGK	CQWCKVGCYN	LCEHYKLIGF	120
HYIGMAEY	VAPAENCFLK	PNNVSFESAA	MLDCIAVAIH	AVNKAIEIRG	DSVLIVGAGA	180
IGLFALQAAL	IQGASDVVVV	GKYEFLQLEKA	RKLGATAVIN	AKNESFSEKI	MEITSGKGVD	240
KVIEAVGGAS	NVISDVLDVT	RKKGTIVTTG	IFTRPLSVDF	FKILTKEITI	TSSWGYCYFD	300
RRKEFGTALD	FLAKEKINTQ	ELITHKYPLE	KVSEAFETAL	NKNKEKSIKV	EIIF	354

SEQ ID NO: 27 moltype = AA length = 350
FEATURE Location/Qualifiers
source 1..350
 mol_type = protein
 note = Lachnospiraceae bacterium
 organism = unidentified

SEQUENCE: 27

MKAWVLHGIG	DIQYEEVKEP	QSAGDEVIMK	VMSVGICGSD	IPRIYKTGT	HFPTIPGHEF	60
SGIVADVAG	VNPKWKNKRV	GVFPLIPCHV	CASCKKGQYE	MCRNYGYLGS	RQNGGFAEYA	120
AVPVKNLIEL	PDSVSMQEA	MLEPMAVAVH	AMRRARPMTG	DRIVVCGLGT	IGMLLVSFLR	180
ETGIRDIFVI	GNKAFQRITI	LKTGLQENCY	CDSRVQDVKE	WLMERTEGKG	ADVFFECVGR	240
NETISQAVNL	TAPGGSVCLV	GNPCSDMTLE	REYWKILRN	QLTITGTWNS	SFTHEIEDDW	300
HYVLDRLLEQ	KISPGDLISH	KFPLKNLESG	LHIMRDKTQD	YIKIMAEINV		350

SEQ ID NO: 28 moltype = AA length = 348
FEATURE Location/Qualifiers
source 1..348
 mol_type = protein
 organism = Alkalihalobacillus ligniniphilus

SEQUENCE: 28

MKAAVLHSEN	LFTYEEVEID	PCKDDEVKIK	VMAAGICGSD	THKMTSKWKY	PLPTIMGHEF	60
SGFIVEIGEK	VTNASIGDRV	VAIFPLPCND	CEYCKRGQFS	LCEDHGMVGS	KSPGGFAEYV	120
NKSTNVLP	GDMDFEAA	IEPLAVALHG	VLNIHPRIGD	IAAVFGIGTI	GILTIQWLKI	180
AGVKEVIAID	ISDNKLADAH	AMGCERTINP	FKENLLEKID	EYTNRGVVDI	AFECAGSKVT	240

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QEQLLVTA	RGKVGYL	YSDVHL	SEQA	FENIFR	REL	LKGF	WNSY	SA	PPGEE	WTKS	300
IDFINKGR	LKELISH	RFL	LQDTQ	KAFDM	MINRE	EETG	K	VLIL	PNKE		348

SEQ ID NO: 29	moltype = AA	length = 348
FEATURE	Location/Qualifiers	
source	1..348	
	mol_type =	protein
	organism =	Jeotgalibacillus soli

SEQUENCE: 29												
MKA	AVLH	SEN	HFI	YEE	IDIA	PCQ	DDEV	KIK	VIAT	GICG	SD	60
SGY	IVEK	GKN	VTNA	AIG	DRV	VAIP	PFLP	CND	CEY	CKRG	QFS	120
NI	ESSN	VLPI	GDME	FEE	AAM	IEPL	AV	ALHG	VLNI	HPRV	GD	180
AG	AK	EIIA	IDS	EKL	ADAR	AMGC	CERT	TNP	LNED	LLEK	IN	240
QE	QCL	LITR	KGK	IGY	L	YSD	VHL	SEKA	FENI	FR	REL	300
IE	FI	KNGR	LK	ELI	SHR	FL	LQDT	Q	KAFDM	MINR	EETG	348

SEQ ID NO: 30	moltype = AA	length = 350
FEATURE	Location/Qualifiers	
source	1..350	
	mol_type =	protein
	organism =	Heyndrickxia sporothermodurans

SEQUENCE: 30												
MKA	AVLH	SEN	EII	YQD	IEIS	ECQ	PNEV	KVK	VMA	AGIC	GSD	60
AGY	VVEK	GKD	VTN	VEI	DRV	AAIP	FIP	CNE	CEY	CLRG	QFS	120
NIP	AKN	VLPI	GEM	SFE	DAM	IEPS	AVSL	HG	VLNI	HPQL	GD	180
AGV	KEII	AID	ISD	KKL	QDAM	DLG	CEKS	INP	LKED	LKKI	Y	240
QE	QCL	LITR	KGK	IGY	L	YSD	VTL	SEKA	FENI	FR	REL	300
IE	FVQ	QKIK	LK	ELV	SHRY	FL	LQDT	Q	KAFNM	MINR	EEYK	350

SEQ ID NO: 31	moltype = AA	length = 346
FEATURE	Location/Qualifiers	
source	1..346	
	mol_type =	protein
	organism =	Clostridium fungisolvans

SEQUENCE: 31												
MKA	AVLH	SVG	DIR	YEN	IEIK	PYG	DDEV	KIK	VKA	AGIC	GSD	60
SGY	IVEV	GKN	VKG	FKV	GDRV	VAV	PLEP	CHQ	CEY	CKGQ	FS	120
NV	KATN	VLKI	DDI	YEE	AAM	IEPL	AV	ALHG	VLNI	NPQL	GD	180
SG	VEKI	IAVD	ISD	KKI	SEAM	SLG	ADIG	INA	LKEN	PVEK	IM	240
QE	QCL	LVTK	KGK	IGY	QGIG	HAG	IEL	SEVA	FEG	IPRK	ELN	300
IEY	VKQ	KKIK	LK	QLI	SHRF	S	LED	TAKAFEM	IGER	KEEY	NK	346

SEQ ID NO: 32	moltype = AA	length = 348
FEATURE	Location/Qualifiers	
source	1..348	
	mol_type =	protein
	organism =	Alkalihalobacillus nanhaiisediminis

SEQUENCE: 32												
MKA	AVLH	SEN	HFI	YQD	MESF	ECQ	NDEV	KIK	VMA	AGIC	GSD	60
SGY	VVEK	GSD	VAN	VEI	DRV	VAIP	PFLP	CHT	CEY	CKSG	VYS	120
NL	KATN	VLPI	GNM	D	FEEA	AM	IEPL	AV	ALHG	VLHI	QPR	180
AGV	KEII	AID	ISD	EKL	AEAK	AMGC	CERT	TNP	LQDN	LIEK	VH	240
QE	QCL	LITR	KGK	IGY	L	YSD	VHL	SEKA	FENI	FR	REL	300
ID	FILK	GR	LK	ELI	SHRF	P	LEET	Q	KAFDL	MINR	KETG	348

SEQ ID NO: 33	moltype = AA	length = 348
FEATURE	Location/Qualifiers	
source	1..348	
	mol_type =	protein
	organism =	Neobacillus cucumis

SEQUENCE: 33												
MKA	AVLH	SEN	QIV	QFE	FEIG	ELK	PDEV	KVK	VMA	AGIC	GSD	60
AGY	IVEK	GKD	VVH	VEV	DRV	AGIP	FVPC	N	CEY	CFRG	QFS	120
NIP	AKN	VLPI	GDMD	FEDA	AM	IEPS	AV	ALHG	VLNI	HPQI	GD	180
AGV	KEII	AID	ISD	EKL	QDAI	ALGC	CERT	TNP	LKEN	LLEK	IH	240
QE	QCL	LITR	KGK	IGY	L	YSD	INL	SEKA	FENI	FR	REL	300
VEY	VKQ	QKIK	LK	ELI	SHRF	P	LEET	Q	QAFNM	IINR	EEYK	348

SEQ ID NO: 34	moltype = AA	length = 341
FEATURE	Location/Qualifiers	
source	1..341	
	mol_type =	protein
	organism =	Staphylococcus aureus

SEQUENCE: 34												
MIN	QVYQ	LV	PRQ	F	DVTYNN	VDI	YGNH	VIV	RPLY	LSICA	A	60
SLV	HEAV	GEV	VFDS	KGV	FEK	GTK	VVM	PNT	PTE	QHII	IAE	120

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VVMAHDRIVP LPNDIDLSTI SYTELVSVS Y HAIQRFERKS IPLKNSFGIW GDGNLGYITA 180
ILLRKLYPEA KIYVFGKTDY KLSHFSFVDD IFTVNQIPDD LKIDHAPECV GKGGSQVALQ 240
QIVEHISPEG SIALLGVSSEL PVEVNTRLVL EKGLTLIGSS RSGSKDFEQV VDLRYKYPDI 300
VEKLALLKGH EINVCTLQDI VQAFEMDLST SWGKTVLKWT I 341
```

```
SEQ ID NO: 35      moltype = AA length = 350
FEATURE           Location/Qualifiers
source            1..350
                  mol_type = protein
                  note = Staphylococcus aureus subsp. aureus 71193
                  organism = Staphylococcus aureus
```

```
SEQUENCE: 35
MLLFEVVLPM INQVYQLVAP RQFEVTVNNV DIYSYVIVR PLYMSICAAD QRYTGSRDE 60
NVLSQKLPM S LIHEGVGEV FDSKGVFNKG TKVVMVPNTP TEKDDVIAEN YLKSSYFRSS 120
GHDGFMQDFV LLNHDRAVPL PDDIDLSTI S YTELVTVSLH AIRRFEKSI SNKNTFGIWG 180
DGNLGYITAI LLRKLYPESK IYVFGKTDYK LSHFSFVDDV FFINKIPEGL TFDHAPECVG 240
GRGSQSAINQ MIDYISPEGS IALLGVSEPP VEVNTRLVLE KGLTLIGSSR SSGSKDFQDVV 300
DLIYQYPDIV DKLALLKGQE FETATINDLT EAFEADLSTS WGKTVLKWM 350
```

```
SEQ ID NO: 36      moltype = AA length = 341
FEATURE           Location/Qualifiers
source            1..341
                  mol_type = protein
                  organism = Eubacterium ventriosum
```

```
SEQUENCE: 36
MINTVYQLVA PRRFEIKYSY IDLNNNKVIV RPTYLSICNA DQRYYQGLRD AEILRKKLPM 60
ALIHEAIGEV VRDPSGNFKT GELVVMVPNT PVEEDDIAE NYLRSSSKFRA SGFDGFMQY 120
VEITPDRLVR LPQNIDRTVA AFTEIVSVSV HAIGREFDKIA HKRRNVVGIW GDGNLGYITA 180
VFFKTMFPET KLYIFGVNPD KLQDFTFADA TFTVEHIPED VKIDHAPECV GGAGSGKAIN 240
QIIDYINPEG TISILGVSEY PVPINTRMVL EKGLRIFGSS RSGVADFEKT VALYKSNPEI 300
INYLNSIVGA VVPVRTIEDM KKAPEMDIQK AFGKTIMMWD K 341
```

```
SEQ ID NO: 37      moltype = AA length = 343
FEATURE           Location/Qualifiers
source            1..343
                  mol_type = protein
                  note = Pradoshia sp. D12
                  organism = Pradoshia sp.
```

```
SEQUENCE: 37
MINQVYRLVS PRQFEVAYND RSLQYHEDFV VVRPTHLSIC AADQRYYTGS RGKEAMAKKL 60
PMALIHEAIG KVVYDPKGEI KNGTMVVMIP NTPTEKDEVI GENYLRSKSF RSGYDGMQY 120
DYVFMGRDRI VELPEGIDPY VVAFTELMTI TTHVLSRFES RSHKRRNVFG VWGDGNLGF 180
TTLFLKMKYK DCKVIFGKT GYKLDNFSFV DEVYTIDEIP EDLRIDHAFE CVGGKGSQYA 240
IDQMIDHLHP EGSIALMGVS EFPVEVNTRM ILEKGITLIG SSRSGREDFV RTVNFLKENP 300
DVVEYLGLTV GEVQEVKTIS DILSAFEKDL TSSWGKTVMW WKI 343
```

```
SEQ ID NO: 38      moltype = AA length = 341
FEATURE           Location/Qualifiers
source            1..341
                  mol_type = protein
                  note = Lactobacillus plantarum EGD-AQ4
                  organism = Lactobacillus plantarum
```

```
SEQUENCE: 38
MLNQVYRLVD PRQIEVQTV AEEITDNDIVV RPKYLSVCHA DTRYFTGQRP QATLRQKLPM 60
ALIHEGVGEV VKDPQNKFKP GTLVAMVPNT PFETDPIKE NYLPTSKFERS SGYDGMQY 120
VTLHRDRAIV VPDNFDHOMS AFIEVSVSGV HALTQLEAVM DADRKIIGIW GDGNLGFITA 180
TLVKQIFPDS QLMIFGRHRS KLDYFSFADK TYLVDDIPSD LQVSQALECT GGRGSESAIA 240
QIIQHIRPMG TAILMGVSED PVGIDTRSVL AEGLTLRGVS RSGRADFQRA IDILTKSPVT 300
RERLQNLVGF TRKVSTIQDI TDFEGALTN YWGKAVMEWD V 341
```

```
SEQ ID NO: 39      moltype = AA length = 337
FEATURE           Location/Qualifiers
source            1..337
                  mol_type = protein
                  note = uncultured Ruminococcus sp.
                  organism = Ruminococcus sp.
```

```
SEQUENCE: 39
MINYVYQLVS PHVFSLYKEE VDFKKEVIIR PKCMAVCHAD QRYFLGRRDA KILRQKLPM 60
LIHECCGQVW WDGTSFQRG QKVVLIPNVP GRDDGVITYEN YAEGRFLSS GCDGFMREYV 120
NLSPDRVVPF EQVPLVAAI TEFVSVAVHA AGRFMRAAHT RREAIGVWGD GSLAYVVCV 180
LRRLLPESRI FVVGKHERKL LQFSFADRTF LFDLSLPEDFH IDHAFFECTGG EGSYYAIDDI 240
ITAVRPQGAV MLMGVSENRV PVRTRDILEK GLTFTGCSRS GREDFVKAVE LMEQPQFQKR 300
LSAIIYEDRP VHTVADLSRV FETDLNTPFK TVFRWEI 337
```

```
SEQ ID NO: 40      moltype = AA length = 338
FEATURE           Location/Qualifiers
```

-continued

```

source                1..338
                      mol_type = protein
                      organism = Kandleria vitulina

SEQUENCE: 40
MINTVYRLIA PKSIRADFVD LKLDEERIII RPTYLSICAA DQRYTGSRG RVVMKQKLPM 60
ALIHEAVGEV VYDPKGELPV GTHVVMIPNT PNEKDDIIKE NYLRSSQFRA SGFDGFMQNV 120
VSMRRDLVIP PDYEPEVGV LELMSVGMNA VSTFKKHAHD RKDAIGIWGN GNVGFVTALI 180
LRKEFPDAKI YVFGVEEHKN RYFSFADDTF IINDIPEDLQ LDHAFECVGG RGSEAAISQI 240
IDYIKPQGTI NLMGVSENPV PINTRMVLEK GLQLLGNSRS NYDDFKKSVD MMANNKDvre 300
YLSTIVSEMV VVNTIDDMIR AFEDDTYNDF KTVMKWEI 338

SEQ ID NO: 41         moltype = AA length = 341
FEATURE               Location/Qualifiers
source                1..341
                      mol_type = protein
                      organism = Lactiplantibacillus plantarum

SEQUENCE: 41
MLNQVYRLVS PRQFEVETVA EEVDADDLIV RPKFLSICHA DTRYFFGQRS QAALRAKLPM 60
ALIHEAVGEV VYDPRNEYKP GTLVAMVPNT PVKDSIIQE NYLPSSRFRS SGYDGFMOEY 120
VCLRRDRAIT VPSDFDPHMA AFIEMVSVGV QALTKLESTM DEDQRVIGIW GDGNLGYITA 180
TLVKQIFPDS QLLVFGRHQS KLDYFSFADK TYLVDDIPAD VKVSALECT GGQGEAAIQ 240
QIIQHIQPMG TAILMGVSEN PVGIDTRSVL AEGLTLRGVS RSSREDFKRA VDILSSSPVT 300
RERLQNLVGY VQKVSSIQDI TNFFDSALTN YWGKEVMEWD V 341

SEQ ID NO: 42         moltype = AA length = 342
FEATURE               Location/Qualifiers
source                1..342
                      mol_type = protein
                      note = Lachnospiraceae bacterium
                      organism = unidentified

SEQUENCE: 42
MINTLYKLTA PLIVESFYEN IDCHSDNIIV RPEKLSICKA DIRYFGMNRN AKVLKEKLPL 60
ALIHEACGSV VYDITGKFKA GDQVVMLPNI PGINMQYHEN YRRDSLFRSS GADGFMQELV 120
SIPQSQVIKF EKMDQNIAAF TEFISVAIHA VSSYLDRSAG KRNGIKRKIG VWGSGAMGYA 180
VSCVLKYFIP SAEITVIGGS KAKLDMFSFV NHTCSLSELP DDAEFDDCFE CVGGAQSGNA 240
INQIDHILP EGLLLLGVSV EEAVPVNTRM VLEKGITLLG RSRSESDFI KTIDILDtNE 300
NIRRRISNLI SGEVEINEVN DIHKAFNQSK IADYKIILNW RI 342

SEQ ID NO: 43         moltype = DNA length = 1599
FEATURE               Location/Qualifiers
source                1..1599
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 43
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 organism = synthetic construct

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 organism = synthetic construct

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FEATURE	Location/Qualifiers
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moltype = DNA length = 4633

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Location/Qualifiers

source

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mol_type = other DNA

organism = synthetic construct

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SEQ ID NO: 57 moltype = DNA length = 4576
 FEATURE Location/Qualifiers
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 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 58 moltype = DNA length = 4645
 FEATURE Location/Qualifiers
 source 1..4645
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 59 moltype = DNA length = 4636
 FEATURE Location/Qualifiers
 source 1..4636
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 59

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SEQ ID NO: 84 moltype = DNA length = 4555
FEATURE Location/Qualifiers
source 1..4555
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 84

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SEQ ID NO: 85      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

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SEQUENCE: 85
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cgttcaaccc ttcaaaattg aaactctcag gctgagcggc tgagcaagct ctgagaatgt 180
gggtcttcga agactttcgc ttgagcatgg tgcattgtga aagagatgga atgtggaaat 240
gttgtaggta ggaacacagc ccggttagagc ggcgcagtggt ctcagcgaag ctgggagaa 300
gactgaaccc gtcttgatag tcatcatctc agcgtgcgtc agcgcgtcag ctacagcgtt 360
tgagaatggc accgactgac ctgtcgtaca tcccaaggag gcccgctgcg tatactcgcc 420
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ctatctttcg ggtcgggtgcg

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SEQ ID NO: 86      moltype = DNA length = 1002
FEATURE           Location/Qualifiers
source            1..1002
                  mol_type = other DNA
                  organism = Moniliella pollinis

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SEQUENCE: 86
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SEQ ID NO: 87      moltype = DNA length = 1038
FEATURE           Location/Qualifiers
source            1..1038
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 87

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gtgcacggta tggacggatc gaccggacgt cgaaaaccac cgctgggtat gggtcacgaa 180
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aacctcagca acaatcgaca agtgttgga gtgtctcctg gcgagtatcg ccgtcatggt 360
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gatattgacg aacgaatctt gcaactcaca aaatcgcgcg gttttgacct tacatttgaa 720
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SEQ ID NO: 88

moltype = DNA length = 978

FEATURE

Location/Qualifiers

source

1..978

mol_type = other DNA

note = Candidatus Dormibacteraeota bacterium

organism = synthetic construct

SEQUENCE: 88

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gtggaggcct atttgggtcg catggctaata cgaacaccgc cggtggtgat gggtcacgaa 180
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SEQ ID NO: 89

moltype = DNA length = 1014

FEATURE

Location/Qualifiers

source

1..1014

mol_type = other DNA

note = Candidatus Falkowbacteria bacterium

organism = synthetic construct

SEQUENCE: 89

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ctccacatct ataacggagg catgaaaaat ctgccgttgc ctctgattat gggtcacgag 180
ttcagcggag ttattgaaga gcttgcccga gacgtgaaga actttaaggt tggagatcga 240
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aatctgtgtg ctaactctcg cattattgga gttcacccgc cggagcgct cgcgaatac 360
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ggtaaaaagg tggcggtcat cggccaagga cccatcgccc tgcgtgttga ccaaatcctc 540
cgcgacgcag gcgctgaagt cattggcatt gacatcgag attggcgatt gaaactcgcg 600
aagaaaaaga agtgggcaga tcatacaatc aactccagcc agaaaacctg gctcaagaag 660
atccagaaga tcgcaccaga aggcgtggac acagtgtttg aggttgttgg ccgtgaggaa 720
actgccgaga ttgctattga aatcgcgctg cgagatgggt acgtttttat cctcggtgtc 780
tttgaagcgc ccgcatcttt gaacatgatg aatattgtta agaaggaact caacgtgttc 840
ggaagctgga cctgtgcggt ttctttcccc ccttcgattg ctctcgccgt tgccaagaaa 900
gttgatttga aatccttgat cacgaatcga tatgacgtga agacgggtgc cctggcattt 960
aaggagtctg cctcttataa aggtaagcga atcaagactg ttatcaactt ttaa 1014

```

SEQ ID NO: 90

moltype = DNA length = 1044

FEATURE

Location/Qualifiers

source

1..1044

mol_type = other DNA

note = Acidobacteria bacterium

organism = synthetic construct

-continued

SEQUENCE: 90

atgaaggcaa	tgttgcttag	ccagtatcga	cgactggaaa	ttgcccaggt	gccgacaccg	60
atccctggto	gtggcgaaact	cttgatccat	gtcgcggcat	gtggatattg	cggaagcgat	120
gtgcacggat	acgatggcag	cagcggacga	cgtatccctc	caatcgtgat	gggccatgag	180
gctgcaggac	gtgttgcggc	actcggtcgg	caggtoacgg	gctggtcgga	aggtgatcgc	240
gtcacattcg	atagcaccat	ctcctcgccg	acttgccctt	actgctcccc	cggcgagatt	300
aaacctctgc	accgtctgta	agtgtctgga	gtttcgtgcg	gtgactatcg	ttgcgctgga	360
gcttttgcgg	agtatgttat	tgtgccgcaa	cgcattgctt	ataaattgcc	ggatgcottg	420
agcttctcgg	aggccgcaac	tttggaaagc	gtcgcgggtt	ctgtgcacgc	tgttagcctt	480
tcccaggttt	cgccaggtac	tcaggcactg	gttattggcg	ctggatgat	tggcctcttc	540
accttgcaag	cgctgcgcgc	agccggttgc	agctccgtgg	tgatccgcca	cggtgatgcg	600
tcccgtctgc	gtttggctaa	agaggttggc	gcaaccgaaa	cgcttaatgc	gacaggagat	660
gaatttggtt	ccgaaatctt	caaacgcaca	aacggagcgg	gcacagacct	tgcgctcgaa	720
gctgtcggta	tcgatgctac	cgtccgaacg	gccattaact	ctgtccgaaa	aggcgggtaca	780
gttacccttg	tcggcaatat	cacgcgggag	gtcactcttc	cgcttcagaa	gggtggtgaca	840
cgccagattc	gaatccaagg	ttcttgcgca	tcgctggtg	aatacccccg	agctatccag	900
ctccttacct	ctggcgctac	taaaagtc	ccctccatta	ccgctgtggc	cccgttgtaa	960
gacggtcttc	agtgggttga	gcgcctttac	agccgcgagc	ctaactctat	gaaggtggtc	1020
cttaccctcg	gcagcgtgtc	ttaa				1044

SEQ ID NO: 91 moltype = DNA length = 1035
 FEATURE Location/Qualifiers
 source 1..1035
 mol_type = other DNA
 note = Spirochaetaceae bacterium
 organism = synthetic construct

SEQUENCE: 91

atgaaagcga	tgggtgtgaa	agaatataag	tatctgga	tcgaagacat	tcctattccg	60
cacatcgaa	cagcggacga	actgtgtgtc	cgcgtgaagg	ctaccgcat	ctgcggttcg	120
gatatccat	gttttgatgg	ttccaccgga	cgcgtaaac	cgccattggt	catgggacac	180
gaggcatcgg	gtgtgtgcca	aagcgtcgga	agcgaagtta	aatcgctgaa	aatcggtgac	240
cgtgtcacct	tgactccac	agtttgggtc	ggcgaatgct	acttttgcg	acagggtcag	300
gttaacctgt	gcgcaaaacg	acgtgtgctt	ggtgtgtcgt	gcgacgaata	ccgcaaggac	360
ggtagccttg	cggaaatagt	cattgtgccc	gagcgtattg	tgtataagtt	gccggataat	420
gttgccctcg	aagaagcagc	attgactgag	ccggtttctg	tcgcgatgca	cgcatattgat	480
atcaccgga	tgaagccagg	agaaaatgcg	gccgttatcg	gtagcggctc	tatcggactc	540
ctcctgattc	aaattatccg	tgtctattcg	ccccacttga	tcacgcgctt	cgatacagac	600
gaaaaaacgc	gaaaagcagc	tctcgaagct	ggagccgata	ttgcattgga	cccccaagat	660
cctgagtcgc	ataaaactgt	cctgagcatg	actaataacc	gaggtgtcga	ccgagcattt	720
gaggctgtgg	gcgcctcgcc	agcgtacggt	acagctattg	aatccgtccg	aaaaggctcg	780
tcgctcactc	tgatcggcaa	tattagcaat	aaagctgaga	tcgcgctcca	atctgtgggt	840
acgcgccaaa	ttgccccttt	gggttcgtgc	gctatttctg	gagagtatcc	tgtctctttg	900
gctttgatgg	cgaatcgacc	tattgacgtg	cgtcaagtta	tctccgcaat	cgccctttg	960
gaggagggtc	ctctgtgggt	cgaacgcttg	tataaacgcg	aacctggtct	tttgaaggtt	1020
gttctcagcg	cgtaa					1035

SEQ ID NO: 92 moltype = DNA length = 1047
 FEATURE Location/Qualifiers
 source 1..1047
 mol_type = other DNA
 note = Candidatus Hydrogenedentes bacterium
 organism = synthetic construct

SEQUENCE: 92

atgaaggcac	tccgctgta	cgctccaggt	gacttgcggt	tggaggaaat	cccaattcct	60
ttgccacgto	ccgatgaggt	gttggttcgc	gtgcttgctg	gcggtatctg	cggttcggat	120
atccccctgt	tttttgaa	cggtagctat	aaatacccat	tgggtccggg	ccatgagttt	180
tcgggtattg	ttgaaaaagt	cggctccgaa	acgggaaaag	attgggttgg	tcgtcacgtt	240
gtcgtctatc	ccctcatccc	gtgtcgacaa	tgctcctctt	gctactctgg	tctttatacc	300
caatgcctca	actacgacta	cttgggatct	cgcctgacg	gaggatttgc	agagtatgtg	360
actgtcccgt	tgaataatat	tatcccggtc	cctaattgaa	tctctggaga	gatctcggcg	420
atcaactgag	catgtgcggg	cgcccttcac	gcggttcgca	aggcccaagc	ctgcgtgggt	480
gacacgggtc	ttattacggg	agctggaacc	attggtatct	tgatccgccc	ctgggcgcga	540
gcagcaggtg	ccgaagtctt	tatgactgac	attgtgggtg	gaaagtgtga	actggctcgt	600
tcgttgggtt	ttcagcatct	gctggacgcc	aaacaaacaa	atgtttctga	atgggttaag	660
caaaagtgtg	aacacggagt	tgatatcggt	ttcgaaacca	cgggacatcc	cgcgggactc	720
gaaacgtcga	ttgactcttg	ttctattaca	ggcaaaatta	ttctcgtggg	caatcccagc	780
ttcgaagaga	tgaaggttcc	ggctaataca	tatggaatga	ttctgcgcga	ggagatccag	840
attactggaa	tttgggaattc	gctcatcaac	gaatatccaa	aaaacgagtg	gcaagttgtg	900
ctgcaagagt	tcaagaagga	gtcgtccttg	tactcgaaga	ttatcacgca	taaggttccc	960
attgagggcc	ttaaggaatt	gtttgtgacg	atttacaaca	aacaggtctt	tccggtgaag	1020
gctatgttta	tcaataagga	gatctaa				1047

SEQ ID NO: 93 moltype = DNA length = 1014
 FEATURE Location/Qualifiers
 source 1..1014
 mol_type = other DNA

-continued

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organism = synthetic construct
SEQUENCE: 93
atgaaggcag tcgtgtacac gcaaccagac gcagtggccg tccttgagcg tgaatgcct 60
gcactggtgc caggagaggt gcttcttcga attgatgcc a gcggaatttg cggttccgat 120
ttgcacgcat atcatggaca tgaccacgt cgtaaacctg gacttggtt gggtcacgag 180
tttgacgga cctgtggttc ttctagcgat gaaagccgcc tccctgtggg tcgacgcgtc 240
acggctaata cctcgtttac atgcggacac tgtgattact gccttcaggg ccgtgacaat 300
ctgtgtgcga atcgtggcat ggtgggtatg tctcgccctg gtgccccttc tgagtatctt 360
gccattccca tgcgatgcgc tattgtctatt cccgaagatc tgccgagcgt gtctgctgcg 420
cttaccgaac ctgctgcgac ggcactgcac gcagtcaatc ttacaatgcg agcgatgagc 480
cgacctctcc cggagggcgc tgttttgatt atcggaggtg gagcgatcgg catggtgacc 540
ggagcacttg tcggtgcgta cggctgtcgc tcccttgccg tcgcggaaac caacgccctg 600
cgccgtgcgt ccgtccgacg tcactcttga tgcgagggtt tggaccctac agcgaacgcc 660
cggctgagtg cttcggttga cgtggtcatt gacgcagtgg gtgcggcacc aacccgccat 720
ctggcactgg cggcaacgcg accaggaggt gttatcatgc acattggact ccaggcatgg 780
tccagcgaaa tcgatatgct taagtcact cttgctgaaa tcaactctct cggtacgtat 840
acatacacga tggcgatgat gcatgcgacc gtgcgagcac ttgcctcttg aagcttcgga 900
tctctggaat gggtcgagac gcgagccctg acagatgcgc catctgcttt tgctgacctt 960
gctgcgggtc gagtcggcgc gggcaagggt gtgcttctcc cgcaacagga ctaa 1014

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SEQ ID NO: 94      moltype = DNA length = 1044
FEATURE           Location/Qualifiers
source            1..1044
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 94
atgaagtcgc tcgttattca cgcgaagga gatgtccgtg tggaaagagc acccattccc 60
catttgcaaa cggatgaaga tgttttggtg aaagtcgtgt cttccggact ttgtggcagc 120
gacattcccc gacattctcaa tcgtggcgct cattattacc caatcacact tggccatgaa 180
ttctcgggct atgtcagatc ctatggagcg ggtgtcactg acttgcaacc aggtgatgct 240
gtcgcttggtg tgccgctcct cccgtgtttt cactgcccc agtgcgagcg aggttacttc 300
tctctgtgta agcagatca atttgcggt tctcgacgc agggaggaaa cgcagagat 360
gtcgttatga aacgagcgaa ctttttcga ctcccgctccg atatgccaat tgaagacgga 420
gctttcattg agcccatcac cgtcggtctg catgccttcc accttgccga gggttgcgag 480
ggaaagaatg tcattatcgt ggggtgcagg accattggcc tccctgccc gcaatgtgca 540
cgagaactcg gagccaaatc gattaccgca atcgacatca acccacagaa gcttgcatcg 600
gcaaaggagc tggcgccgac ttacgtgttt aattcccag aaatgtcggc gaccgaaatt 660
cactctgatt tggaaaaatg gcagttcgac caacttatc tcgaaacggc tggaaacgct 720
cagactgtca cgctctccat cgaattgcg ggtccccgag ccaattggc gcttggttgg 780
acgttgacac cgatttggc gcttgcgtcg ggtgtctttg gtcaaatctt gcgcaaaagag 840
cttactatcc tcggatcttg gatgaactac agctctccct ggccaggcga agaattgggtc 900
actgcggcac gtctgctgac ggagaaacgt ttgcaattgg aaccacttat tgcgcactcc 960
ggagatgcgc aatcctttgc gtctgcagtg aagtctctca atggcgacc tatgcaagggt 1020
aaaattctgc tcaagctttc gtaa 1044

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SEQ ID NO: 95      moltype = DNA length = 1041
FEATURE           Location/Qualifiers
source            1..1041
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 95
atgaaatctg ttgtgaacga tacggacggt attgttcgta ttgcagagtc ggtgattccc 60
gaaattaaac accaagatca agtgcggtgtt aagatcgctt ctccggcctt ttgtggttct 120
gacctgccgc gcatttttaa gaacggagca cactattatc ccatcactct cggacacgaa 180
ttttcgggat acatcgatgc agttggctcc ggtgtcgatg atcttcactc tggtagcgcc 240
gtggcatgtg ttcaccttct gccctgtttt acctgtccag aatgtctgaa aggcctttac 300
agccaatgtg ccaagtacga ttttattgga tcccgctcgt atggtggttt cgcggagtat 360
atcgtggtta agcgaaaaaa cgtgtttgct ttgcccaactg acatgtctat tgaagatgga 420
gcctttatcg aaccaattac ggttggcctt caccggttcc acctcgcgca aggttgcgaa 480
aataagaatg tcattattat tggcgccggt actatcgcc tgccttgcaat ccagtgtgcc 540
gtcgctctcg gtgcgaagtc gtttaccgac atcgacatct cgtctgaaaa actcgccctg 600
gccaagtctt ttggagcgat gttgacgttc aattccttgg agatgtccgc cctcagatg 660
caaggagtc tccgagaatt gcgattcaat cagttgatcc tggaaacccg tggcgtgcct 720
aagaccgtgg aactggcggt cgagattgct ggaccccacg ctcaactggc cttggttggc 780
actttgcacc aagacttgca tcttacgagc accacttttg gaaaaatcct gcgcaaaagag 840
cttactgtca tcggatcttg gatgaattat tctgtcccct ggcctggtca agaattgggag 900
actgcactct gtctcctcac cgagcgaaaag ttgtctctcg aaccggtgat tgcctcatcg 960
ggcagctttg aatcgttcgc tcaggctgtc cgagacattg ctcgcaatgc catgccagga 1020
aaagtgtctc tgattcctta a 1041

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SEQ ID NO: 96      moltype = DNA length = 1020
FEATURE           Location/Qualifiers
source            1..1020
                  mol_type = other DNA
                  organism = synthetic construct
                  strain = K12

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-continued

SEQUENCE: 96

atgaaatcta	ttctgattga	gaaaccaaac	caattggcca	ttgttgaacg	tgaaatocca	60
actccatccg	ccggtgaagt	tcgtgtgaaa	gtgaagctgg	cgggaatctg	cggatccgat	120
tcccatattt	accgtggtca	taaccattt	gcaaaatc	ctcgtgtgat	cggacacgaa	180
ttttttggtg	tcattgacgc	agtcggagaa	ggtgttgaaa	gcgcccgcgt	tggtgaacga	240
gttgacgtgg	atcccggtgt	gtcgtgcggc	cattgctatc	cctgctctat	cggaaagccc	300
aatgtttgta	ctaccttggc	ggtgctgggc	gtgcatgcgg	acggaggctt	ctctgagtac	360
gcggtgggtc	cggccaaaga	cgcgtggaag	attcccgagg	cagttgcoga	tcaatatgcc	420
gtgatgattg	agccgtttac	aatcgccgcg	aacgtgacgg	gacacggtca	gccaactgaa	480
aatgacactg	ttctggtgta	tggcgctgga	cccattggcc	tcacaattgt	tcaagtctcg	540
aaaggagtg	ataatgtcaa	aaacgtgatt	gtcgcgcgat	gaattgacga	gcgacttgag	600
aaggccaagg	aatccggagc	tgactgggct	attaataata	gccagacacc	gctgggcgag	660
attttcactg	aaaaaggcat	taagccgacg	ctgattattg	atgcgcgttg	tcacccctcg	720
attctgaagg	aaagcgtgac	gctggcttct	ccagcggcac	gtattgttct	gatgggattc	780
tcgtctgaac	catctgaagt	tattcagcaa	ggaattacgg	gcaaggagct	ctctattttc	840
tcacgccgcc	taaocgcaaa	caagttcccc	atcgtgatcg	attggctctc	caagggcctg	900
atataaccag	aaaagctcat	tacgcacacg	tttgattttc	aacatgttgc	ggacgctatc	960
agcctttttg	agcaagacca	gaaacactgc	tgtaagggtg	tgctcacttt	ctccgagtaa	1020

SEQ ID NO: 97 moltype = DNA length = 1074
 FEATURE Location/Qualifiers
 source 1..1074
 mol_type = other DNA
 organism = synthetic construct
 strain = cp400

SEQUENCE: 97

atgaacattc	cgaccaagat	gaaggcagtc	gttaaatcgg	aggcaggtta	tgacaatatg	60
tcctcgcaag	agattcctgt	ccagaaagtt	aagggaatac	agggtgtgat	gaaagtgcgc	120
tatacgggta	tcctcgctgt	ggacattcat	acctttaaag	gcgagtagcg	caatgcctgt	180
accccaactg	tcctcgccca	tgaattttct	ggcaaggctg	ttgcgcctcg	agatgaagtt	240
accaagggtc	aaccgggtga	cggagttacc	tccgaaccaa	catttgatac	atgcggagag	300
tgtgtgtact	gtaaaaataa	ggaatataac	ctgtgcgaaa	accgttcggg	tattggtacg	360
aaagccaatg	gatcgatggc	taattatgtg	cttacgcgag	aggagtctgt	gcatgtcctt	420
cctgacaatg	tttctctaaa	acttgccgcg	atgtctgagc	cgtcggtctc	ctgtgtgcac	480
gcaatgtacc	agaaaacccc	gttcacccct	catgatacag	ttctcattat	cggccccgga	540
cctatggggt	tgctctctct	gcagattgca	aaggaaattg	gagcctttac	catcgtctcc	600
ggtattacga	aagacagcga	ccgcctggaa	atcgcaaaag	aacttgagag	cgacatcggt	660
gttgacacgc	agaagggaag	cttgccggaag	atcgctcatg	aggctactaa	cggagtggga	720
gttaccaaag	tctatgatgt	tagcggcgct	acaccagcgg	tgaatgtctg	cctccccctt	780
ctgaagaagg	gaggtacttt	tcagcaggtt	ggcctgtttg	caaagcctct	gaacgaattg	840
gatgaacgta	caattattca	gcacgaaatt	acatacgtgg	gatcccgctc	tcagaatccg	900
tatgatggcg	ctatcgccat	tcacctggag	gccaaagggt	ctattaatga	agataaaatg	960
gttacgcaag	tcttcgattt	ggatcattgg	cgtgaggcat	tcgaggctat	gatggctgga	1020
aaagaactca	aggttcttat	tgccagcaat	ccagacgacg	aaactcttaa	ttaa	1074

SEQ ID NO: 98 moltype = DNA length = 1053
 FEATURE Location/Qualifiers
 source 1..1053
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 98

atgaaagccg	cggctctcca	tggaaacgaac	gacatgcgct	tcgaggatat	tgaaatcaag	60
cgtgcgcaaa	gcgcagaggt	caagatcaag	gtcatggccg	ctggcatttg	tgatctgat	120
cccccccgag	tcgtgaaaca	ttggaagtat	cctgttccgg	ctattctctg	acatgagttc	180
tctggtgtta	tgcgagaggt	cggaaaagat	gtgaaaaatg	tcaagggtgg	tgaccgagtt	240
gtggcgatcc	cctttatttc	gtgtaatgaa	tgtgagtact	gtaaacgagg	tcttttttag	300
ctttgtgacg	atcacggaat	gctgggtgcc	aaatcctttg	gagcatttgc	cgagtatgtg	360
aacattaaag	cgactaatgt	ccttcccatc	ggcgatatgg	atttcgaaga	cgcagcgatg	420
attgagcccc	ttgcgcgtcg	catgcacgga	gtgttgaaca	tcggagttca	agtgggagat	480
acggtcgcgc	tcatgggctc	gggcacaaat	ggtcaacttg	tgattcaggg	tcttaaaatt	540
gctggtgctg	gcactatcat	tgctgttgac	atttctgata	acaaactgcg	agagtccaag	600
gaattgggcg	ctgacattat	cattaacgca	aaggacatta	accctgttga	aaaaatcaaa	660
gagcttacag	gcggaagagg	agtggacatc	gctctggaat	gtgcgggttc	caaaattacc	720
caagagcaat	gcctcttgat	cactaaaaaa	aagtcgaaga	tcggtttcct	gggaattgcg	780
tactctgaca	ttacctgtgc	ggaggaggcg	ttcgagaaca	tctttcgcaa	agagctcgaa	840
cttaaaggat	tctggaacag	ctattcggca	ccctttcctg	gccaaagagtg	gaccaagggc	900
attaatcttg	ttaacgaggg	aaaaaattaag	ctgaaggaaa	tggtttcgca	cgccttttct	960
cttgaggata	cgtataaagc	gttcgagatg	atccgagatc	gcaagggaaga	gtttaacaaa	1020
atcctgatcc	ttccccaggg	cgtggaaaaa	taa			1053

SEQ ID NO: 99 moltype = DNA length = 1053
 FEATURE Location/Qualifiers
 source 1..1053
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 99

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atgaagtcgg tccgatttta tggaaattcgt gatacacgag tcgaagacgt tgacgtgccc 60
aagatttcgg aaaaggatga cgtcattatc aaggtaaaagg tggcgggcat ctgtggttct 120
gacatttagca aataactcgaa aacaggcccc caccatgggtg gcgagatttt gggacacgag 180
ttctccggcg aggtcgctca ggtcggcaag gaagttcgct ccttcaagat cggagatcga 240
gtcgagtttt gccctgcgat gccctgcttc gagtgtgatg agtgcaaaaa aggttttgtat 300
tcccggttgcg ataattgtgc tatcatcgga aataaggagc tcggaggatg ttttgcggag 360
tacacgaaag ttaaaagagcg caatctgatt aaaaattccag atgagattag ctacgaaacc 420
gctgcggccc ttgaaccogt ctgtatcgca ggacatggtc ttttcgctc tgaagccaag 480
gtcggagata cggtcgttgt gctgggaacg ggtcctatcg gattgttcag cattcagtgg 540
gccaaaatct ttggttccac gaagattatc gcggttgacg tgtttgacga aaaactggac 600
ttggcgaaag agctcggtgc ggatatttgt attaatgcta aggagaaaaa tatcgttgaa 660
gagatcaaac gactgacaga cggagatggg gcagatattg tcattgagtc tgcgggtacc 720
ccgcttaccat gtggccaggt gcttctgctg gcgaaaaaag gtggtactgt cttgtatgca 780
ggcgctccat acggcgagct cgccttact cgagaacaat ttgagaaaaa cgttcgaagc 840
gagcttacgg tgaaaggaaac ttggtttggc aattctttcc catttccggg aaagggaatgg 900
tctgtggcc tgtaccacat gcagaaagga gacatgaacg tggaaaagct tgtcactcac 960
cgtatcaatc tggaaagaag tcacgataac ttcgagaagg ttataaacg tgatatcttc 1020
tttgtaaga ttatgattaa tattgacaat taa 1053

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SEQ ID NO: 100      moltype = DNA length = 1050
FEATURE            Location/Qualifiers
source              1..1050
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 100
atgaaagctt ccatgctgga ggacttgaac aaattcagcg tcaaagagat cgacatcccc 60
tccccaaaaa aggatgaggt cgtcgtgaag gtgatggcgg caggcacggt tgatccgat 120
tcgcataaaa tgattagcgg atggaaaatg ggctatccc cggtgatggg acacgaattt 180
tctggaattg tgacacaact gggagaaaaa gtgtccaacg tcagcgctcg tcaacacgctc 240
gccgtgggctc ccttcattcc ctgcttcaag gtgcattatt gtcaaatcgg actcttcocag 300
atgtgtgaaa attattccat gctcggtcag caaaaatttg gaggtttcga gcaatcgtt 360
tccgtcccg cagctaacgt tctggacatc ggcaagatgt cgttcgagga aggtgccctt 420
attgaaccga tggcagttgc gggccacgct gttatgggaa ttaaacccga gctgggtgat 480
acagttgcgg tcttcggcct cggtaacgct ggagatcttg ttgtccgtct tcttatttcc 540
tcgggagcga caaacgctat tggcattgac attgacgacc agaagcttga aaagggactg 600
gatgagggtt gcacacacgt gctcaactct gccaaagagt ctcttgaaga aaagatcatg 660
gaatacacca acggtctcggg tctcgatatt tccatggaat gcgcggcgag caaaatcact 720
gaggagcaaa cactgctcgt tacgaaacgc cgtggaaaagg ttggatttgt cggatttgtt 780
tatagcgacg tctcgtctca tcagaaagca ttgagaaca tcttcgccca cgagcttacg 840
gtgactggat ttggaattc ctttccgca ccatttccgt gacgcgagtg gaccaactcg 900
atccaattgg ttaatcgtgg tcgtatcaag atcaaaagat tcattactca tcgatttgag 960
cttgaagaca tgcaaaaagc attcaatatg atcaaccactc gttcggaatc gtttaataag 1020
gtcatgtttt tccccaaagg catcaattaa 1050

```

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SEQ ID NO: 101      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 101
atgaaggctc tcaatttgta cggtatccag gacctccgct ttgaggaaac acccgctccc 60
tctatcgaa atgatgacga tatcattatc aaagtcaagg ctgtcggaat ctgcggatcc 120
gatttgtcgc gttataagaa actgggacct tatgtccctg gaatgacttt tggacacgaa 180
ttcgcggaag aggtcgtcaa aattggtcgt tccgtgacgg gattttctat cgggtgactg 240
gtcgtcgtct gccctactta tacttgcggc cagtgcctgt actgtcaact cggcgagccg 300
acacgatgtg agcgaactgc tgtgatcggc gcacgccacc caggtgcata cgcagaatac 360
gtcaaaactc ccgcgaagca tgtcatcccc ttgcccaacg ttgttaacta tgacgaggcc 420
gccctcattg aaccggcttc ggttgttgcg cacggctttt atcgtaacta cattaagcca 480
ggtgcaagcg tcgtatttat gggcggttgc tcgatcggtc tgcctcgctt ccaatgggca 540
aaaaatcttc gtgcgacaac ggtcctttgc attgatattg acgaacagaa acttaacgctc 600
gctaatacac ttggagcaga tgtcttgatc tccagcctcc aacgaccgcg acacaagcaa 660
atccttgaat ataccaacgg aatcgggtgt gacgtggctg tggagtctgc tggtaacact 720
tctacctcgc cccagggtgt tgcgtgcggc aaaaaagggt gtgaagtgtt ctttcttgtt 780
attccttatg ccgatgtcca gattgaacga ttttacttcg aaaaaattgt tcgtaatgaa 840
ctgcacgttt acggatcttg gaatgcactc tctcctccat tcccggttaa agagtgggcc 900
acgactatcc attatatgtc gtccggacaa ttgaatgtcg cgcccatgat ttctaccga 960
ctccctctcg ctaagggccg cgagacgttc cagcagatcg caaagggtga actcaagccc 1020
acaaaagtc tttctatccc tgagaagttg tctgagcgca agtaa 1065

```

```

SEQ ID NO: 102      moltype = DNA length = 1056
FEATURE            Location/Qualifiers
source              1..1056
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

```

SEQUENCE: 102

-continued

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atgaaagtcc ttagcctgca taagcctggc gatattcgat ttgaggacgc tccaatgcct 60
gagccaggag agggcgaagt gctgctgaaa gttatggcat gcggcgctcg ttcgtcggac 120
gaggctcgaa ttttcaagga cgggtgcccc attatgccaa ttgttccctg ccacgagttc 180
tgtggccaac tgggtgaaaac gggagcagga gtggacgaaa gcctcatcgg ccgaaaagcg 240
agcgtcttcc caatgcttcc ctgtttcgaa tgcccttctt gtaaacgagg tgactacgca 300
acctgttcta actataaata ctatggttcc cgcaacgatg gtggaatggg agaataccga 360
gcaatcaagg cgtggaacct gaaccttctg gatgattccg tcgattattg cgtcggcgca 420
ctgtccgagc catccgcggg tggccatcac gccactgaag tgggaggagt gaaagccgga 480
gacaaggttt tcgtcattgg tagcggaaac atcggacttt tgatcggctt cttcgcctcag 540
gctaagggtg cagagggttg gatcgggtcg cgtcgtaaa agagcaacga tttcgcggcg 600
agcctgggtt ttaaaacgat cgacaccaat aatacgaag acgaggttgc aaaggccacg 660
gatggtaacg gcatggatgt tgtcttcgaa cgacttggtg gcaataaagt tcttattgaa 720
gctattggat ctgtcacggc gggaggcgca ctgtccaccg tgggcaatcc tgtgggagac 780
atgaatcttc ccgagatgt gtactggcgc atcctccgaa agcaactcac ggtccgaggt 840
acttggaaact ctactactc tgacaaggcc aatgactggc acgaagtggc cgcgttgatg 900
aaagaaaaaa aaatcccgtt cgaaaaactc atcaccgta agtacaact tctcaatgc 960
gatgaggcct tcgattttt cegtgaccac tctattcaga agtcccgcgt gatgtttgtt 1020
atgcacgaaa aagaatccga agcttacaac ggataa 1056

```

```

SEQ ID NO: 103      moltype = DNA length = 1011
FEATURE            Location/Qualifiers
source              1..1011
                    mol_type = other DNA
                    note = Faecalibacterium sp. OF03-6AC
                    organism = synthetic construct

```

```

SEQUENCE: 103
atggtaacaa tgatgaaagc cctcgtgttt actgcgcccc atacgttcga gtatcgtgac 60
gttccccgac ccgaaccggc ccccgacgag gtccgtggtg aggtgaaagc atgcgcaatt 120
tgtgtgtctg atgttccagg tagctcgggc gaaaacggtc gacgcattcc tcccttggtg 180
atgggcatg aggcaagcgg cactatcgca caattgggcg aagctgtgca gtcgtgggca 240
gtgggagatc gcgttacatt tgactctact gactactgcg gctactgttg gttttgcacg 300
cacggtatgc acaatttgtg cgcccagcga cgtatcgtgg gcgttgctg cacagagtac 360
cgtaaaaatg gtgcaatggc tcaatacatt cgactgaaag cccgtaccct ttatcgtttg 420
ccgcgcggcg tctcctttga acagggttgc cttgtcgaac ccttgctgtg gggatgcat 480
gctgtcttgc gatctggact ggcagcggga cagacggcgg cgtgtgattg agatggaaca 540
atcggctctg tgactcttat tgcggccaag agccaggag cacaagctcg tatcagcgga 600
caccacaccg gtcccatgca gctcgagcc cagctcggag ccctggattg tgcccaactg 660
ccccagctga cagctgagcg aggtgcggac gttgtttacg acgctgttgg ctctcaatcc 720
agctttacgc aagcgatgga gcttgtccgt aaggaggcgc gaattatggt tatcgccaat 780
catgcacgca cgtcgtgatt tcctttgcaa cgttgtatca cggaacagat cgacggttcag 840
ttttcgtaca gctctgcggg tgagtatcct gtttgtcttc aagccattgc tgagggacgc 900
gcagatttga caggactgtg gcaatcttcc ccactggcgt agggcgctca agttttcgac 960
cgctcgttcc accgagatga gtccctgacg aaagccggtt tgatcccata a 1011

```

```

SEQ ID NO: 104      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 104
atgaaggcct atgttttggg ttgtattggc cagctgaatc aacatgagta tgtcctccga 60
gacttgacgc ctggagaagc cctcgttcaa gtggctgcgg ctggtatctg cggtagcgac 120
ctgccccgca tctttgagac tggaaacttat cattttccaa ctatcccagg ccatgaattc 180
gcaggtgttg tcaagaactt ttatgctgat gccgacgaag aaatgctccg acaatggatg 240
ggtaagcgcg ttgcggtgtt tccactgatt ccctgtcgaa cctgtcgaag ctgcgcgaat 300
catcagtagc agatgtgttc taactataat tacctcggct cgcgtacgga cggaggcttc 360
gctgaatact gtatcgtgcc tgtttggaac ttgttcgagt tgccggacga ggttagcttc 420
gcacaggccg ctatgatgga accggcagct gtggcattgc atgcaatccg acgtggcgcg 480
ctgcaccccg accacccgga gaccggtatc ggcaagcgtg ttctggttat gggacttggc 540
actattggcg gattgatcgc ccagtggtct agccaatgtg gcgttacgga gctcaccgcc 600
actggtcacc acgcccccca ggaagagctt atgcaccgat atgctaacga gaactaccga 660
ttcattaata ctaatcaaac tgcttgagc gatgacgaaa acacctacga tcttattttt 720
gactgcatta atgacagcga atccgttcag caggcggtgc gctacgtcaa acccggtggt 780
cagattgttg ctgttgccaa ccgaaaaaaa gatcttactc tcaaaaagga catctattgg 840
aaactcctgc gaaaacaagt gtgtatgacc ggaacatgga actcgagctt cagccatgat 900
tcgaacgatg actggcagat cctccttcga taccttcagg agcaaaagct gcatccgcag 960
gcacttatca cacatcgctt tccgtttgcc caacttccca agggcgttga gatcatgcgt 1020
cttcgtgaag agttttacaa caaaattatg atcgacttga ctgtctgttc ttaa 1074

```

```

SEQ ID NO: 105      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 105
atgaaggccg tgacttacac tcccagaccg ggttttgcac ttggtcgagg tgagacaccc 60

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-continued

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cagcttggac ctgacgaagt tcttctccag gtggaggctt gtggcgtctg cggaagcgac 120
cgtcaagtcg ttgcaggcga atcgggttcct accggtaccg ccttccctct ggtgatgggc 180
cacgagattg ccggacgtat cgcgcgcttg ggcgaaactg tgacagattg gcatgttga 240
gactctgtga ttgtccatcc ctctcgtggca tgtggtacgt gcgcaccctg tgttcaaggc 300
cagcctaact tgtgcgtccg acagggtgtg attggttacc aacgacctgg aggatttgca 360
gagcaagtgg ctgtcccggc tcgtcaactg gtctcagctc cagcagaatt ggcaccggcc 420
gccgcagcat tgggttgtga tgcgtttgcc acgccatacc acgcgatgcg agcggcgggg 480
gttgagagcg gccaaacggt ttgattattt ggaacccggc gacttggctc cgcggcactt 540
atgtcggcca tgggcttcca tgtcggccgt ctgggcccgg tcacgcgacg ccaagagggt 600
attgcaatcg cggaggctca tggcgcgcag atggttgtcg tcacggacca ggatgagcgt 660
acaattgcgc gccagcttcg ccgctggctc ggcgcacccg gcattgacgt cgtcatcgat 720
acggtcgcta ctgctgacac tattacattg gcagctgaaa gcgttcgccc aggaggcact 780
atcgtatttg tcggaatgtc ggccgacacc gccctctatc ctgtcgttaa gaccgttcgc 840
cgaggtctga cattcgttgc gagctacgga tcctccgag aggatgtcga gcagttggtc 900
agctgggttg cagaaggtcg actggaccgc gtctcagctc tcgcgggagc cctcccactc 960
catgaagcgc ctctgtgcatt cgcgccggac cgaccgcccg gacgttgggt tattgatcca 1020
tctgaataa 1029

```

```

SEQ ID NO: 106      moltype = DNA length = 1074
FEATURE             Location/Qualifiers
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = KGMB03662

```

```

SEQUENCE: 106
atgaaaagct tgattaatat cccggaaaaa atgaaagccc tcgttttcta tgggtgttgg 60
gatttgcgtg tcaactgaagt ggacgttccg aagctcgaaa aaggaaacgt cctccctaaa 120
attaaagctt cggcctactg ttccctccgac atgcctcgtt gtttcgttac tggcacttac 180
cattttccga ccattccagg acacgagttc tctggacaaa tcgtttcggg ggggtgatgat 240
gttgatgaat ccttgttggg aaagcgttcg tgcgttttcc ctatgcttcc ctgcaaacac 300
tgcaaaagctt gcaagatgga ggaatatgct cagtgtagcg gctattctta ctccgctct 360
cgccaaagtg gtgcctgggg ggaataacct gtggtcccgg tgtggaacct ggtgcccttt 420
gacgatacac tcgattataa aactgcagcc ctgtgcgaac cagctcgggt ctctatgcac 480
gcggttaata ttggagacat taaagaaggt cagaatgttg tgatcatcgg aacgggtacc 540
atcggattca tgattgtctc ctttgcctaa gaacgtacga aaacaggaaa ggttgtgac 600
gcgggacgct cgaataataa gctggaatac gcgaaaaagt tgggcttcga tgcgttgac 660
acctcggttg acgatttgcc tgcccgaagt aagaaaaatca tcggtattga tggagctgac 720
gtcacattcg aagcgggtcg ctccaatgag gccattgcct ctgctattga atctacgggc 780
gctttggggt cgattgttct tgcgtgtaat ccttccactg atctcatctt gccaaagaat 840
atttattggt ccatcctccg aaagcaaat acagttaaag gttcctggaa ctctgtctat 900
aactctcgcg ttaatgactg gaagaaagcg ttggaggtgt ttgagcatag cgacgtgaat 960
tttgacagct tgatcacaca cacattcact atcgaagaaa acgagaaagc tttttctgcc 1020
atccgtaaca ctggagaatt cactctcaaa gtcattgtta ccttcgatga ttaa 1074

```

```

SEQ ID NO: 107      moltype = DNA length = 1047
FEATURE             Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    note = Clostridiaceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 107
atgaaaagctg tcgtccctgca tggtaaaaaa gacatttgct atgaagaatg gcctgaaccc 60
atcctgaaaaa caggaacagt taagattcat gtcaaggctt gtggcatttg cggctctgac 120
attcctcgcg ttctcggaga ttctgccccat tactaccga tcactcctcg ccattgagttc 180
tcgggttgtg tcaactgaggt ggctgatgac gttgacaaga tccagggttg ccatcatggt 240
gcaggagtcc cccttctccc gtgcattgaa tgcgatgatt gtcaacaggg aaactattct 300
cagtgtaaac attactcttt tatcggcagc cgcgagcaag gtggatttgc agattatggt 360
gtgatccag ccgataacgt ggttaaagtc gacccaactc ttcttacga acaggtcga 420
ctcttcgagc cttgtactgt tgccttgcat gcaactcaaag tctcctcgta taaaggagga 480
aaacgagtggt cgatcctggg tggcggcagc attggtcttt ttgcactcca gtgggctaaa 540
atctatggcg cgtcccaagt cacagtcttc ggtcgcagca aggaacactt gaagatcgct 600
gctgctctgg gtgcccagac aactatttcc actctggacg ataattttat gacagaggct 660
atgcgagtcg cagagaaaaa gggttttgac tacattttac agtcgcgcgg ttccacaatc 720
accatgaagc ttgcttttga actggcagcc aaccacgcga aaatctgttt cattggcacg 780
cccacgaaaa acctgacttt cagccctaag ctgtgggaga acatgaatcg aaaggagttc 840
cagcttatgg gttcctggat gtcgtgttcg gcgccgttcc ccggcagatg atggaaggag 900
acagcccatc tctttaacac cggacaactt cgatatgtgc ctgagatggt ttccgcacag 960
tatcccatgg acgatgttaa aaagcgttc gagcagttcc ataacaagaa taaagttaa 1020
ggacgtatcc tgcataaaaa caagtaa 1047

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```

SEQ ID NO: 108      moltype = DNA length = 1014
FEATURE             Location/Qualifiers
source              1..1014
                    mol_type = other DNA
                    note = Thermoplasmata archaeon
                    organism = synthetic construct

```

-continued

SEQUENCE: 108

```

atgaaagtgg cggttttgta tggtagacaa aatattgtct tggaggaccg tccgatccct 60
tctatcgggtg aaaaggacgt cttgggtccgt gtcaagtcgt cttccatctg cggagctgac 120
cttcatgcct acaaaggcct ctctaagcgc cgcaagcctc cattggtgat gggacatgaa 180
ttttccggtg tgggtgaacg cctcggcgat gagacaaagg gccttaagggt cggcgatcgt 240
gtcgtggttg agcccatat tagctgcgga aagtgcagg catgcaaaaa gggaaaaacg 300
aatgtctgcg aaaatcgtaa gctgatcggt ctgaacctcg acggtgcttt cgtgaatat 360
gtttctgtcc cttatgaaaa gtgccatgaa ctcccccggt accttttcgt cgatgagggt 420
gcgattatcg agcgtttggc cgtcgcggtt catgcggtca atattactcc catctcgtcg 480
aacgataacg tgcctatcat tggtagcggg gtgattggcc tcctcatcac acaattgctt 540
cgaagccgca cttcgggcga cattttgggt accggacgtc gcgattacct ccttgaaaag 600
gcccgtgaat ttggtgccac agatgcaatc aacgtgatcg actcgcgtgc tgtgaaaacg 660
ctgctcaaaa aacacggctg cgtcgatggt gtcttcgaag cagtgggctc taacgagact 720
gtgaaccagt ccatttccct ggtcgaaatc ggaggttcta tcaactcgtg tggcatgctc 780
gatcgcgaga tgaccgttga tatgatggag gttgttgccc gtgaaatcac tctcgcgagg 840
gattatggtt atacaactga cgactttaag gaggcgatcg agctcgtctt ttcgggtaaa 900
gttgatctca aaggcctcat cacgcatgtt ctcccgtcgt atgagatcaa aaagggtctc 960
gatactctgg cagaacgcgc agaaaatgtg atcaaagtgt ttattaaccc ataa 1014

```

SEQ ID NO: 109

moltype = DNA length = 1032

FEATURE

Location/Qualifiers

source

1..1032

mol_type = other DNA

note = Lachnospiraceae bacterium

organism = synthetic construct

SEQUENCE: 109

```

atgaaagcac tgggtgtgga ggaatcggt aaattgacct ttaagagggt tgagacgccc 60
gtgccaggac gcggagagggt tttgtgctc atccatgcct gtggtatttg ctcttcggac 120
aaagcccgct ctttcaagaa cggtaactac accatgccgc ttatcccagg tcacgagatc 180
tcgggtgaga ttgttgacgg aggtcccgcc attcgagaag agcttatcgg ccgacgcgct 240
gcgggtcttc cgttgattcc ttgtcgccaa tgcgactcct gccgaatcgg caagtaacgt 300
caatgcgagc attacgacta tttcggatcg cgtcgagacg gaggccttgc tgagtacctc 360
gctgtcccggt tgttaatat caatcttctt cccgatgata tggattatca tacggctgca 420
ttgtgcgaac cagccacgggt tgcactctcat gcacttaagc gcgccaaaaa gaaatctggt 480
gatacgttta gcatcattgg cactggcacg gtccgacctc tggctgctgg catcgctcgt 540
attcgacgag catctaaat cattatcgtc ggacgaagca aagaaaaaat catgaatgca 600
gctcatttgc ctgattcatg gatcaatgca gagggatggg acgtgaacga ggcgctggct 660
tcgctgacgg aggggtcacat gacgcatgtg gtcttgagga ctggcggcaa cagcgatgct 720
tgacgctacg cgcttatggc tgcggcgaac ggtacgactg tgatgtttat tggcaatccc 780
gaagaaccga ttgtttttta cgtgatatt tattggaaga tcctccgaaa cgaactgaca 840
ctcaagggtt cttggaattc tggatatgct cctgaggata atgactggaa tgacgtcac 900
cgttatatgg cagatggaca gattccggca gataaaatta tcaccgatag ctttccattg 960
gaagattata ataaggcatt cgagcttctt ctctcggatc gcttttacac caaatcatg 1020
tttgacatct aa 1032

```

SEQ ID NO: 110

moltype = DNA length = 1047

FEATURE

Location/Qualifiers

source

1..1047

mol_type = other DNA

organism = synthetic construct

strain = DSM 16973 = JCM 13450

SEQUENCE: 110

```

atgaaagcca atgttctgta ccacattggt gacctgcgat aactgacgt tgccttgccg 60
gaactccacg acggtgaggg cctcgtgcga gttcgtgcat gcggtatttg tggtagcgat 120
gttgcccagag tcttccaccac tggtagctat catttcccca ccattatttg ccattgattc 180
tcgggcgaag tctgtgaacg atattctgct gctgacgcgg agtgggtggg tcgccgcgtg 240
agcgtttttc cctccatccc ttgtcgccat tgtgatgcct gtcacatggt tcagtacgaa 300
atgtgcacat gttacaacta tctcggcagc cgtaccgatg gagccttcgc agagtacgtg 360
gctgtcccggt tctggaatct ctttctgtct ccagattccg tgtcttacga ggagggcgca 420
atgatggaaac cggccgctgt cgtatgcac gcgctcaaga ttgcccggtt gggatagggt 480
gacactgtgg caattacagg tcccgaaact attggtatga ttttgaccca attgatccaa 540
cgtattggcg gtacaaaagc ggttctgac ggacgcaccc aagcaaaatt ggattacgca 600
cgacgcacat gcacccgtca tgtgtgcaat tcgggaacgg agcagctcgc ggaccgcgtt 660
ggacaactta caggcggacg tggagccgac atcgcaattg aaggaacggg tgcgtcggaa 720
cccatgaatc tgtgcttgaa ttgggtgcgc tctgaaggac gtattgtcgc tatgggaaac 780
ccccaaagtg acatgctggt ccgaaaggat gcgtattgga agcttctgct caagcagctg 840
acccttctgt gaacctggaa ttgcacatat gggattgggg gagatgattg gcaggatatt 900
tcgcccttct ttgctgaggg agcattgaac ttgcgcggcc ttatcacaca ccgactccca 960
ctggccgagc tgcagcaggg actcagcttc atgcgagacg cagcagctca cactaacaaa 1020
gttatgatcg tcgacgagca aggttaa 1047

```

SEQ ID NO: 111

moltype = DNA length = 1014

FEATURE

Location/Qualifiers

source

1..1014

mol_type = other DNA

organism = synthetic construct

-continued

```

strain = OM05-6
SEQUENCE: 111
atgcgcgcct atgagcttca cggagtcac gacttgcgctc gcgaggacat cgaataacct 60
gagattccct ctggatgggt tcttggtcaa gtcaaaagcga gcggcatttg cagctccgac 120
atccctcgta tcttcacgaa cggcacttat cactttccta ccattccggg ccacgaattc 180
tctggagtcg tggccgccta tggcgaaggt gtcccgggag aacgcgttgg caaacgagtg 240
ggatttttcc cgctgatccc gtgtcgtaaa tgtccccaat gccgccagaa gaaatacga 300
atgtgcgagc actatgatta ccttggtctg cgtcgtgacg gcggattcgc cgaatacgtt 360
gcagtcgccg actggaacct cctggaattg ccggagaaacg tttcgttccg cgaggcgccg 420
atgctcgaa cctgtgctgg ggcacttcat gccgtgaaac gcagcggcgt gaaacctggt 480
gatacagcgg ctgtgatcgg cacaggaatg atcggttttg ccgcagctgc ttgggccaac 540
gcactgggag cagaacacgt gtctgtgatt ggacgcggcg aggcacaacg cgctattgct 600
gaacaaatcc ccggaatttc ctatgtgact gaagaggaat ccaaggccat tcaggcggac 660
gtggctcgct aagcagttgg tacaccacag gctgtttcct ccgcggtgtt gcttgccaga 720
ccgggaggtt ccattgttct gatgggcaac ccattccggtg atcttgcat ggaataaac 780
gtctactggc gtatcctctc taagcaactt aacttgatgg gaacgtggaa tagctcgtat 840
gaaaaagacg aagcctgtga ctggaccgaa gcgctcgacg cattgtccaa gaaaaagatt 900
cctgcccacg cacttatcac tcattgtttt cccagcgaaa aacttatgga tggactggaa 960
ttgatgaaga atcataagga accatactgt aaagtattgg tcagctggga gtaa 1014

```

```

SEQ ID NO: 112      moltype = DNA length = 1065
FEATURE
source              Location/Qualifiers
                    1..1065
                    mol_type = other DNA
                    note = Candidatus Bathyarchaeota archaeon
                    organism = synthetic construct

```

```

SEQUENCE: 112
atgaagcgta tgcgagctgc catcttgat ggacaaaag acatccgaat tgaaacatt 60
gagatgcccc aggttaagcc ggagtggtg cttattaagg ttccgcaac aggtatctgc 120
ggttctgac tgcactttta caaagagaag agctttatcc ctattattag cgacgtgggc 180
aaaggacgct acgtcccggg tcatgaattt tcgggcgagg tctacgagtt gggttcgaag 240
gtcaaaagcc tgaagaaagg agatcgctgc ggcgtcgaa ccatcattgg ttgcggcaaa 300
tgtcagtggt gcaaggtcgg atgttacaac ttgtgtgagc actacaatt gattggcttt 360
cactatattg gaggtatggc tgagtattgt gttgctcctg ctgagaattg ttttaagctc 420
ccaaacaatg tttcggtcga aagcgcggca atgcttgact gtatcgccgt ggctattcac 480
gcagtgaata aagcagaat tcgcggcggt gacagcgctc tcattgtcgg ccgcggtgcc 540
atcggaactc tcgcctccca ggcagcgctc atccaggagg cgagcgacgt gtacgtcgtt 600
ggaaaaatcg aattccagtt ggagaaagct ccgaagtggg gagcgacagc tgtgattaac 660
gctaaaaacg aatctttctc cgaaaaaatc atggaattta cttcgggtaa aggagtcgat 720
aaggtcattg aggcgtcggc gggagcctcc aatgttattt ctgacgtcct tgacgtgact 780
cgaaaaaagg gtactactgt gacctgtgt atctttacac gacctctc ggctgatttt 840
tcaaaaatc tcacaaaaga gttgacgac acttcgtcgt ggggatactg ttacttcgac 900
cgtcgcaagg acgttggcgc ggcgctcgac tttctggcaa aagaaaagat caacactcag 960
gaacttatta catataagta tcctttggag aaggtttccg aagcctttga gacggcactc 1020
aacaaaaata aggagaagag catcaaggtg gagattatct tctaa 1065

```

```

SEQ ID NO: 113      moltype = DNA length = 1053
FEATURE
source              Location/Qualifiers
                    1..1053
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 113
atgaagcct ggggtgctga cggtatcgga gatatccaat acgaggaagt taaggagccg 60
cagtcggtcg ggcagaggtt tattatgaaa gtgatgagcg tgggaatctg tggatctgac 120
atccctcgca ttataaaaac cggcacctat catttcccaa caatcccggg ccacgaattc 180
tccggtattg tggccgatgt ttgtgcccga gtcaaccgga aatggaagaa taagcgcgtc 240
ggcgtgttcc cattgatccc atgtcacgtt tgtgcttcgt gtaaaaaagg tcaatatgaa 300
atgtgtcgca actacggata tctgggtagc cgcacaaatg gtggttttgc agagtatgcg 360
gcggttccgt tcaagaatct catcgaaact ccggactctg tctctatgca ggaagctgcc 420
atgcttgagc cgattggcgt ggcagtcact gctatgcgac gagcccgctc catgacggga 480
gatcgtattg tcgtgtgcgg acttggaact atcggcacgc ttctcgtgct ctttttgcca 540
gagacgggaa ttccagacat ctttgtcatc ggcaataagg cattccagcg aatcaccatc 600
ctcaaaactg gccctgcagga aaactgttat tgtgattctc gcgtgcagga tgcgagaag 660
tggctcatgg agcgaactga aggaaaggcg gcagacgtgt tctttgagtg tgttgccga 720
aacgaaacca tctccaaagc agtcaacctg acggcacccg gaggttcggt ctgtctggtt 780
ggtaatcctt gttcggatg gacgcttgag cgtgaggtct actggaatat tctccgtaac 840
cagcttacta ttactggcac ctggaattcc tcgttcaccc acgagatcga ggaacgactg 900
cattacgtgt tggatcgctt ggaacaaggt aagatctcgc ctggtgatct catttctcac 960
aaattcccgc tcaagaactt ggaatctggt ctgcatatta tgcgtgataa gactcaagat 1020
tatatcaaaa tcatggccga gattaatgtg taa 1053

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```

SEQ ID NO: 114      moltype = DNA length = 1047
FEATURE
source              Location/Qualifiers
                    1..1047
                    mol_type = other DNA

```

-continued

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organism = synthetic construct
SEQUENCE: 114
atgaaggccg cagtgtctca ttccgaaaat ctgtttacgt acgaagaagt ggaatccgat 60
ccctgcaaaag acgacgaggt taagatcaag gtgatggctg cgggtatctg tggttccgat 120
acacataaga tgacgagcaa atggaaatat ccccttccga ccatcatggg ccatgagttt 180
tcgggcttta tcgtcgagat tggtgaaaag gtccactaac cgctcgattg ccatcgagtt 240
gtggcgatcc ctttcttgcc gtgcaatgac tgcgaatact gtaagcgtgg acaattctcg 300
ttgtgcgagg atcacggcat ggttggatcc aaaagctttg gtggtttcgc ggagtcagtc 360
aatatcaaaa gcactaatgt tcttccgatt ggtgacatgg attttgagga agcagccatg 420
attgaacctc tcgcagttgc gcttcacggc gtgttgaata ttcacccgcg tattggagat 480
attgcagccg tctttgggat tggcacaatc ggaatcctca caatccaatg gcttaaaatt 540
gcaggagtga aagaggttat tgcgatcgac atttccgata acaagctcgc agacgcccac 600
gctatgggct gtgagcgtag tatcaatcca tttaaagaga acctccttga gaaaatcgac 660
gaatatacta acggtcgcgg tgttgacatc gcctttgagt gtgctggatc taaagtcaca 720
caagaacagt gcctgctcgt gacagccaag cgaggaaaag tcggctatgt gggtagtcgt 780
tactctgatg tccatttgtc ggagcaagct tttgaaaaca tttccgccc cgagctcaca 840
cttaagagat tttggaatc ttactccgca ccatcccgcg gagaggaatg gacccaaaagc 900
attgacttca ttaataaggg tcgtattaaa ctgaaagagc tgattagcca ccgatttaag 960
ctccaagata ctcaaaaagc gtttgacatg atgatcaacc gagaagagac atacggcaag 1020
gtgcttattt tgcccaacaa agagtaa 1047

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SEQ ID NO: 115      moltype = DNA length = 1047
FEATURE             Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 115
atgaaggccg cggtgcttca ttctgagaat cacttcattt atgaagaaat tgatattgct 60
ccttgtcaag atgatgaggt taaaatcaaa gtcatgtcga ctggaatttg tggatcggac 120
acgcacaaga tgacaggcaa atggaaatat cctctccccg ccgtgatggg acacgagttc 180
tcgggataata tcgttgagaa aggaagaac gtccactaat cagccatcgg tgaccgcgtc 240
gtggcgatcc ctttctctcc gtgtaacgac tgtgagattt gtaaacgtgg ccaatttagc 300
ctttgcaag accacggtat ggtcggctcg aagtccttcg gaggttttgc tgagtacgtt 360
aatattgagt cttcgaatgt ccttcccatc ggagatatgg aattgaaga ggcggccatg 420
attgagccgc tcgcgggtggc ccttcatgga gtcttgaata ttcacccgcg tgttggcgac 480
gtcgcggccg tgttcgggtg cgttaccatt ggtatgtcga ccatccatg gctgaaaatt 540
gccccgtgcca aggagatcat gccaatcgac ttgtcggatg aaaaacttgc tgacgcgcgt 600
gctatggggt gtgaacgtac gatcaatccc ctgaaacgaag accttcttga aaaaatcaac 660
gaatacacca acggaatggg agttgacatc gccttcgagt gtgctggctc gaagattaca 720
caagaacagt gcccttctat caccgcgaag caaggcaaaa ttggatacct gggcattgct 780
tattccgatg tccacctgtc tgagaaggct ttcgagaaca ttttccgacg cgagcttacg 840
tgaaaggtt tctggaatc gtattccgct ccatctccgg gtgaagaatg gacaaaatcg 900
attgaattca ttaaaaaagg ccgcatttaag ttgaaggaaac tgatctctca ccgatttaag 960
ctggaggata cacaaaaagc atttgatatg atgatcaacc gtgaagagaa atatggtaag 1020
gttttgattc ttccgaacga ggagtaa 1047

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SEQ ID NO: 116      moltype = DNA length = 1053
FEATURE             Location/Qualifiers
source              1..1053
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 116
atgaaggccg cggtcctgca ctccgaaaat gagattattt accaggacat tgaaatttcc 60
gagtgtcaac caaacgaagt gaaagttaag gtcatggcgg ctggaatctg tggttccgat 120
acccataaaa tggtagagcaa atggaaatac cccctccctg ccatcatggg tcacgaattt 180
gcccgtctacg tcgttgaaaa aggcacaagc gtgacgaacg tcgaaatttg ccatcggtgt 240
gcccgtatcc cgttcacccc atgcaacgag tgtgagattt gcctgcgagg tcagtccagc 300
ctttgcgaaa atcacggaat gatcggcagc aagtcctttg gcgcatttgc cgagtttggt 360
aatattccag ctaagaacgt tctcccgatt ggcgaaatgt ctttcgaaga tgcagctatg 420
atcgaaacct ctgcggtcag cctgcatgga gtgctcaaca ttcaccaca gctcggcgat 480
gtcgttgctg tcttcggtgt gggaactatt ggaatcctcg caattcagtg gttgcgtgtc 540
gcccggcgtga aagagattat tgctattgac atttccgaca agaagctcca ggacgctatg 600
gatctggggt gtgagaagtc gattaaccct ttgaaggaaag acttgattaa gaaaatttat 660
gaatacacga atggcctcgg cgttgacatc gcatttgaat gcgctggatc taaaattacg 720
caagaacagt gccctgctat tacccgtaaa caaggaaaag ttggttattt gggaatttgcg 780
tacagcgatg ttactctctc tgaaaaagcc tttgaaaata ttttccgccc agagctcact 840
ctcaagggtt tctggaatc gtattccgcg ccgtttccag gaaaggagtg gatgaacacg 900
atcgagttcg tccaacaggg caaaattaa ttggtgtcga tcgctacaag 960
ctcgaagaca cccagaaggc gtttaacatg atgatcaacc gagaagaaga atacaataag 1020
gtgatgatca tcccaacga ggaactggag taa 1053

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SEQ ID NO: 117      moltype = DNA length = 1041
FEATURE             Location/Qualifiers
source              1..1041
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 117
atgaagcgag cgggtgcttca ttcggtcgga gacatccgct acgaaaacat cgagatcaag 60
ccttacggag acgatgaagt gaaaatcaaa gttaaagccg ccggtatctg tggatctgac 120
ccgccacgtg cactcaaaaa gtggaatat cccgtgccag ccattttggg tcacgagttc 180
tcgggttata tcgttgaagt gggaaagaac gtgaagggtt ttaagtgagg agatcgtgtc 240
gttgccgtcc ctctcggaacc gtgtcaccaa tgtgaatact gcaagaaagg ccaattttcg 300
ctctcgcaaa gctacgatat gttgggtgct acttcctttg gcggttttgc agagtacgct 360
aacgtgaagg caacaaatgt tttgaaaatt gacgatatcg actatgaaga ggcggccatg 420
attgaacctc tcgcagtcgc actgcatggt gtgcttaata tcaatccgca actcgggtgat 480
actgttgctg ttttgggagc tggaaacgat ggtcaactca cgattcaatg gctcaagggt 540
tcgggagtcg agaagattat cgcagtcgac atttcggaca aaaaaattag cgaagctatg 600
tccttgggag cagatattgg tattaatgct cttaaagaga atccggtcga aaagatcatg 660
gaacttacaa acggttttcgg cgtggatatc tgcattgaat gcgcgggctc taagatcact 720
caggaaacat gctctgtggt cagcaagaaa aaaggttaaga ttggatatca gggtatcggg 780
cacgcccgaat ttgagttgtc tgaagttgct ttcgagggtta ttttcgtaa agaattgaac 840
atccaagggtt tttggaatag ctacacgccc ccgttccccg gacaggaatg gttcaaaagc 900
atcgagtatg ttaagcgaag aaaaatcaag ctcaaacagt tgatcagcca tcgcttcagc 960
cttgaagaca cagcgaaagc ttttgagatg attggtgaac gtaaggagga gtataacaag 1020
atcatgattg tccaggacta a 1041

SEQ ID NO: 118 moltype = DNA length = 1047
FEATURE Location/Qualifiers
source 1..1047
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 118
atgaagcgag cagttttgca ttcgagaat cacttcattt atcaggatat ggagagcttc 60
gagtgcgaga atgacgaagt caagatcaag gttatggctg ccggcatctg cggaagcgat 120
acgcataaaa tgggtttcgaa atggaaatat cccctgacct caattatggg acatgagttc 180
tcgggttacg tcgtcgaaaa ggttagcgat gtggcaaacg tcgagatcgg cgaccgagtg 240
gttgccattc cattcttgcc ctgtcacact tgtgagtatt gcaagtcggg tgtttattct 300
ttgtgtgaag atcacggtat gatcggttcc aagagctacg gtggcttcgc ggagtatgtc 360
aatctcaagg caaccaactg ccttccaatt ggaacatggt attttgagga ggcgctatg 420
atcgaaacgc ttgcagtcgc cctccacggc gttctccata ttcagccgag tctcgggtgac 480
gtggccgccc tttttggagt cggcaccatt ggcctgttgg ccacccaatg gctgaagatt 540
gcgggagtg aagaaattat cgctattgat atttcggatg agaagttggc ggaagcgaaa 600
gccatgggat gcgaacgaac cgttaatccc ctccaagaca atctcatga aaaggccac 660
gagatataca acggtctggg cgtggatgct gccctcgagt gcgcgggac taagattaca 720
caggaaacat gtcttttgat cactcgcaaa aagggtgaaa ttggttactt gggaaatcgcg 780
tattctaattg tgggtttgag gaacaagct tttgagaaca tcttctgtaa agagcttacg 840
ttgaagggtt tttggaattg ttaagcgcct cccctccctg gcgaggagtg gacgaaatct 900
atcgacttca ttctgaaggg ccgcatcaag ctgaaggagc ttatcagcca ccgctttccc 960
ctggaagaaa cgcgaaggcg attcgacctg atgattaatc gaaaagaac ctatggtaag 1020
gtgatgatca tcccccaataa ggaataa 1047

SEQ ID NO: 119 moltype = DNA length = 1047
FEATURE Location/Qualifiers
source 1..1047
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 119
atgaagcgag cagtgtctca tttcgagaat cagatcgtgt ttcaagaatt tgagatcggc 60
gaacttaagc cagacgaagt taaggtaaaa gtgatggcag cgggcatctg tggctcggat 120
acccacaaga tgggttcgaa gtggaagtat cccctgccag caatcatggg tcatgagttt 180
gcggctata tttcgaaaaa gggaaaaagt gtggtgcagt tggaaagtcg tgatcgtgtg 240
gcaggcatcc cttttgtgct ttgtaatgac tgtgagtact gttttcgcgg tcagttttcc 300
ctttgcgaga atcacggtat ggtgggttct aagcttttcg gagcctttgc ggaatatgtt 360
aacattccag ccaaaaactg gctgccgatt ggagacatgg attttgagga cgcggcgatg 420
attgagcctt cggccggttc cctccatgga gtcttgaaca ttcaccccca aatcgggtgat 480
gttggtggcg tttttggagt tggcaccatt ggaatccttg cactccagtg gttgaaagt 540
gcgggtgtta aagaaattat tgccattgac atctcggaca agaagctcca ggaatccatc 600
gccctcggat gtgagcgac gattaatcca ttgaaagaga atttgctgga aaaaattcat 660
gagtatacga atggtcttgg agtgacatc gccattgaat gtgcggcgag caagattaca 720
caagaacagt gtcttcttat tacgcgcaag cagggcaaa gcggtatct tggcatcgct 780
tactccgata ttaactctgc ggaaaaagca tttgaaaaca tcttcgcggc gaaactcaaa 840
attaatggtt tctggaattc ttattcggct ccgttcccag gtaagagtg gacaaacacc 900
gtggagtatg tgaagcaagg caagattaag ctgaaggagc ttatttccca ccgattcggg 960
ctggaagaca cgcagcaagc atttaatatg attatcaacc gagaggagga atataataaa 1020
gttatgattg tcccaaacgg cgagtaa 1047

SEQ ID NO: 120 moltype = DNA length = 1026
FEATURE Location/Qualifiers
source 1..1026
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 120

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atgattaatc aggtttacca gctggtggcg ccgcgccagt tcgacgtcac gtataataac 60
gtggacattt acggcaacca cgttatcgtc cgaccgctgt atctctcgat ctgcgcgggc 120
gatcaacgat actataccgg acgtcgatgat gagaatgtgc tgcgaaaaaa attgccaatg 180
tctctggtcc atgaagccgt tggtagaggt gtttttgata gcaagggagt gttgagaag 240
ggcacaaggg ttgttatggg cccaaatacg cctacggagc aacaccatat tatcgccgag 300
aactacctgg cgtcttcgta ttccgcctcg agcggttacg atggtttcat gcaggattat 360
gttggtatgg cgcacgacgg tattgtcccg ctgcccacac atattgacct gagcactatt 420
tcctatacac aattgggttc ggttagctac cagccatttc aacgtttcga gcgtaagtcc 480
attccgctca agaatagctt cggtagtgac ggagacggta atttgggcta catcactgca 540
atccttcttc gcaagctcta cccggaggca aaaatttatg tcttcggaaa aacggattat 600
aaactttctc atttcagctt cggtgatgac atcttcacgg tgaaccaaat ccccgatgat 660
ttgaagattg accacgcctt cgaagtgcgtt ggccgaaagg gctcgcaagt tgcgttgca 720
caaatcggtg agcatatctc cccgaaggt tctattgctc tcttgggcgt gtccgaactc 780
cccgtcgaag ttaacacccg actcgtggtg gaaaagggct tgaccctgat tggatcctct 840
cgttccggtt ccaaggactt tgagcaggtt gtgcacctct atcgaaaata tcccgatatt 900
gtcgagaaac tcgcactttt gaaaggccac gaaattaatg tttgtacact tcaggatatt 960
gtgcaggcat ttgaaatgga cctctctacg tcttggggca agacagtttt gaagtggagc 1020
atctaa 1026

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SEQ ID NO: 121      moltype = DNA length = 1053
FEATURE            Location/Qualifiers
source              1..1053
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = 71193
                    sub_species = aureus

```

```

SEQUENCE: 121
atgttggtgt ttgaagttgt tctcccaatg atcaaccaag tttatcaact tgtcgacccc 60
cgccaatttg aggtgacgta caataacgtg gacatttaca gcgactatgt tattgtccga 120
cctctttaca tgagcatttg cgtgcccagc cagcgatact acaccggatc gcgtgacgag 180
aatgtccttt cccaaaaaat gctatgtctc ctcatccacg aaggagtggg tgaggttgtt 240
ttcgattcca agggagtctt taataagggc actaaggttg ttatgggtcc caataacacc 300
accgagaaag acgacgtgat tgacagagaat tatcttaaaa gctcgtatct tcgttccctc 360
ggccatgacg gtttcgatga ggaacttcgt ctcccttaac acgatcgagc tgtcccgtcg 420
ccggacgaca tcgatctctc catcatttcc tacacagagt tggtagacgt ttctctccat 480
gcccacccgc gatttgaaaa aaaatccatc agcaataaaa acaccttcgg catttgggga 540
gacggaaatc ttggttatat taccgcaatt ctcccgcgca agctctaccc agaactctaa 600
atttacgtgt ttggttaaac tgattataaa ctgagccatt tctccttcgt ggacgacgtt 660
ttcttcatca ataagattcc tgaaggactt acctttgatc acgcctttga gtgcgtcggt 720
ggacgaggtc ctcaatccgc aattaaccaa atgattgact atatttcgcc cgaaggttagc 780
atcgccctgc tcggtgtttc cgagttcccg gttgaggtca atacgcgcct cgtcctcgag 840
aaggagctga cctgatcggc ctcttctcgc tctgtttcta aagactttca agacgtggtt 900
gacctctata ttcaataacc ggatattgtc gacaaattgg cgtctcctaa aggacaggag 960
ttcgaaattg ccacaattaa tgacctgaca gaggcgtttg aggtcgacct ctccacctcc 1020
tggggaaaga cggttttgaa gtggattatg taa 1053

```

```

SEQ ID NO: 122      moltype = DNA length = 1026
FEATURE            Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 122
atgattaata ccgtctacca actggtggca cctcgacgat tcgagattaa gtattccgac 60
attgatctca acaataataa ggttatcgtc cgtcccacct acctctctat ctgcaacgca 120
gaccaacgat actaccaagg actcccgcat gccgaaattc tgcgtaagaa acttccgatg 180
gcgttgatcc atgaagccat tggagaggtc gtgcgagatc catctggaaa ctttaaaacc 240
ggcgagttgg tcgttatggg gccgaacaca cctggtgaag aagatgacat catcgctgag 300
aattatttgc gatcctcgaa atttcgtgca tcgggttttg acggcttcac gcaagaatat 360
gtggaaatca caccggatcg cctcgttcgt cttccccaga atatcgaccg tactgttgca 420
gcttttacgg aaattgtttc tgtcagcgtg catgcaatcg gtcgttttga caagatcgcg 480
cacaaacgac gaaacgttgt cggcatctgg ggagacggca acctgggata tattactcgc 540
gtttttttta agactatggt tccagagacg aagctgtata tttttggagt caaccagat 600
aaacttcaag attttacggt tgcgcacgcg acgtttacgg tggagcacat tcttgaagac 660
gttaagattg accatgcttt tgagtgcgtg ggtggcgagc gctcgggcaa agccattaat 720
cagattatcg accatatcaa tccgaagggt accttttcca ttttgggcgt ttctgaatac 780
ccagtgccaa ttaacaccgc catggtcctc gagaaaggac tgcgcactct tggatcgagc 840
cgtagcggtg tcgcgagatt tgagaagacc gtccgcgtct acaaatccaa tccgtgagatt 900
atcaattatc tttcgaatat cgtcgggtgca gtctgccag tgcgaacgat cgaagatatg 960
aaaaaggcgt tcgaaatgga catccaaaag gcttttggtg agacgattat gatgtgggat 1020
aaataa 1026

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SEQ ID NO: 123      moltype = DNA length = 1032
FEATURE            Location/Qualifiers
source              1..1032
                    mol_type = other DNA
                    organism = synthetic construct

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-continued

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strain = D12
SEQUENCE: 123
atgattaatc aagtctatcg tctcgtctct ccacgccaat tcgaggctcg gtataacgat 60
cgcagcctcc agtatcacga ggaactttgt gtggttcgac ctacgcctct gagcatctgc 120
gctgcggatc aacgatatta cactggctcg cgtggaaaag aagctatggc aaaaaagctt 180
cccatggctc tgatccacga gcccatgtgt aaagtggctc acgaccccaa gggtgagatc 240
aaaaatggca ctatgggtgt catgattccg aacacgccta ctgaaaagga tgaagtcatt 300
ggtgagaatt atctccgac tagcaaatc cgtcgtccg gatatgatgg attcatgcaa 360
gattacgtct tcattgggtc cgcacgtatt gtggaactcc cggaaaggat tgaccctgat 420
gttgttgctt ttaccgagtt gatgaccatc acaacccatg tgttgagccg atttgaatct 480
cgttcccata aacgacgtaa cgtttttggc gtttggggag acggcaacct tggtttcac 540
acgacactct tcctgaagaa gatgtataaa gattgtaaa ttatcatctt cggaaaaact 600
ggatataaac tcgacaattt ctctttcgtt gatgaggtgt atacgattga cgaatccct 660
gaggacctgc gcctcgacca tgcctttgag tgcgttgagg gtaagggttc tcagtatgca 720
atcgacccaa tgattgacca tctccaccct gaaggttcca ttgctcttat ggggtgttagc 780
gaatttcag ttgaagttaa tactcgtatg atcctggaaa aggaatcac actgatcgga 840
agctctcgat ccgttcgaga ggaactcgtg cgcacggcca actttctgaa ggaaaatccc 900
gatgttgttg aataccgtgg tacccttgtg ggcgaggtcc aagaagcaa gacgatctcc 960
gatattttgt ctgcatttga gaaagacctt acatcctctt ggggtaaaac cgtcatggag 1020
tggaagatct aa                                     1032

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SEQ ID NO: 124      moltype = DNA length = 1026
FEATURE             Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = EGD-AQ4

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SEQUENCE: 124
atgtcaacc aagtgtatcg cctggctgat ccacgtcaga ttgaagttca aactgttgca 60
gaagagatca ctgataatga catcgtcgtc ccccccaaat atctgtcggg ctgtcatgca 120
gacacgcgat actttacagg tcaacgtccc caagcaacct tgcgccagaa gctcccaatg 180
gctttgatcc atgaaggagt tggcgagggt gttaaggacc ctcaaaataa atttaagccc 240
ggtagcttgg tcgctatggt ccccaatacg cctttgaaa cgcaccctat cattaaagaa 300
aactacttgc caacgtctaa atttcgcagc agcggttacg atggtttcat gcaggaaatc 360
gttacacttc acgtgtatcg cgcgatcgtc gtgccagaca acttcgacca ccaaatgagc 420
gcatcctatg agatgggttc cgtcggagtc cagccctcca cccagctgga ggcctgatg 480
gatgcggatc gcaagatcat tggaaatttg ggagatggca acttgggttt tatcacagca 540
accctcgcca agcaaatctt tccagactct caactgatga tctttggacg acatcgctcc 600
aaattggact attcagctt tgcggacaag acatacctgg tggacgacat tcttagcgat 660
cttcaagttt ctcaaggctt ggaatgcacc ggaggccgag gctcggagtc tgcatttgct 720
caaattatcc aacacattcg accaatgggc accgctatcc ttatgggagt ttccgaagat 780
cccgttgcta tcgataacag cagcgtgttg gcagagggat tgacctctcg cggcgtgagc 840
cgtagcggtc gcgcggagtt tcagcgcgcc atcgacattt tgacgaaaac ccccgtcaca 900
cgtgaacgct tcgaaaatct tgtgggtttc actcgaaggg tcagcacaat ccaggatatt 960
accgatttct tcgagggtgc attgacaaac tactggggca aggccttat ggagtgggat 1020
gtttaa                                     1026

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SEQ ID NO: 125      moltype = DNA length = 1014
FEATURE             Location/Qualifiers
source              1..1014
                    mol_type = other DNA
                    note = uncultured Ruminococcus sp.
                    organism = synthetic construct

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SEQUENCE: 125
atgatcaatt acgtgtatca gctggctcag cccacgtct tttccctcaa atatgaagaa 60
gtggacttca agaaagaggt catcatccgt ccgaagtgtg ttgctgtgtg ccacgcagac 120
cagcgatatt ttctgggacg ccgcgacgct aagatcctcg gacaaaaatt gcctatggcc 180
cttatccacg aatgttgccg tcagggtggtc tgggatggca caggttcctt ccaacgaggc 240
caaaaggctc ttctcatttc aaatgtgcca ggacgcgacg acggtgtcac atacgaaaac 300
tacgtgtaag gatctcggtt tcttagctcg ggctgcgacg gtttcatgcg cgagtatgtc 360
aacctttccc cagatcgtgt ggttccttc gaacaggttc cgttccgggt ggcgcgcatc 420
acagagttcg tgtccgtggc ggttcacgcg gccggtcgtc ttatgcgagc ggcgcacaca 480
cgccgcgagg ctatcggagt gtggggagat ggctctctgg cttatgctg ctgctgcgtt 540
cttcgtcgac tgcctccgga gtcgcgtatt tttgtggtgg gtaagcatga acgcaagttg 600
ctccagttct cgtttgcga tcgcaccttc ctttttgata gcctcccgga ggacttccat 660
attgaccacg cgttcgaatg cactggagga gagggctctt actatgctat cgatgatatt 720
attaccgcgg tgcgtcccca aggcgctgtt atgcttatgg gtgtctctga aaaccgcgtt 780
ccagttcgca cccgtgacat tctggaaaaa ggcttgacgt tcacccggtt ttctcgttct 840
ggacgcgaag actttgttaa ggcagtcgag cttatggagc aaccccaatt ccaaaaacgc 900
cttagcgcaa ttatttacga ggatcgctcg gtccatactg tcgccgatct gtcccggtt 960
ttcgagacgg atttgaatac gccgttcaag acagttttcc gttgggagat ttaa 1014

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SEQ ID NO: 126      moltype = DNA length = 1017
FEATURE             Location/Qualifiers
source              1..1017
                    mol_type = other DNA

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organism = synthetic construct
SEQUENCE: 126
atgattaaca cgggtgatcg ccttatcgcc ccaaagtcca ttcgagctga ctttgtggat 60
ctgaaattgg atgaagaacg cattatcatc cgcctacat atttgtcgat ctgtgcagca 120
gatcaacggt attacaccgg atccccgtgt cgcgtcgtga tgaagcaaaa acttccaatg 180
gcacttatcc atgagggtgt cggagagggt gtttatgacc ccaaagggtga gctcccagtg 240
ggaacgcacg ttgtcatgat tcttaatact cccaatgaga aagatgatat tattaaagaa 300
aattacctgc gtatgtccca gttccgtgcy tccgatttg acggttcac gcagaatgtc 360
gtgtctatgc gtgcagatct tgtcatccca ttgattacg agccgggaag cgggtgtgctg 420
cttgaactca tgagcgtcgg aatgaatgcy gtgagcactt tcaaaaaaca tgcccatgac 480
cgcaagacg ctattggaat ctggggaaat ggcaacgtgg gctttgtcac tgcactcatt 540
ctgcgaaaag agtttctcga tgccaagatt tatgtttttg gtgtggaaga gcataaaaat 600
cgatactttt cttttgctga tgatacattt atcatcaatg atattcccga ggacttgcaa 660
ttggaccacg cctttgagtg cgtcggaggc cgaggctctg aagcggccat ttctcagatc 720
attgactaca tcaaacgcga aggaacaatc aatttgatgg gcgtgagcga gaatccagtt 780
cccattaata cacgaatggt cctggagaaa ggccccaac ttctcggaaa ttcccgaagc 840
aactacgatg attttaaaaa gtctgttgac atgatggcga acaacaaaga cgttcgtgaa 900
tacctttcca ctattgtctc tgagatggtt gtggtgaaca caattgatga tatgatccgc 960
gctttcgaag atgatactta caatgatctt aagaccgtca tgaagtggga aatttaa 1017

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SEQ ID NO: 127      moltype = DNA length = 1026
FEATURE            Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 127
atgcttaacc aagtttatcg actcgtttct ccccgtcagt tccgagtgga gacagtcgca 60
gaagaagtgg atgcgggatga ctgtatcgtg cgtcccaagt ttctcagcat ttgccacgcg 120
gacactcgat acttttttgg acagcgtcgc caagccgcac tccgagccaa acttccgatg 180
gctttgatcc acgagggcgt cggtaagtc gtttacgac cccgtaatga atacaagccc 240
ggaaccctgg tcgcaatggt gccaaacacc ccggtcaaga aggatagcat cattcaagaa 300
aattacctgc ccagctcccg ttcccgatct agcggttatg atggcttcac gcaagaatac 360
gtgtgtctgc gacgtgatcg agctattact gttccttccg actttgaccc gcataaggcg 420
gogttcattg agatgggttc cgttgagtg caggcaacta ctaagtggga atccacaatg 480
gatgaggatc aagcogttat tggcatttgg ggtgacggaa atctcggtta catcacccgc 540
accctcgtaa agcaaatctt tcagactctc cagctcctcg tcttcggacg ccatcagttc 600
aaactggatt atttctcggt tcagataag acgtatcttg tggacgatat cccggccgac 660
gtcaaaagtt cgcaggccct cgaatgtacg ggcggtcagg gctccgaagc cgcgatccaa 720
cagatcatcc aacacattca accgatgggt acggcaatc tcatgggtgt ttctgagaat 780
cccgtcggaa tcgacactgc ttcggttctt gctgaaggtc tgaactctcg aggagtgtcc 840
cgtagctcgc gcgaggattt taagcgtgcy gttgacatcc tgtcctcttc gccagttacg 900
cgtgaacgat tgcagaatct cgtcggctat gttcagaagg ttagctccat ccaagacatt 960
accaacttct ttgatagcgc cctcactaat tactggggca aggaggtgat ggagtgggac 1020
gtgtaa 1026

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SEQ ID NO: 128      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

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SEQUENCE: 128
atgattaata cactgtacaa acttacggcc ccactgattg tggaaatcctt ttatgaaaat 60
atcgactgtc attccgacaa tattattgtg cgcctccgaaa aactttctat ctgcaaggct 120
gacatccgat actacttcgg aatgcgtaat gctaaagtgc ttaaagagaa gttgcccttc 180
gcactcatcc atgaggcatg tggctctgtt gtctatgaca tcaccggcaa gtttaaggct 240
ggtgaccagg tcgttatgct tctaatatt ccgggtatta acatgcagta tcatgagaat 300
tatcgtcgtg atagcctttt ccgcagctct ggagcagacg gattcatgca ggagtgtgtg 360
tcgatccctc agtcgcaagt catcaaatat gagaagatgg atcaaaacat tgccgcgttc 420
acggagtcca tctctgtcgc tattcatgca gtcagctcgt atcttgatcg cagcgcggga 480
aagcgaaatg ttatcgatca taaaatcggt gtgtggggta gcgcgcgaat gggctacgcg 540
gtgtcgtgtg tcttgaagta cttcattcca tctgcagaga tcaactgtcat tggaggttagc 600
aaagcaaaagc tggacatggt ttcccttgtc aatcatactt gttccctgag cgagcttctc 660
gacgatgcag agttcgatga ttgcttcgaa tgtgtcggtg gagcccaatc tggcaatgca 720
atcaatcaaa ttatcgatca tattctgcg gaaggcctcc tgtgtgtgtt gggcgtttagc 780
gaagaggccg tgctgtgcaa cactcgaatg gtgctcgaaa aaggcatcac tttgttgggt 840
cgatcgcgct ctgagcgttc ggactttatc aaaacaatcg acatccttga tacgaacgag 900
aatattcgac gccgcatttc taacttgatc tctggcgagg tggagattaa cgaggtgaac 960
gacatccaca aggtctttta tcagtcacaa attgcagact ataaaatcat cttgaatttg 1020
cgtatttaa 1029

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SEQ ID NO: 129      moltype = length =
SEQUENCE: 129
000

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SEQ ID NO: 130      moltype = DNA length = 1002

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-continued

FEATURE
source

Location/Qualifiers
1..1002
mol_type = other DNA
organism = Moniliella pollinis

SEQUENCE: 130

aacacgtaga	acacagggat	cgccatttgt	ccgaaatgca	atctttgtct	tcgaagcttg	60
aagcttgaat	agcttgaagc	atgagtgcac	gaggggtgct	tatgatcttc	caaccaaact	120
tgcttgggca	ttgcactttg	gctggatagc	tatcatttag	ctgaccccca	aagagactgc	180
caatgatatg	atatatttgt	aggtgcacga	gtggatgagt	ggcgcgacta	cctgtcacaa	240
gatagatgct	cgatgcactt	catgtgaagt	ggctagtggg	tgctccggaga	ctggagtaca	300
gtaaagagtt	tggagttggg	aacagcgcca	atgaaagagg	ctcagagaga	ggtaggtttg	360
atgtgacgtc	aagcgggcta	gtcagttagc	aagggagtat	ggaatgcagc	gcgcaacgcg	420
cagcgcgagg	cccagtgcgc	ctgcgctcag	gtactcatac	tcagtactca	gtactcaggc	480
gccatgcccg	agcttacatg	caatcaatta	accctgacgt	ctggcgcggc	ctgacgcctg	540
aggcctgagg	cctgacgcgc	ccaagctcca	agcgcaaccc	caggctggct	tgcgctggcg	600
ggcgcgagac	gcagcagggg	gcaagtgtcg	cagagtcgcc	gatcggatct	gtggaacagc	660
acaggagggc	cgtgcgcagt	gctgcgcagt	gcggaggaga	agctggaagc	tggaaagctg	720
aaggtggaac	ctggaacttg	gaacttgga	aggaagaaga	gcgggaaggc	gatgcacgat	780
gcacgatgca	cgatgggggt	ggattgcgta	cgagtggcgc	cacagcgcac	agcgacacac	840
actcacccca	tgcctccacg	ccccagcccc	tgctcgggcc	atgagctgct	ggcaaccccc	900
cgaggcagcc	ggcctttaac	tcaagtagca	ttcgccgccc	gtcggttctt	cttagcctcc	960
caccaccacc	acttgcccgt	gcttggaagc	taattacaat	aa		1002

SEQ ID NO: 131

FEATURE
source

moltype = DNA length = 679
Location/Qualifiers
1..679
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 131

agctcagacc	ttgcatgtta	gccttgacct	tgccctctatg	accactaatt	cgacaagat	60
tgacggttac	cattgtcaac	ttgccatgta	tccaatgttt	cctgtttgcc	atagtaagta	120
gtagatacat	cgatcatcga	tgcatgatgac	gcagatggca	ttggttcgct	ttgccctatg	180
catccatcat	gcacatgcga	tcatgcatca	tgccgcaacta	tccatattaa	tcattgccct	240
ctgtactagg	ccatgacacc	tgacagacct	tcccagggga	tgcttcttgt	aatagccaat	300
atactacaca	actttaccca	ttacgatcga	gatcgatctc	actatagttt	gtcgaaactc	360
gatgcacatc	gaggggtgcg	aaaaaagctg	acggaagcgc	aagtggttga	attggttaat	420
ttagttagat	tggcaggggt	gttcccaaga	ttcaacgcaa	gctgagaaag	aaaacagaaa	480
cacgagggat	aagggcagtc	agaaaaatga	ttacgagggg	agttgaagtg	actgtggggc	540
agtgcctggg	tcgtcatgtg	caaatTTTTT	cttcgggttag	actgcgcagc	catgcagcaa	600
ctctgcgggc	cctatataac	tagtgcttca	tcttcttgct	attcttgtct	ctttaccaac	660
acacgggtac	tcctttcac					679

SEQ ID NO: 132

FEATURE
source

moltype = DNA length = 1002
Location/Qualifiers
1..1002
mol_type = other DNA
organism = Moniliella pollinis

SEQUENCE: 132

catcggtcca	gcttgccctg	ggtgctagcc	agcgcaaaag	tccgacgaac	agacgaattg	60
accgtagtcg	atgacagtcg	atgacagtcg	atgacagtcg	atgacagccg	acgacgcagc	120
aacaacaacc	aaccgcacgc	agcatgaggg	cgagaggctc	aggctgagcc	gctaaggcgc	180
tgaagcctga	agctcgggtg	tccagtcaga	tctcatgato	tcacaccgcc	gcgctgctgc	240
ggtcgtatgt	ttcaatttgc	tgtgcctagc	ctagcctggg	tacgtcggta	actagccact	300
agccactggc	cattatttgg	ctactgggca	cactaggagg	cgcactagtc	catcgcaagc	360
ggtaacctac	ctacgggcca	ctacctgatg	atttccctct	tgccctggcg	cctggcgggc	420
tggcggcgca	ctcttttgcg	tcggacccgc	aaattgcaaa	ccaacaaacg	actagtgtcc	480
ggtacgtgtg	cccaagtaac	ctgggcagac	atagaaccac	attgatcgta	gggaaccgca	540
atggaccagt	acgctattct	cagtaacatt	tgtaacattt	tgccctagatt	ggaatctggt	600
gaggatcaag	atcaaatctc	aaaccaactc	cacggcactg	gttcccgcga	cttgactgtg	660
agtactgagg	tactgaggta	ctgaggtact	gaggctcaac	tgagtctcaa	ctgaccgtga	720
ccgcgcttgt	tggttggcgt	ggattctagt	gttgatgtg	cgagtggtg	tgtgcgggtg	780
tgcgttcaga	ttcgtcggtg	gaacttgact	attcacgtct	tgtgctgcgc	tgacgcttgg	840
aacctggaat	gttaaacctt	gggtcctgag	ggtctccacc	ccccaaagtc	actgggttct	900
atataaagtc	gggtaggcaa	ttggctaggg	agaatctctc	ttttctcttt	ccccagaaca	960
ctgttctact	cacttctcac	ctcttactt	acctacttca	ca		1002

SEQ ID NO: 133

FEATURE
source

moltype = DNA length = 1000
Location/Qualifiers
1..1000
mol_type = other DNA
organism = Moniliella pollinis

SEQUENCE: 133

cctcgttctt	gattgctttc	aaaagtaatt	taaggagaga	acctggctgt	ttgttgcctc	60
actgaatgat	cttacctata	ccagcatacg	acaccttgct	gcccctcccc	ttcaccacca	120
ccatcaccac	tttccctctt	ttctttgggt	ttttctagtc	gttgcaattta	tggttgattc	180
gctcttttag	aaccgtatgg	ctgttattcc	ttcccaactc	ttttgtctgt	tttatcgttg	240

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catccaaacg tttgggctct tctgacctct tttttctctg tttgaccttt atttttggtt 300
tttctatccg tttgatgggt gtgactcttg ttttctcttg ggaagggttg agctcaactt 360
ctcttcgggtc cttgttgagg tttgacttcc ttttctcttt cagctgtggc cgtacattac 420
gatcaataat ttgtgaatat gatatcgact acatcctgaa tcttttagcg aacctagca 480
gttttagcgtc atagatatgc atgcatgatg tacatcttac acgcaccacg gatgatgctc 540
agcgtctcagc caccgcagcg ctcagccgct cgctcggtag gctcgggtcat gatgtgcatg 600
atgcaagcct aatgattgac gggtaactga tctgtgtccc aattcgatcg agatcgatct 660
cactatagtt tgtcgaaact cgatgcacat cgagggtgcg gaaaaaagct gacggcagcg 720
gaagtgggtg aatgggttaa tttagttaga ttggcagggt tgttcccaag attcaacgca 780
agctgagaaa gaaaacagaa acacgaggca taaggccagt cagaaaaatgc attacagggg 840
gagttgaagt gactgtgggc cagtgcctgg gtctgcatgt gcaaattttt tcttcgggta 900
gactgccgca gcattgcagca accctgccgg ccctatataa ctagtgtctc atcttcttgt 960
cattcttgtc tctttacca caccgcgta ctctttcacc 1000

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SEQ ID NO: 134      moltype = DNA length = 1007
FEATURE
source              Location/Qualifiers
                    1..1007
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 134
cttgacgggt tcaatattca atactcagtg ttcaagcagt ggtaggttgc tccgtggatg 60
gaacgacggg ttaatttct tccgccatgc accatgggtc atgatcggtt cagatggaca 120
gttggtatga gaaagtggta acagataaat agaatttctg acaagtggc aagagcctgg 180
ccgatgggtc atggctcgat ggctcgatgg ctcgatggct cgatggccac acaccatgct 240
atctactttc tgtgcttcta tgcggacgag tcggaacagc tgtcaaatcg cgtttggggt 300
atttgagcgg aatttgaaac ctgtgtgaaa ctacattgtc tctctggga 360
tctgtttgat cgaacggcgt catgcaatag acacgcgatg aggcccaacg acggtgacgt 420
tgaacgcctg gccacactgc taccaacacc tagaacgtgt tgaagttgaa cgtcggacgt 480
tggacgttga gttgacgctg gatgctgtaa ccgtggatgt agcagcgaca gacaagccct 540
ctaagcaatc tagtcgattt tcaaaccaac taacgcagcc ttagttcgga caaccctagt 600
ccagtgcggg ggcagcgtgg ggtcagttgc cagcgaaggt caccacagct agctggggtg 660
gctagcttga ctggctcaga agtatcgatg ctggagcttc aggccagtac gaaaaacac 720
tgtctagatg ctcttccggt agttctgggc tcttgagccc ggcggttggt catcggtgca 780
cagtgcaacg accggcgggt gggacaacaa actgtctggg cgtgaaatcg tgcgagtgtt 840
gatgcaaacg ggaagtgacc agcacgcagt tcccagtag cagacatact cagtacctca 900
gtggttgagc ctcagcatcg cctccgtatt tctacaaagc tgttgagtcg ccttttgact 960
ttttactacc accatccctt cttcttctaa accccttaca ttccaaa 1007

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SEQ ID NO: 135      moltype = DNA length = 1000
FEATURE
source              Location/Qualifiers
                    1..1000
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 135
tcgacgacag tgagtatagt caactatgct atctagtagc caaagtaagt ctacattccg 60
ggctttgcag gggtttagcta ttctattaga caatacttac aaccttcgaa gggcttcccc 120
agaattcaat tctatattga ttcaatatcc acccatcgga cccgctctcc gtaccccttg 180
tctgcttcgc cactacaacc atgtcaagac ctgagaatcg tagcaaatg cccactacg 240
gccgagtgcc tctcggctgg ttaaaagtaga atgacttggg tagccagatc gagacgcctg 300
acattgatac gcagacagcg cagtcagttt caaactgaca actgacgcgt actggattga 360
caggactcct tggccggtgc catctgaatc gctttagtag ctcttgccag catcttgcca 420
gagaagcttg aaggctggat cgctcagcct gagctcacac ttagtagctc cactagtcac 480
tcagggtgctc agcctctcct cagcgctcctc agctcaagcc tcaggggcctc agctgcacg 540
agctctgggt ccagtgccag tgccagtgcc agtgccagtg ccagtgccag tgccagtgaa 600
gagagtggtg tgagcccaagc tatacccaaa agcaaatgat ggttattgct agcttccatg 660
caccctctgg gcgaaggaca acgaacaaca gacagagttt gaattattca ctggcctagt 720
actaccaatt taaatttggt ggttcagcgc aggaacggcg cgagttagcg taagggcaaa 780
ctccgcagtg ggcgtgacgt cagtgtccct ctggctgagg cactccagca ggctctgggt 840
aacgcataca cgtgtgtgtg tgtctaccaa gcacattgtg cgctcgtgtg ccgtgcgccc 900
aaatcgaccc agtcagaacg tctataaaac cctcttgtct tggagggtta ccgggcccgt 960
ttttcccaac gctccggctc actcactact ctacacaaca 1000

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SEQ ID NO: 136      moltype = DNA length = 997
FEATURE
source              Location/Qualifiers
                    1..997
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 136
tcggacccca agctctctag cttcaagctt tcaagctagc ttccagccag ggtcatggaa 60
ccgtaggagc tgggaaacgt gacgttgtcg ctatgcctat cccatataag aattaacgtc 120
gaatcatgag gcgcattgga ggtgcattg gtggtatgat acataacgtt atgattctat 180
atgagcgatg gagcgcggcc gtctgcgctc agtgtggaag ttcgatacgc ccagtccatt 240
tcgttcagat gacgttttgt ccagaagtac ggtacatacc gagcgcaaac gtcgaaacta 300
tggccttggt cgctcgttgt cggcgctcgt cgacactttc atcaacggac gaaaggtaga 360
gccgaacgcc agcggccaat ggcatggcg atagtagaat gaatagtcgt cagatctagg 420
tcttaggcta caggccaccc tgccctgaggc ctgggggtct tacagcgagt ctaaaaaacg 480

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aaaagccaac gtgccaactt gggcctgccc atgcaccctg ccagagagcc ccaagctggt 540
ctgatgcaag cggagcaaaa tgttggaaga atagtttcaa cgatgggttg gttccagtac 600
gcacagtcga gagatgaggt actgcgcaac gagttgttgg cgacggaacc gaagatggat 660
tgaactatth gatctacagg tgatggcgtg gagtcgggcg tggggccgct gtgtgatatg 720
tagaagtggt gctattcgag ggctgcgccc gtctggatgg atgcagaatg aagcttgcat 780
gcccagctcg gcaggatggg cgcagaaccc ggtgaaacat agttaaaagc cggggccagt 840
gtcatatttc gcttgagcca ttctactcgt tggcagccca ggcgcgccta attacgcca 900
actcttggtt ctcgagagct cgggcaatgt ggtatataag cttgaggttc cctacagcaa 960
agcctggcct cttaccacaa accgcactct agcaaac 997

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SEQ ID NO: 137      moltype = DNA length = 1002
FEATURE            Location/Qualifiers
source              1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 137
gagcgtagag agctocactg atggcctggt tacaagttgc ctgaagacca ccaatacggc 60
cgcacagcca acggcaggcc aagaaaggat aattgacaac cgtgaactcg aaagcttcgg 120
cgctcaggag ctcatgtggt ttgtttaga tataaagggt tctgaatacc agttgggtat 180
tgacacgggg agttggactg aatctctcag atctcccaga tctcaagtac tgagtcccaa 240
gcgtgccata attaaactta ggttgcgcgg gacacgccac ctgagatcag ggtgtacgcg 300
cataactaat cattccacca ccaaagcaag actgagggta ccgcctgagg ttctggctat 360
tccaacagtc acaagtcaac gcgattgccc tcaaatccaa aagcgtagta ggcgctcgac 420
ggagggtatg tggccagtac ccaatttcga gcaactcagt agatctgttc gtgtttagt 480
gtcctgcgag atctatctta aatgtatact gtgccaggta ggatctctcg gcacctatag 540
atgatataaa agtatatgat atatgccaa tgccaaagcc tcaaaactgac aaaatgaaag 600
aaacagcagg caagctaggc aagcgtctat cgggggatta aacggatgag tcagcagcac 660
aggagcagtc agtcgacaat tcgggctttt tcccgcgcgc aaggcccccag accccagacc 720
ccagccgcca ccattgctcgt atccgcgcgg gttagcatga ccacggcaga ttctccgct 780
cagcttgctc ggcagactat atataaagaa tcctttcttc gcagtcctgc cagagccctc 840
ctccacccca tcggtttacca ccaacgacat cacttggtct tgacgcgcgc ctgcgttttc 900
ccgcttcgcg gtctccctag tgatttagct ggtgttttcc tctaccgcgc actctactgt 960
ttcatcgctt gttccaagt ttctcagtt cattctcatc aa 1002

```

```

SEQ ID NO: 138      moltype = DNA length = 1002
FEATURE            Location/Qualifiers
source              1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 138
tggacgtgga caagcacctt gaagtcgctc agggcttcca gatcaaggct ttgcctaccg 60
tctacggcct gtatgagggc aagcctgttc gtcagtgtaa gttttaaaa atattgttct 120
tgtctgttcg ggttttacat cagttggtga ctgacacggc tttttgtttt attgtgcag 180
tccaagggtg acctcctgag gatattcctg gccaatttgt ccaagatctt cccagcttgg 240
cgcgccaaag tgcccagaac ccgcctgctg aaagtagcga ggctgccaac taagtatagt 300
ccatcctatg agctgtccta ccccaaacaa ctttttcacg cccaacgtcc aaacgtcact 360
tcttattgag gcaagaagaa gctgcttccc cactcattca ttaccctttt ctgatccacg 420
agggtgaccc atggcctttg tacattccat gaaacaggta tccttttccc caccattaca 480
tctgaactta caaagacttt gcttctcttt gtacacagct cgtcgtccat cgtccatcat 540
cgatgacgat catcctgcat tcgggtatcat gcaccatcat gcagcagtg caacgagggc 600
catgacccat gctgcataa ctatgtgtat agccaatagc caatottcaa tctccaact 660
cctacagaca cagcctgtgg ggaatcgctt aatcgctgtc catgcgtcca tgctccgct 720
tcctcgcgct cgtcgcgccc tcgcgacgat taagtcacag tcacgcccac acagcgcgcg 780
agcgcacgag ccaccatcat cactgtatga gacaaagagt ccacttcctg aaaagcaatc 840
ttaataatag atgtaacct agcaatgtag ccctaataat ttgtttggcg tttattttat 900
tgcgcccccg cagcttttgc accccggaga gtaaaagtat tccgggtgct cgtcagagca 960
tccctttctt tctccatcag caaagcgtgc gaccgccaag aa 1002

```

```

SEQ ID NO: 139      moltype = DNA length = 921
FEATURE            Location/Qualifiers
source              1..921
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 139
tttgatgag gcattcaatg tctccatggt tgaagaccag agaaagtgtg ggtaacacaa 60
tagccatgga caattagcga atccgaatat gtgtactaac attgtgcctc caactgacag 120
ctggtcgaat aatgattgaa atcatcaaac atatcagaga tatatcccag aaccatttta 180
caactgttat tgtatgtatg cattactggt ttacattgtg tctttccaaa tactaacatt 240
gccattatcc aatccataca tatatatata tatagattac acaacgcaaa cctcgacagt 300
gatttggttt acatcgtgta agcagccttt tacggaatat gatggacaac cagatcctca 360
tagatggaga caatgtgcag atagattcca tttatgatga cattgatgtg tttttaaatg 420
caaaataatc tcctatataa gtgagttccg agtctcccca tgcttatctt atcgtgtttt 480
ttctgtaaat taggtcaaa gattcaaaac attccctcaa tcttcccat tttttgtgta 540
gccccatat catcaaatg acgtatcaaa tttcctaatt agtagccccg tatgttaaat 600
tagacatgtg tgactgccat ccgggacagc ctacccaatg acaccgcgc accgcacata 660
ttgtgtttag tgcgcgcgtg ctgctgaagc gactcctgtt ttggaggagg cagagggcgg 720

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-continued

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gttgcccgat cccctgcccc tcgctcctcc tcctgggctc ccccttgccag agggacaccg 780
aggggatccc tcgtgtgaga gcttgagggt ggatgggtgt caattttctc atttgattga 840
aagatttggt atattgaaaa ttcagtttgt ggaagtgtgt attaaaaggt ttactgttt 900
gttctgtaga cacattcaat a 921

```

```

SEQ ID NO: 140      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 140
acggatcaac aaaaatgtgc gccagctttg ttgtgtgccc ttttctccta gcggatccta 60
tccacatttc gtgaatgaac cgattaagct ctgacgagat ttccttgttt tctgtagata 120
cctctttagt aggtgtcaaa gttatgttgt gatcgagaga gttacctgca aggtattatg 180
caagcgggctt gatgaggaca ctcaactacg tgtgagttag agtgcggtgc tcagcagttg 240
acgcgtgagc cgttgagca gcttgataaa ggtcctaata gtcttcagc tgttggcaga 300
cgacaatgtg ctgtagtagt aggatagccc cgtaaaagtg tagaagcaat aggttgagac 360
tgactgtaat gggactactg gttcaaaagc tgaaagtcca aaggattgct caaaggcccc 420
tgaggtctga gcctctgagc gcgctcttcg tcaaaattcg ccggggccat cggtcatcct 480
ctcgagggca gcatcttctc 500

```

```

SEQ ID NO: 141      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 141
gtactaacca agtctattgt aaagctggtg tctgctgctg cttcagcaag tgccattgct 60
accatcgcaa gcaaatgtga aaaaaattcg ttcaatggtc taatttaagt aaaatactca 120
taatttataa aatgccgctt gtccctacct tttgtcttcc tccttattat agtagctgta 180
gttatcgtag tcctcaggac ttcgtaacg ctcggttagt atcatatata cttcacctca 240
ttcattcata aagcatcctt taggattgta tgatccggca gtgcatcatg catcatgcat 300
catgcatcag gacatcatgc atcatgccgt atcggctgat cattgcattt gattggataa 360
tacgatcgtg tttaggcatt ggcagtcgaa acgaggataa agaagaggtc agtaccagtg 420
cacagcacgc gctcgcgctc tatgcgcact ttgacgcttg aaaaaaactt ttcagccaat 480
cggatataga cagcggaccc 500

```

```

SEQ ID NO: 142      moltype = DNA length = 493
FEATURE            Location/Qualifiers
source             1..493
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 142
atggctctgag ccgcccacga ttccttccct catgagcgta tgacgacctt ttacttagc 60
atgtgcctct aaagaatacgt ttttgcatac ccacatatcc gtaaccctgt cctctatagc 120
ttctacaggt catagtcact ggctgccagt ataaacgtgt caatatctaa tcctcataag 180
atgcctcgct ctgtcgtcca caataaaaatt cagtaacgca aggactgcgc gactgagtga 240
ttgccgatga tatgagaaaa ctaatttcaa gccatttgaa cttcaataaa tgcttccgct 300
aacatcacgg atcatcgatc atcaaccagc ctccattaat aggcctgcca ccaactcgcg 360
aatcatgaat ctgatcatac ctggctagcg attctcataa caccagaatt acctgcccat 420
taaagaaacc ctattctgcc tgtaaccccg caccagccgg tccgaattca cgacagacta 480
aggctctcgc gca 493

```

```

SEQ ID NO: 143      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 143
atcagttgga atcatcaacc tttttttttt gtatcgattt cttcctaccc tcacctccct 60
tttccccctt aactcctcgc tttgaacctc cttaccggag tgcctacctt tacctttttc 120
tttttctctc tccccctgctt aatattctag gcttgctact atactgctct ctctaacacg 180
atttttgcgc ttgtaagcat ttttctcttt ttaaatctag gttttatcat ttgttgcat 240
tatccaagtt tttttttttt ttgttttcta gtgtttttac tgtttttgtg tttttagct 300
tgactaggtt cttattctat tgtagatgta tatgatcgga tgcattcccc atctaaaagg 360
aaaatgttct cgactgatat ttatttttac cttattgtct tcctaaatgg ttgactcct 420
gcctctctcc ttgggacgct ttgcaacggt ttgtacgtac gtaagggttc atcgtttga 480
taagcgacac agttgtcaag 500

```

```

SEQ ID NO: 144      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 144
attcacctcg gagtaactgc gcttcacgtt ttctttgtta cgacaataat accacgctgc 60

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atgcttttgaa ttgagtatat attttgatga ctgggttcaaa tagatctttc ctccatagcc 120
ttcgagtact cgtgatctca gcattatcag atttgtgtcc agggcccccga ggacgatgcg 180
gttttccctgg agtcactagt tgggaagatt tgtactgtgt ggggtgtagg agagagaacc 240
attgattttg gtgtagctgt tcgttttata aggcgagat tacgtaatgc acatgctgcc 300
ctagcaaatc gcgtcgcgtc acgc 324

```

```

SEQ ID NO: 145      moltype = DNA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 145
atgtgcttag attaacatac acttgacttc cttacgcgtt gatataact tggatctcca 60
taactttctat tgtatagttt aatcaatgga ctccatgttg acatccatt atgcatctca 120
tagtagctgg aaatagcaaa attgaagttc ggcagacgac tgcaccagga ggggtagatt 180
ctgtccata actccctacc acaggccgtg ctctgggtcg attacctta gcctgatgcg 240
ctcaatgtaa cgcgtaattc tgagtgggtg tcggtcgccc aaggtttcag tatgcgctgt 300
gcagatagta ctgtatcgtg aaactcgacc gttgcacatg cactttatca agtgcgtag 360
gcccagagtc gtaggccaag agtcgtaggc caagatggaa ggcctgaaga aaaagaaagc 420
tttgatcggt ctgactctat caatggattc atgagctaag tattagctat acaaatgatt 480
cgagcatagc agaagg 496

```

```

SEQ ID NO: 146      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 146
gttgetccct ttgetgttct atgatccaaa agatgcttag tgetccactc tctactcccg 60
actacctttt acgcattctg acatgactag tagtgcaatg ttacccctgg tgttttctcg 120
caatcacctt ctgctcgcta tagcttttga tcgtcagtag ttttttgga tagctaaagt 180
agacttttgt acctggcggt caatgtaatt ttactccttt tttgtcgat gcaacagatg 240
gtgtaaaagt cgtcgtaaaa gttttggcgt gatgcttgat tcggtgacgg ttgcttgggt 300
gatccttata tcgtctctcg gtgttctcat aagccagggc tcttcttagc tatgtacaac 360
aggtccacca ctgactatgt tatgtagagg catgcattta acttacggaa agatttaagt 420
gagaagaacg tacctagtca gatgcttgac atgaaaacgt attcctgcaa ttcacaaatc 480
agtatcctgc ctgtccactc 500

```

```

SEQ ID NO: 147      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 147
gcgggttcgca cttttatccc aggcataaaa aatgacgcac ggctgtcgca cgctacaccc 60
agtagtcgcc ttttcaggca gtctatgagg ggagcttttc attgccaaa attactagcc 120
tgcaattcat gatacagggt tcactttctc cggatgtcta tagatttctc atagcgcttt 180
gaaaagacaa tttagctact cagtttataa gcaaatgatg tggaggtatt tattaggtag 240
aagatcacga aacgttttaca gcgggggaag caagtggcaa atcgctgaca acaccgcatg 300
gtgggtgggtg aatcgctact catcaaaagt tgcctgtcac tttgggaggg accaggttga 360
ggccatcgcc cgaagggtgtg tgtggccagg gtttcgatat ttcatgaatc aacgtgttga 420
tggtattgga ttgggtaaga ggtcgatccg gctgttggtc agagtgtcga tcacattcgt 480
cgttgtcaag cccgagttgc 500

```

```

SEQ ID NO: 148      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 148
gtggcggttga ttcgatatct attattgata tgtcctgttc ccacggtgc caagtgattg 60
ttcatgctat cctctctcca ttctgtttgt gttgcacagt cggtagtgtg acgtagtagc 120
tcgtcacaaa atatgaagtg agccttgcaa ttaggcatt cacaagctgc taggcaggct 180
atgatctaag ctaagtggca aaaattctct tgattcaatg ccaaactata ctattgttcc 240
tcaggcaaaa gcagctatat aagttgcatt caacgcaacc gccatacaag caaattagag 300
agaaacggga tataaggtgc tttatccctg aaggagtagc cagcgtctac ttgggttgcta 360
cacaatacag caaacagtgt acttaatgag ccattggctg aactggagct cgagaggtt 420
cgagaagggc gggtcatacc agctgccctc catggaagct ggggtgtcga tgatgaaaag 480
tgtcggaacy tacgtagta 500

```

```

SEQ ID NO: 149      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

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```

SEQUENCE: 149

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-continued

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acgaagggtt tcccactgtg gtgacctgat cctcttttac gtttttttgt tttgtttttt 60
cttttggtca tgatcaactt gggagtgatt cctcttggtg atgtcgaaaa ttgaactgtt 120
ataccaatat tctgttgctt ttgtttcctg tttgtagtgt actacttttt acagtagtgc 180
ttgttcttcg cttgcttgcg tttgtcaggg ttcccttta gcttagtacc aatatctagc 240
cttaattttg cttattcttt ctcaccgccc atgattgatt tcaatcatga ttaatgattt 300
atgcatgaaa actcaaaata ctcaattgtc ttgtacaagg caatcacaaag ctgggcagcc 360
attctcagcg cctataagag ctttcagcca ggccggcccc aaaacgatct accctgtgga 420
cctggctagg ctaaaccttg gaaagtcatt caattatgaa cgtttgaagc tcaagcgctc 480
ctaccctctc ctggagcagt                                     500

```

```

SEQ ID NO: 150      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 150
tgggtgtcac cactgttgct tcccctttgt tggagcgctc gtcgttctga tggcgctctg 60
agcccaaaat atcccttctt cgcgcgagag caagcctcgg agtgtgcttg caatctactt 120
ttcctactaa agttatcttc ttgtggcttc tgaatttgaa tagaacgtgt tcagacttgt 180
tagctaaatt tgttgatgcg agatacttga cagaatcccg ttgatgaact catctgtgtg 240
accaccgcct ggtgatggaa tctcgcgcgg aagccgtggt gttactgact ctcaagcaat 300
tcagtagata aatttgtata acatcattgc aaggggtgct agcaatcaat tgtccaatac 360
caaacggcgg caagtgcaaa gggggcगत agagtttctg cgaggaaggg aaggaatccg 420
tatgcctgaa aacttttgta caagtgtctt tgtaagggcg aggtctgctc tgacatctct 480
ctgagatctg agatctgata                                     500

```

```

SEQ ID NO: 151      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 151
gtttcttggc tgttgacgaa atcttatgaa atgcacgggt ctaatatctc ctgccaatgg 60
tgtaccatcc tcgttttgta atgtcatagc tagctctgta gtttctaaac gaccgtctta 120
ggccgggggt tgtgatttgg ggattaagga taggatactt ggccagtttg ggccgatcaa 180
tatcaatctg ctaggcgctc cttagggtcaa cttgaattat gatatatgag gtatgaggaa 240
agtggggtca atctgtggat taggatcaga aaaaacgtgg cagaccctcg atctaggcct 300
acgtcagcaa tttcagcaat ttcagcgctt tgccccttag atagggttca gccgctcagc 360
gcgcgcaatt tcagcatgga tgagatcaaa ccaaggggca gcagtgtggc accctgtagt 420
tccaatcgca tgcttagtgc ataacgcgca catcaacgtg tcagattctc agagatctca 480
acaagcccac tagaccctag                                     500

```

```

SEQ ID NO: 152      moltype = DNA length = 504
FEATURE            Location/Qualifiers
source             1..504
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 152
aatcgccgtg gtagcgaaac gggccagctg ggtaaggcca gcggccggta gctaggacta 60
tgaggctacc aggtctgtgg gtaactaggg agctaggcac ctaggcaact gagcaactga 120
gcaactgggc aactgagcgg tataggtttg gcttggggcac agttgatatt gatatcaacc 180
agtcgctcgt tcggttcacc agaaccaggt gtgtggtagc gagcgtagaa tagccaaaag 240
gttttgagct gaggcgtgta ggcgcacagg gagggacatca ccgcctgagc acacaggatt 300
tcaaactgtc agacggcggt ggtgccagtg ttggaagtgg gcgaagcacc tcgcatctaa 360
gacatgtaat taaactctcg acagccatcg tccattgaca ccaacgtgca ccaacacacg 420
cgagagctgc ttcgttgtgc ctgtcgacga cattgcctcg ttccccagcc aagtcgcccc 480
accgctacac caagcgccact cgca                                     504

```

```

SEQ ID NO: 153      moltype = DNA length = 499
FEATURE            Location/Qualifiers
source             1..499
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

```

```

SEQUENCE: 153
catccgaaat tgggtggttg acaaatgcac cttcttctcc aggggctcgt gcaaacgcag 60
tagcccgagaa gaggcgtacc ggagaggaag acccaacaac cactgatcgg tttgaagttg 120
gatgagaaag ttgtgccaat cgcgggagcg tagcaacgct acgagcgcta cctacaacga 180
ctccacgcgc aatcggtacg cgcaaggaa cggtatgagc accgagcagg ttgcgcgaag 240
ccatgacgat tgccttacc ccaacgagct ggatcaagtt gggcagaaca taatttatct 300
cgcttccccg cgccacaagt agctccacca ttcttctcaa aatagccgga ttccgctttc 360
gcgggggcaaa acttccgect ataccgatcc gattgaaacg ccaaaaaaat taagtgttag 420
tacgcctggt ggtcgagccc gactggagca ccatttgaga cactaggaga taattttattc 480
ttttccttct tggcccca                                     499

```

```

SEQ ID NO: 154      moltype = DNA length = 300
FEATURE            Location/Qualifiers

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-continued

```

source                1..300
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 154
cgggtaagac caagtcggtt ggtgtttcga acttcagcac tcgtctggtc gacttggttg 60
aggaagcttc gggcgaaagt cctgcggtta accagatcga agctcaccct ttgttgcaac 120
aagacgagtt ggttgctcac cataagagca agaacattgt catcactgct tacagtcctt 180
tgggcaacaa tgtcgctggt aaaccacctc tgactgagaa ccccggtatt gtggatgctg 240
ctaagcgttt gaaccatact cctgctgctg tgctcattgc ttggggattt caacgcgggt 300

SEQ ID NO: 155        moltype = DNA length = 453
FEATURE              Location/Qualifiers
source              1..453
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

SEQUENCE: 155
gcgaggagta gttactaggt agcggatatg caattccgcc cataatgcat aaccgtttctc 60
cctaccaaaa caaaacaatg gctacgtggc cggcgtgcgt ggccgacatg gtgcgggtgc 120
ggggggcggt gtacgagttg ggggagttca ctagtctgct ggctagctga ctagtgcacc 180
ggcgtgtggt tgtggcgagt cgagtggtga aagattaaat atgaaatatt gaatatcaaa 240
atgacgtttg ggggtggcag tgcgcgcgcc aacaacaaca agagcagcca ccaagccaaa 300
gcaccaagca ccaaccccgc acaccatttt gttacaccgc cggcagaaat ggcgtcagcc 360
agcagtcact attactcaat attcaatagg ggttcactt gactcacttc ctcatgtata 420
tatatatcca tctaattctt ccaatacata acc 453

SEQ ID NO: 156        moltype = DNA length = 500
FEATURE              Location/Qualifiers
source              1..500
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

SEQUENCE: 156
gcggcctatg cagacgcttg gagcttgggc acattggata ttggatatgt gctggctgcc 60
gttcgagatc ccgttgctgt cgcacccgcc accggtcacg ttgtcacttg gtcacgtgga 120
ccactgcac  tgcgcttagc tacaagagtg catgatgcat gatgtcatgc atgatatgca 180
tgatacaagc atgaatatgt atttcaatat tgaacattta gtattgatat ctcgatatgc 240
gtaacacgtg acacgcacca tatcaatcag ccgcaccggc actaaatgtt tcaatttcaa 300
catcccacac gctcgaatgc tccctcgagc taatcggatt tcggagctcc gacatggccg 360
gcctcgagcg gagtttgaat ctcggttcgc cccgaccccg ctggctgcgc cgatgaaatc 420
ccgagtcaag tcatagcaag ccaagtggct ggctgactca aaagctaata aaagtctgag 480
caagtaatac tccgcacgct 500

SEQ ID NO: 157        moltype = DNA length = 500
FEATURE              Location/Qualifiers
source              1..500
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

SEQUENCE: 157
tcgcgtcttc aaaccgtggc gcgcgcacac tgcgcgcacc aaaatttgtt ggtgtatggg 60
aatagggttt tctaaaattc tgttgaggat gtgagtatca ctggcatggc accagtgtac 120
tgctcatcga taatcataaa cgtatcaacc ctctcggtctt gcgatctgta tctgctactg 180
gagggcacag caagccttct gcttggttgc gctctgcgct ctctgtgggt ctttgccgct 240
cgcgcccaag gtcgcgagcg ggcattcacc accatcaggg ttaagccacc agacgagtgc 300
tgctcgccgg gccagctcgg cgctgtcac gcccccacat tggtttgatg ctcggtctct 360
tctttcatct tggaaaccggc gggcaatggc gcttatcacc actcctgggc cgggccccgc 420
gtatactctc ataactagtg cacgctttca aataaatacc tcgcctcctg gctggccgac 480
cgacgtttac ccaccgcgcc 500

SEQ ID NO: 158        moltype = DNA length = 499
FEATURE              Location/Qualifiers
source              1..499
                   mol_type = genomic DNA
                   organism = Moniliella pollinis

SEQUENCE: 158
tggcatgtag gccgtgtagg gctatgccta tatgctcatg cactcatgtc gcgatgttgc 60
gatatcgaga tggaccaacc aagcaagcgt gcagcgtggg cgcttggggc aaggggtgta 120
agctaagtgt aagcactgaa gtgctgaagt gctgaagtgc tgaacgtgga acgtcaggat 180
gacacatcct ctcccttcgg ctctgcacca gaaagggttg tcagccatct gagatctgag 240
atttgaaatg aatataatga tgaattgagt cgatgagatc tgactatctt gggagattgc 300
accaatgcgc cacggtgacc agatgggtct ggtcttagct ggtatagtgg caccgtgatct 360
gccttgacag atctcatggt catagcttca tgcattcaat ctatctgctt agcctgtctg 420
gatctgcccc actagacgag gacaaatgta attgatcctt gtaacctctg tgctggtcgc 480
agcagcgaga tacgatgac 499

SEQ ID NO: 159        moltype = DNA length = 500
FEATURE              Location/Qualifiers
source              1..500

```


-continued

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mol_type = genomic DNA
organism = Moniliella pollinis

SEQUENCE: 159
aaatcttcat ccagcgtagc aagcatatcg cgcagtgcg agttctctga gtgtgcatac 60
cgaacccaaa aagccatgga tgtcattgac ctgctgtatt catcaatcga gatattctta 120
ctccggagcg tacagtggca gatattatgt aggagatgat gggacaggat gcagagatac 180
ataatccatt tccagttaat agggccgcta gatacttgat attagcctgg cagcgagcga 240
gcgagggtag agtagaatag agtccagtat agatccacc ccatgggtgcg ggaagtggat 300
gacgtaatag cgcgatagct ccccgctccc ttgtcaccta ggccacggga ccctacgtga 360
ttgcatcgty cgtgtcgtcg cctgtcgacg cctgggtgcg gcatacgtcg cgtgtctcga 420
cgatcaacaa caccaccaac cagactcca gtccgggcca gccagtaacca tggcgtcggt 480
ctttgggttt agtgcaccg 500

SEQ ID NO: 160      moltype = DNA length = 400
FEATURE            Location/Qualifiers
source              1..400
                    mol_type = other DNA
                    organism = Moniliella pollinis

SEQUENCE: 160
aaccagatcg aagctcacc cttgttgcaa caagacgagt tggttgctca ccataagagc 60
aagaacattg tcatcactgc ttacagtcct ttgggcaaca atgtcgctgg taaaccacct 120
ctgactgaga acccggtgat tgtggatgct gctaagcggt tgaaccatac tctgtctgct 180
gtgtctcattg cttgggggtat tcaacgcggg tacagcgtct tggtaagtc agttacaccc 240
tcacggatca agagtaactt tgaacagatc actctgtctg atgaggaatt ccaacgggtt 300
accaacctca tcaaggagta cggtagagag cgtaacaacg ttcttttcaa ttacaagcct 360
tcgtgggtcca ttgacgtctt tggtaaccag gacgaggcta 400

SEQ ID NO: 161      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

SEQUENCE: 161
atcctgtgct cgtctcttac attcatcttc ctgccataa aactttcaag ctatttctc 60
caacgtactc aacaacctat ttttttgggc cccgctcaaa acgcacctta ttgcgtctaa 120
gtgtattcta accacgcgac ctgaattgca aagcctgtta ccccatcgcc accttcgtct 180
taagatagtt gtttaacgag ccttgccag ttttagcctt gaccatgact acagatcata 240
atgggttagat gctctctgat cgataatgat atattgcatg tcgacgattg aagattttat 300
ggatgtcgat gacgactgtc gagctcttat tcttagcaac aacttaataa ttactatcgc 360
aatgcacgcg acatcatgca tctgtctcat gaatatcact ggatcactct cgatcgacgc 420
ccactaggtc agcataaacac gcttgaagct gacttgaagc tttggaacaa atggcagcaa 480
cagtggcggt cacggtgacc 500

SEQ ID NO: 162      moltype = DNA length = 486
FEATURE            Location/Qualifiers
source              1..486
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

SEQUENCE: 162
attgttgcac cgtctcatca gatcgctcct cattgtagtg ataattgggt caaccttgta 60
ttagactgta tagcagaatc cctcgctata aaaagtatta tgcatacgca gttccaagca 120
ggaattagta tggtagtggg acttggtcga agggctgacg acatggcgag aaaaacgtat 180
caattcggaa cacactgtgc aatgtcgatc acgttttcgg tggttcacgg taggcttctc 240
agtccatggg aaagtgtgta catctatccc ttgagcgctt gagcttaatg ttaccttgtg 300
caaaagtcat ttgaagcttc acttgaagct tgaagctttg aagttttgaa gcttgaagcg 360
caaacagcag cggatcatgt atactggcgt aacctgccta cagtgggtgca tcagtacgtc 420
aagcgggtag cccgaaatct cattaaggtc atttgccctc taagcttagt acctagtgat 480
ctgttc 486

SEQ ID NO: 163      moltype = DNA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

SEQUENCE: 163
cctgaacgtc ctgaactctg tcttattaca taccattaat ttccctgcct tccctctcta 60
cgtcatcgga attttatctc ttctgagca gcaaagattt ggtcctgtgc cgggatacaa 120
cggagagtcg tgattgggtt tatggaagag aggcgatagc acttgcatth tgaccattcg 180
gggattaagc gaacccccaa gtggcacaaa gagctcgatg gagggtcagt aagtatagag 240
ggacatgcaa aggagagaaa aggaacacac atattgacaa atagagacaa gtacagggtg 300
tcgcaatagc atgatgtagg ggtatgcgca agtcacaatc gttcagggcg gccataacat 360
cagcgtaccc gagcacaatg gttaatgatt aatgagcaga ggaaaaata atgattatgc 420
cgggatggc aggaactaact ggcgcagctc ttcagcttcg gcaacgcgtg cacatagtgg 480
tgtagacca cgagattgag c 501

SEQ ID NO: 164      moltype = DNA length = 493

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-continued

FEATURE
source

Location/Qualifiers
1..493
mol_type = genomic DNA
organism = Moniliella pollinis

SEQUENCE: 164

```

cctttgcctc acgcctctcg tttgcacgct tcttcacggc ttatcacgcg ttatttatc 60
catggaatcc cctccaaagt tttcttcaac atagcttcat atcttctacc ataacatcta 120
tcccaccaag tagcacatga attcaacgta gcttgagaac ttttctcgt cagctgattg 180
cgttcttttag gcagtttgct tctaccccaa ttgcttctac cccaattgct tctgcttatt 240
tcaccagtcg gctcggagat ctctgacaac tctgagatct gatgatcaac ctcttggtga 300
acttgtaaaa tttgatcgat gagacgatta gggacttgaa gctttaagtt agacgcgatg 360
ggcggtaato tatatgctta gactttgttg ctatgtttgt cctataagat ctgatgcgatg 420
atgtatcatt acccagacgg gctatcgcac aaggtgcaga tgcattctat catgacagtg 480
ccacactgtt ggc 493

```

SEQ ID NO: 165

FEATURE
source

moltype = DNA length = 567
Location/Qualifiers
1..567
mol_type = genomic DNA
organism = Moniliella pollinis

SEQUENCE: 165

```

gcttgctggg cacattattg tcagtaccta ccatgaagaa ttcaatatta tttacattg 60
tcaaccatta catggatccc atgttggtga tagcctctat gtcatccttt ggtcttttat 120
gtctgtagcg tgctagtagt attcaagttg tagcactgct gtgaaacagt acgttcaagt 180
agtgtgcagc agtaagttca gttagtggac tggggcagcc tcgccttcag caaaggttag 240
gtacatgagc cagttgtcca ctcttgcttt ctacgaatcc cataggctac ttgccccct 300
aacgtgacg acccctgtg tttgagaagt ttgagttctt ttacggtacc gtgaccaagc 360
gagcagagta gaaactcttg aagtgtgtga tgtaagaata caaattgaga acacaaggtt 420
acctaccttt atagttggta aacagcaagc aataccagag attttcatcg caacgtgact 480
ctatcattca ctatattaag ttgcacacgg atgtgtgaca aactacacg taaaagagag 540
aaaaatacta atgtcgatcg ctcaggt 567

```

SEQ ID NO: 166

FEATURE
source

moltype = DNA length = 504
Location/Qualifiers
1..504
mol_type = genomic DNA
organism = Moniliella pollinis

SEQUENCE: 166

```

ttgctgttac gacctcattg atcgatgac tccgacactt tgagtagctt tccatcctct 60
ctttactgta ctttgatttc ttgctacttt gtgttcttct ttaatgggtc cgaaccggtg 120
attttgtcca tgactttttt tttttttttt cgtcaactgc aagaagatga aaaatctact 180
gttcgatagc acctcaatac cgttccagag tcccagacaac cacaaggatg tttcttagat 240
agggtgcagc aagatccaca aaaaagtgtg ggtattagat ggtttcaagc aaggtacaat 300
attctaattc aatactcatt tgaataatca atattcaaca gcttctagcc tacaagcttc 360
aatgtttccc gtcgggtaac tcagtatact tagtactcag ttcaagtcag tgttggttaa 420
ataaaatcac gtgatccaaa aacttgaaac ttttctttt tctcctcaga tgttggtgat 480
tcgtcccata gcggttacca caac 504

```

SEQ ID NO: 167

FEATURE
source

moltype = DNA length = 1679
Location/Qualifiers
1..1679
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 167

```

aagacctctg tcccacaac cttgatggtc tgtgaaaccg tcaacaacat cattggtaga 60
actgtcaatc caagaaacaa aaattgggtc tgtgggtggt cttctggtgg tgaaggtgct 120
attgttggtg ttagagggtg tgttattggt gtcggtactg acattggtgg ttccattaga 180
gtcccagctg ctttcaactt tttatacggg ttgagaccaa ctcacggtag attgccatat 240
gctaaaatgg ctaactctat ggaagggtcaa gaaaccgttc actccgctcg ttgtcctatc 300
actcactccg tcgaagactt gagattgttc accaaatctg tcttggtgca agaacccttg 360
aagtagcact ctaaggctcat ccccatgcc a tggagacaat ctgaatctga catcattgcc 420
tctaagatta agaattggtg tttgaacatt ggttattaca atttcgacgg taacgtcttg 480
ccacaccac caattttacg tgggtgcgaa actaccgttg ccgcttggc caaggctggt 540
cacaccgtta ctccatggag tccatacaag catgatttct gtcagtactt gatttcccac 600
ctctatgctg ctgatggttc tgcgacgct atgagagaca tttctgctc ttgtgagcca 660
gcatccctca acattaagga cttgttgaac ccaaatatta aggctgttaa catgaacgaa 720
ttgtgggaca ctcatttaca aaagtgggac tatcaaatgg aataacttga aaagtggcgt 780
gaagctgaag aaaaagctgg taagggaattg gacgctatta tgcctccaat tactcctacc 840
gtcgtgtgca gacacgatca attcagatac tacggttacg cctccgttat tagcttattg 900
gatttcaact ctgttgctgt cccagtcact ttcgctgata agaatttga taagaagaac 960
gaatctttta aagctgtttc cgaattggat gctttggttc aagaagaata cgaccagag 1020
gcttatcacg gtgctcctgt tgtgtttcaa gttattggta gaagattgtc cgaagagaga 1080
actttggcta tcgccgaaga agtcggtaaa ttgttgggta acgtcgtcac tccataatgg 1140
tgttcaccac cttgtcttcc cctttgttgg agcgtccgtc gttctgatgg cgtctgagc 1200
ccaaaatato cctctctcgc ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc 1260
ctactaaagt tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgttag 1320

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ctaaatttgt tgatgcgaga tacttgacag aatcccgttg atgaactcat ctgtgtgacc 1380
acgcctgggt gatggaatct cgcgcggaag ccgtgggtgtt actgactctc aagcaattca 1440
gtagataaat ttgtataaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1500
acggcggcaa gtgcaaaggg gcgatagaga gtttctgcga ggaagggag gaatccgtat 1560
gcctgaaaaa ttttgtacaa gtgtctttgt aagggcgagg tctgctctga catctctctg 1620
agatctgaga tctgatacct gcaggataac ttcgtataat gtatgctata cgaagtatt 1679

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SEQ ID NO: 168      moltype = DNA length = 876
FEATURE            Location/Qualifiers
source              1..876
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 168
agatgtagcc ggcgctgtag agttttggac ggataggttg ggtttcagtc gagacttcgt 60
tgaagatgac tttgcaggcg tcgtgcgtga tgatgtgacc ctgtttatct cggccgttca 120
agatcaagtt gtccctgaca atacacttgc atgggtctgg gtacgcggat tagatgaatt 180
gtacgcggaa ttgctgaaag ttgtatccac taactttcgt gacgcttcag gtcctgctat 240
gactgagatt ggtgaacaac cctgggggcg tgagttcgca ctccgagatc cagctggaaa 300
ttgtgttcat ttcgtagctg aggaacagga ttaatgggtg tcaccaactgt tgcttccct 360
ttgttggagc gtcgctcggt ctgatggcgc tctgagccca aaatatccct tcttcgcccg 420
agagcaagcc tcggagtggt cttgcaatct acttttctca ctaaagtatt cttctgtggg 480
cttctgaatt tgaatagaac gtgttcagac ttgttagcta aatttgttga tgcgagatac 540
ttgacagaat ccgctttagt aactcatctg tgtgaccacc gcctgggtgat ggaatctcgc 600
gcggaagccg ttggtgttact gactctcaag caattcagta gataaatttg tataacatca 660
ttgcaagggg tctagcgaat caattgtcca ataccaaacg gcggcaagtg caaaggggcg 720
atagagagtt ttgctcgagga aggggaaggaa tccgtatgcc tgaaaacttt tgtacaagtg 780
tctttgtaag ggcgaggtct gctctgacat ctctctgaga tctgagatct gatacctgca 840
ggataacttc gtataatgta tgctatacga agttat 876

```

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SEQ ID NO: 169      moltype = DNA length = 1087
FEATURE            Location/Qualifiers
source              1..1087
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 169
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag 120
attgcacggt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtacg 180
tagtagatag atcgatcatc gatgcagatg acgcagatgg cattgggtcg ctttgcccta 240
tgcatgcacg atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgacg gagatcgatc tcactatagt ttgtcgaaact 420
tcgatgcaca tcgaggggtgc ggaaaaaaagc tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaaacaga 540
aacacgaggg ataaggccag tcagaaaaatg cattacgagg ggagtgtgaa tgactgtggg 600
ccagtgcctg ggtcgctcatg tgcaaathtt ttcttcgggt agactgcgcg agcatgcagc 660
aactctgcgc gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catggcaaaag ttgacgagcg ccgtgcccgt gctaaccgcg 780
agagatgtag ccggcgctgt agagttttgg acggataggt tgggtttcag tgcgacttc 840
gttgaagatg actttgcagg cgtcgtgcgt gatgatgtga ccctgtttat ctggccggt 900
caagatcaag ttgtccctga caatacactt gcatgggtct gggtacgcgg attagatgaa 960
ttgtacgcgg aattggtctga agttgtatcc actaactttc gtgacgcttc aggtcctgct 1020
atgactgaga ttggtgaaca accctggggg cgtgagttcg cactccgaga tccagctgga 1080
aattgtg 1087

```

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SEQ ID NO: 170      moltype = DNA length = 1257
FEATURE            Location/Qualifiers
source              1..1257
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 170
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag 120
attgcacggt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtacg 180
tagtagatag atcgatcatc gatgcagatg acgcagatgg cattgggtcg ctttgcccta 240
tgcatgcacg atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgacg gagatcgatc tcactatagt ttgtcgaaact 420
tcgatgcaca tcgaggggtgc ggaaaaaaagc tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaaacaga 540
aacacgaggg ataaggccag tcagaaaaatg cattacgagg ggagtgtgaa tgactgtggg 600
ccagtgcctg ggtcgctcatg tgcaaathtt ttcttcgggt agactgcgcg agcatgcagc 660
aactctgcgc gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catgggaacg actttagatg ataccgccta ccgttatagg 780
acatcagtac ccggcgatgc agaagcaatt gaagcccttg atggatcatt caccacagac 840
acagtttttc gcgtgacagc tactggtgac ggatttactc tccgtgaagt gcctgtagat 900

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ccgccattga cgaaagtatt tccagatgat gagagtgatg acgagagcga tgatggcgaa 960
gatgggtgatc ctgaactcgcg cactttcgtc gcatacgggtg atgatgggtga cctgggcgga 1020
tttgtttgttg tgtcctactc ggggtggaac cgtagattga ccgtcgaaga tatcgaagtc 1080
gcgcagaaac atcgtgggtca tggcgtggga cgagcactga tgggtctagc tacggaattc 1140
gctcgagagc gtggggcgggg tcacttatgg cttgaagtta ccaatgtaaa tgetcctgct 1200
attcacgctt atcgccgaat ggggttcacc ttgtgtggcc tcgacacggc attgtat 1257

```

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SEQ ID NO: 171      moltype = DNA length = 1049
FEATURE            Location/Qualifiers
source              1..1049
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 171
attgaagccc ttgatggatc attcaccaca gacacagttt ttgcgtgac agctactggt 60
gacggattta ctctccgtga agtgccctgta gatccgccat tgacgaaagt atttccagat 120
gatgagagtg atgacgagag cgatgatggc gaagatgggtg atcctgactc gcgcactttc 180
gtcgatacag gtgatgatgg tgacctggcc ggatttgggt ttgtgtccta ctcggggttg 240
aaccgtagat tgaccgtcga agatatcgaa gtccgcgcag aacatcgtag tcatggcgtg 300
ggacgagcac tgatgggtct agctacggaa ttgcgtcgag agcgtggggc gggtcactta 360
tggcttgaag ttaccaatgt aaatgctcct gctattcacg cttatcgccg aatgggtttc 420
accttgtgtg gcctcgacac ggcattgtat gacggtactg cttctgacgg ggagcaagcg 480
ctatacatgt ctatgcctcg ccctaatggg tgttcaccac tgttgccttc ctttgttgg 540
agcgtccgtc gttctgatgg cgctctgagc ccaaaatata cctctctcgc ccgagagcaa 600
gcctcggagt gtgcttgcaa tctacttttc ctactaaagt tatctcttgg tggctcttga 660
atttgaatag aacgtgttca gacttgttag ctaaaattgt tgatgcgaga tacttgacag 720
aatcccggtg atgaactcat ctgtgtgacc accgcctggg gatggaatct cgcgcggaag 780
ccgtgggtgt actgactctc aagcaattca gtatataaat ttgtataaca tcattgcaag 840
gggtgctagc aatcaattgt ccaataccaa acggcgggcaa gtgcaagggg gcgatagaga 900
gtttctgcga ggaagggaag gaatccgtat gcctgaaaac ttttgtacaa gtgtctttgt 960
aagggcgagg tctgctctga catctctctg agatctgaga tctgatacct gcaggataac 1020
ttcgtataat gtatgctata cgaagtatt 1049

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SEQ ID NO: 172      moltype = DNA length = 1199
FEATURE            Location/Qualifiers
source              1..1199
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 172
cttaaacacg gcaagggtag cgttgcaaac gacgtaaccg acgaaatggt ccgtctgaat 60
tgggtaactg agttcoatgc actccccacc atcaagcact tcattccgtac acctgacgat 120
gcgtgggtgt tgactacggc catccccggt aagacgcgtt ttccaggtcct cgaggaatac 180
cctgactcgg gcgagaacat cgtggatgcc ttggctgtgt ttttgccgag ccttcaactc 240
attccggtct gcaactgcgc ttccaacagc gaccgtgttt ttaggcctgc ccaggcacia 300
tcacgcataa acaatggtct agtcgacgct agtgatttcg atgacgagcg caacggcttg 360
ccggtggaac aagtgtggaa agagatgcat aagctcctgc cctttccccc agactctgtc 420
gtgacgcata gtgattttct gcttgacaac ctcatctttg acgagggcaa gctcattggc 480
tgcattgacg tcggccgggtg ttggcatgcc gatcgctacc aagatcttgc gatcctctgg 540
aattgtctgg gcgagttctc ccttagcctc cagaagcgtc tgttccagaa atatgggatc 600
gataaaccag tatcttcttg tggcagttc catttgatgc tggatgaatt tttctaattg 660
tgttcaccac tgttgccttc cctttgttgg agcgtccgtc gttctgatgg cgctctgagc 720
ccaaaatata ccttctcgcg ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc 780
ctactaaagt tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgttag 840
ctaaaattgt tgatgcgaga tacttgacag aatcccggtg atgaactcat ctgtgtgacc 900
accgcctggg gatggaatct cgcgcggaag ccgtgggtgt actgactctc aagcaattca 960
gtatataaat ttgtataaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1020
acggcgggcaa gtgcaagggg gcgatagaga gtttctgcga ggaagggaag gaatccgtat 1080
gcctgaaaac ttttgtacaa gtgtctttgt aagggcgagg tctgctctga catctctctg 1140
agatctgaga tctgatacct gcaggataac ttctgataat gtatgctata cgaagtatt 1199

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```

SEQ ID NO: 173      moltype = DNA length = 1621
FEATURE            Location/Qualifiers
source              1..1621
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 173
tattcttgta cgagccatcc caaaaatgga tcatgacagc tgccaagagc caggactata 60
agattgagat ctacagcagc gatgacctca agtccctgaa gctcgaagag gcctttgcga 120
acgaaggctt tcttggtctat cagtatgagt gccccggcct catcgagggtg cctaccgagc 180
aagatcctag caagagctac tgggtgatgt tcatctctat caatccggga gcaccgcgg 240
gcggctcctt caaccagtat tttgtgggtt cgttcaacgg aacgcacttt gaagcttttg 300
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SEQ ID NO: 175      moltype = DNA length = 1475
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SEQ ID NO: 176      moltype = DNA length = 1982
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gaagataact ttagtaggaa aagtagattg caagcacact ccgaggcttg ctctcgggcg 420
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agcaacagtg gtgaacacca 500

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1. A genetically engineered yeast cell capable of producing arabitol, the engineered yeast cell comprising:

an exogenous polynucleotide sequence encoding an arabitol-phosphate dehydrogenase (APDH) enzyme comprising a sequence at least 80% identical to SEQ ID NO:11.

2. (canceled)

3. The yeast cell of claim 1, wherein the yeast cell is an osmotolerant yeast cell.

4. The yeast cell of claim 1, wherein the yeast cell is a cell of the subphylum Ustilaginomycotina.

5. The yeast cell of claim 1, wherein the yeast cell is selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*.

6. The yeast cell of claim 1, wherein the yeast cell is a yeast cell of the *Moniliella* genus.

7. (canceled)

8. The yeast cell of claim 1, wherein the cell is a *Moniliella pollinis* cell.

9. (canceled)

10. (canceled)

11. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

12. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is operably linked to a heterologous or artificial promoter.

13. The yeast cell of claim 12, wherein the promoter is a constitutive promoter.

14. The yeast cell of claim 12, wherein the heterologous or artificial promoter is selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:86), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:132), translational elongation factor 1 promoter (TEFp; SEQ ID NO:133), modified TEFp (SEQ ID NO:131), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:134), 3-phosphoglycerate

kinase promoter (PGK1p; SEQ ID NO:135), enolase promoter (ENO1p; SEQ ID NO:136), asparagine synthetase promoter (ASNSp; SEQ ID NO:137), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:138), and RPL16B (SEQ ID NO:139).

15. The yeast cell of claim 1, wherein the arabitol-phosphate dehydrogenase enzyme has a sequence at least 85% identical to SEQ ID NO:11.

16. The yeast cell of claim 1, wherein the arabitol-phosphate dehydrogenase enzyme has a sequence at least 90% identical to SEQ ID NO:11.

17. A method for producing arabitol, the method comprising:

contacting a substrate comprising dextrose with the engineered yeast cell of claim 1, wherein fermentation of the substrate by the engineered yeast produces arabitol.

18. (canceled)

19. (canceled)

20. The method of claim 17, wherein the fermentation temperature is at or between 25° C. to 45° C. and the volumetric oxygen uptake rate (OUR) is between 0.5 to 40 mmol O₂/(L·h).

21. The method of claim 17, wherein arabitol is produced at a rate of at least 0.1 g L⁻¹ h⁻¹.

22. The method of claim 17, wherein arabitol production is at least 10 g/L when the fermentation is run at 35° C. for 96 hours.

23. The method of claim 17, wherein erythritol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

24. The method of claim 17, wherein erythritol production is less than 50 g/L when the fermentation is run at 35° C. for 96 hours.

25. The method of claim 17, wherein glycerol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

26. The method of claim 17, wherein ethanol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

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